Insurance and Asymmetric Information in Wage Contracts: Evidence from an Online Experiment

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Link to paper



Link to slides



Insuring Labor Product

- Most workers' labor product depends on uncertain factors
- What kinds of contracts might reduce that uncertainty?

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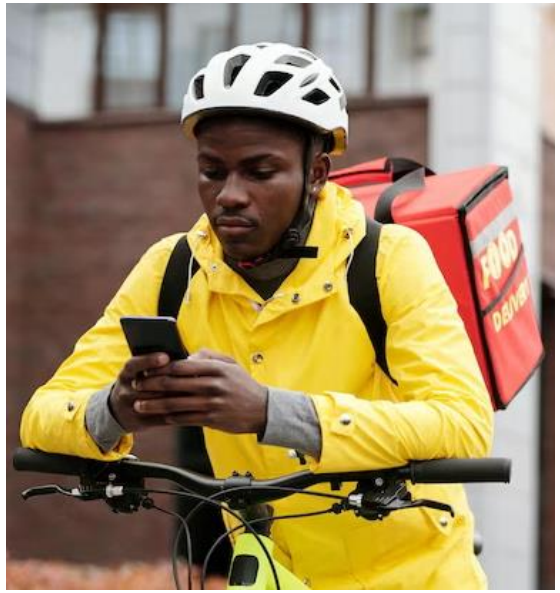
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- Employers provide *implicit insurance* through **fixed wages**

Piece-rate Pay

- Paid by unit of output
- Worker sells labor product at market price
- Examples:
  - Rideshare driver paid per mile
  - Self-employed freelance photographer

← Earnings Risk Insurance →



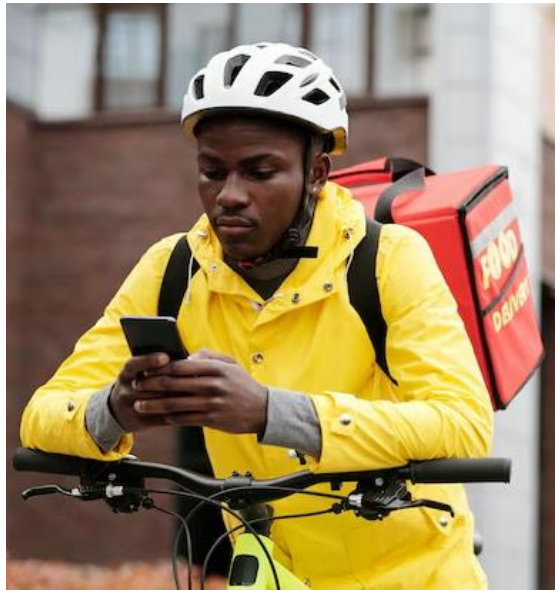
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  - Limo driver paid per hour
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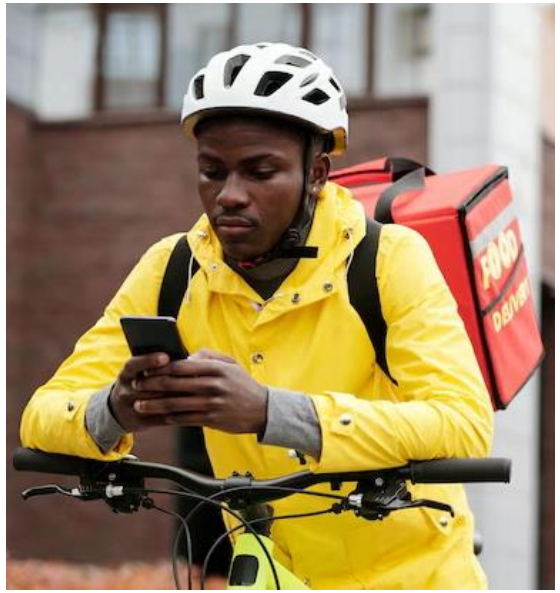
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Hourly wage + tips

Base pay + commission

Annual salary + bonus





Are these contracts socially optimal?

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Earnings Risk Insurance

Asymmetric information ⇒ market distortions:

- **Moral hazard:** fixed wage induces less effort
- **Adverse selection:** less productive workers sort into fixed wages

⇒ Suboptimal wage contracts

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Earnings Risk — Insurance

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Public policies can mitigate these distortions:

- Taxes on tips, commissions, or bonuses
- Hourly wage subsidies
- Employment classification rules
- Portable benefits programs
- Minimum wage

⇒ Promote implicit insurance in wage contracts

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Earnings Risk **Insurance**

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How do moral hazard and adverse selection influence wage contracts?

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay
   - Two-stage RCT separates moral hazard, adverse selection, and wage effects
     1. Offer workers a *choice* between randomized hourly wage or piece rate
     2. Randomly increase hourly workers' wages to match highest wage offer
   - Find evidence of both moral hazard and adverse selection into hourly wages
     - MH ≡ Treatment effect: take-up of hourly contract ⇒ output ↓ by 8.7% relative to the mean
     - AS ≡ Selection into treatment: wage offer ↑ by 10% ⇒ marginal worker's productivity ↑ by 1.9%

       (2nd stage top-ups ⇒ estimates are independent of potential wage effects)

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay

2. Develop model of fixed-wage labor markets under asymmetric info
   - Clarifies roles of AS and MH in shaping equilibrium, allowing for monitoring costs
   - Equilibrium and efficient shares of hourly work are determined by three curves:
     1. $\bar{w}(\theta)$: **reservation wage** for an hourly position at quantile θ
     2. $MV(\theta)$: **marginal value** of output among workers at a given reservation wage
     3. $AV(\theta)$: **average value** of output among workers with lower reservation wage

This Paper:

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3. Use MTE framework to estimate equilibrium and efficient hourly wages
   - Let experimental wage offers serve as **instrument** for hourly contract take-up ⇒
     - Hourly labor supply at given wage, $S(w)$ = propensity score
     - Marginal value under hourly pay, $MV_1(\theta)$ = potential outcome in treated state
     - Marginal value under piece rate, $MV_0(\theta)$ = potential outcome in untreated state

   ⇒ MTE of hourly contracts map directly into model

This Paper:

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3. Use MTE framework to estimate equilibrium and efficient hourly wages

4. Calculate DWL of inefficient wages and welfare impact of tax/subsidy
   - Underprovision of fixed-wage jobs ⇒ Welfare loss of $0.05 per worker hour

     ⇒ Annual loss among online clerical/data-entry workers ≈ $1.6 billion globally
   - Piece-rate tax ≤ 18% ⇒ raises $1 of govt revenue at social cost of $0.88 or less

     ⇒ *Eliminating* taxes on output-based pay (e.g., tips) likely has low social returns!

Existing Literature

- Asymmetric Information in Labor Markets
  - Spence (1973), Weiss (1980), Gibbons and Katz (1991), Farber and Gibbons (1996), Shimer (2005), Ales and Maziero (2016), Pallais (2016)
- Selection and Incentive Effects of Wage Contracts
  - Lazear (2000), Nagin et al. (2002), Gaynor et al. (2004), Shearer (2006), DellaVigna and Pope (2018), Kantarevic and Kralj (2016), Angrist et al. (2021), Brown and Andrabi (2021), Emanuel and Harrington (2024), Kelley et al. (2024)
- Measuring Adverse Selection & Moral Hazard
  - Chiappori and Salanie (2000), Karlan and Zinman (2009), Bryan et al. (2015), Einav et al. (2010), Kowalski (2023), Beaman et al. (2023), Herbst and Hendren (2024)
- Selection and Marginal Treatment Effects
  - Black et al. (2022), Mogstad et al. (2018), Huber (2013), Carneiro et al. (2011), Heckman and Vyctlacil (2005, 2007), Björklund and Moffitt (1987)

Outline

1. **Experimental Design and Results**
2. **Model of Asymmetric Information**
3. **Estimates of Marginal Value and Welfare Loss**
4. **Optimal Piece-Rate Tax**
5. **Extensions**

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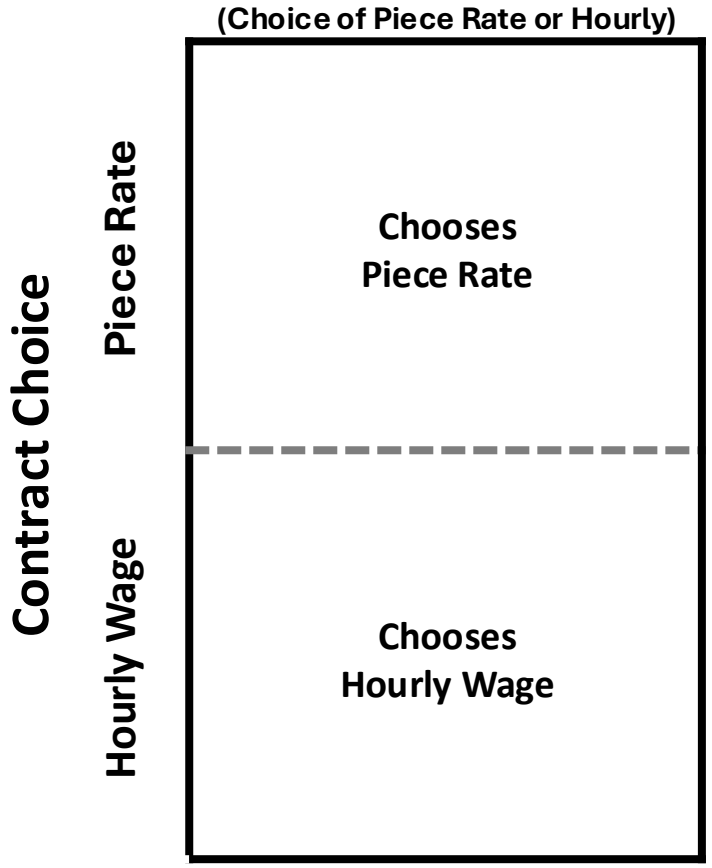
Detecting Moral Hazard and Adverse Selection

- Suppose worker i is offered a choice:
  - Remain on piece rate ($H_i = 0$)
  - Switch to an hourly wage ($H_i = 1$)
- Potential outcomes:

  Y_{1i}: Worker i's potential output under hourly wage

  Y_{0i}: Worker i's potential output under piece rate

$$Y_i = H_i Y_{1i} + (1 - H_i) Y_{0i}$$



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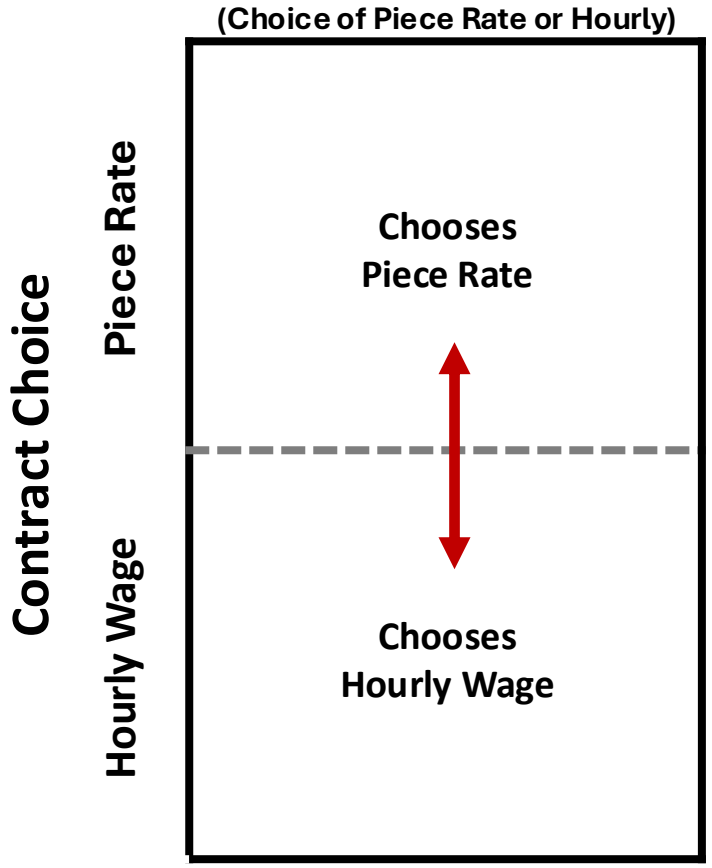
  Y_{0i}: Worker i's potential output under piece rate

$$Y_i = H_i Y_{1i} + (1 - H_i) Y_{0i}$$

- Comparing across self-selected groups combines treatment and selection:

$$E[Y_i|H_i = 1] - E[Y_i|H_i = 0] =$$

$$\underbrace{E[Y_{1i} - Y_{0i}|H_i = 1]}_{\text{Treatment on the Treated}} + \underbrace{E[Y_{0i}|H_i = 1] - E[Y_{0i}|H_i = 0]}_{\text{Selection on } Y_0}$$

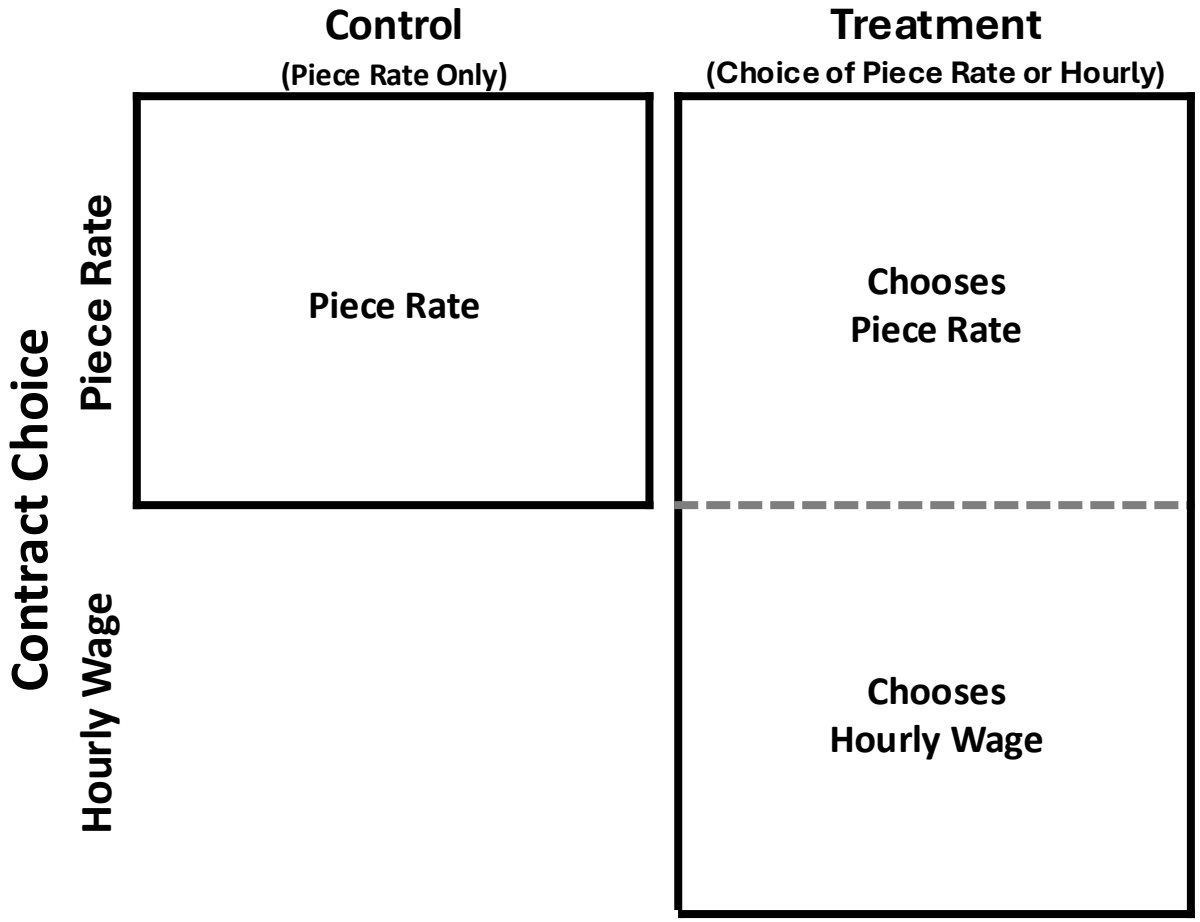


Experimental Design: Single Wage Offer

Simplified Experimental Design

Randomize workers into two offer sets:

- **Treated workers ($W_i = 1$):** offered choice
  - Remain on piece rate ($H_i = 0$)
  - Switch to an hourly wage ($H_i = 1$)
- **Control workers ($W_i = 0$):** no hourly offer
  - Remain on piece rate ($H_i = 0$)

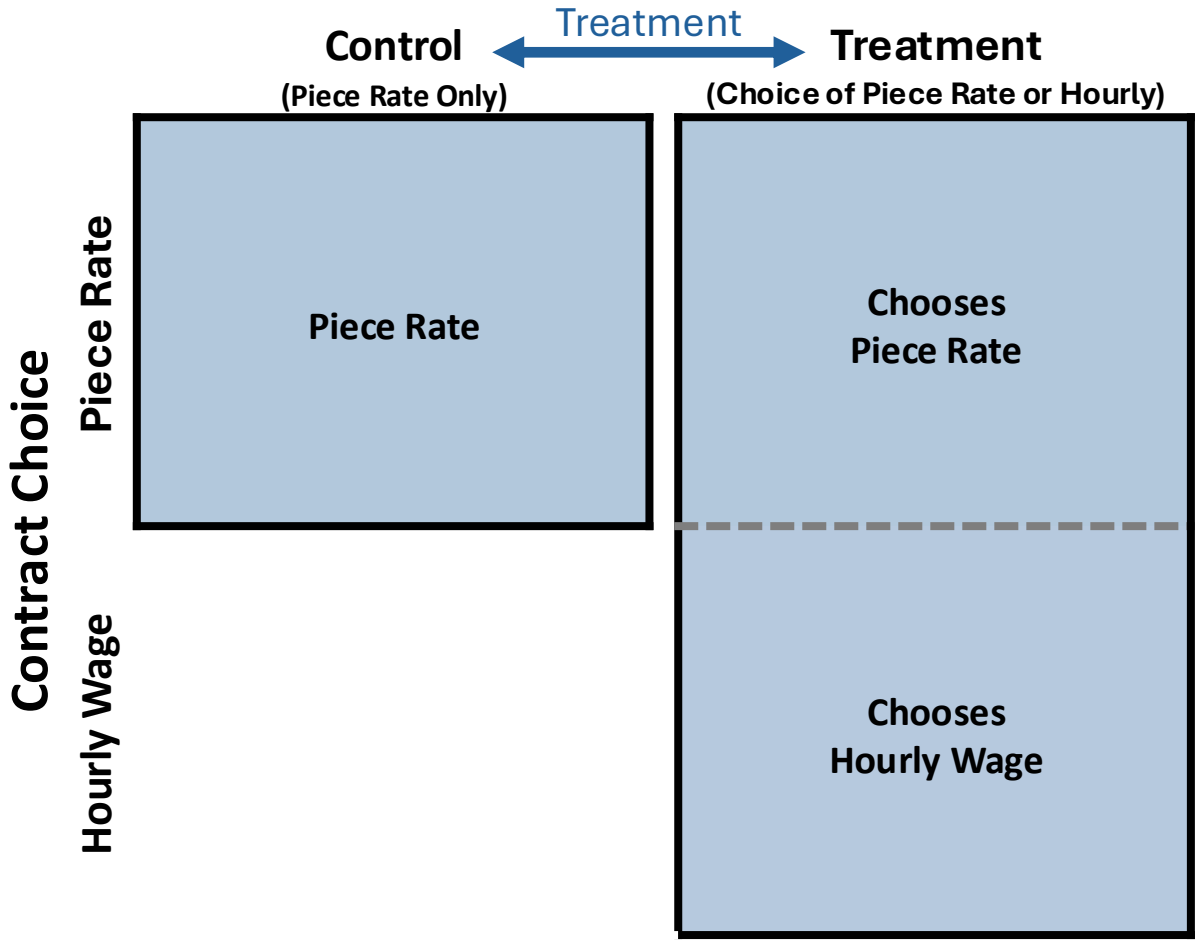


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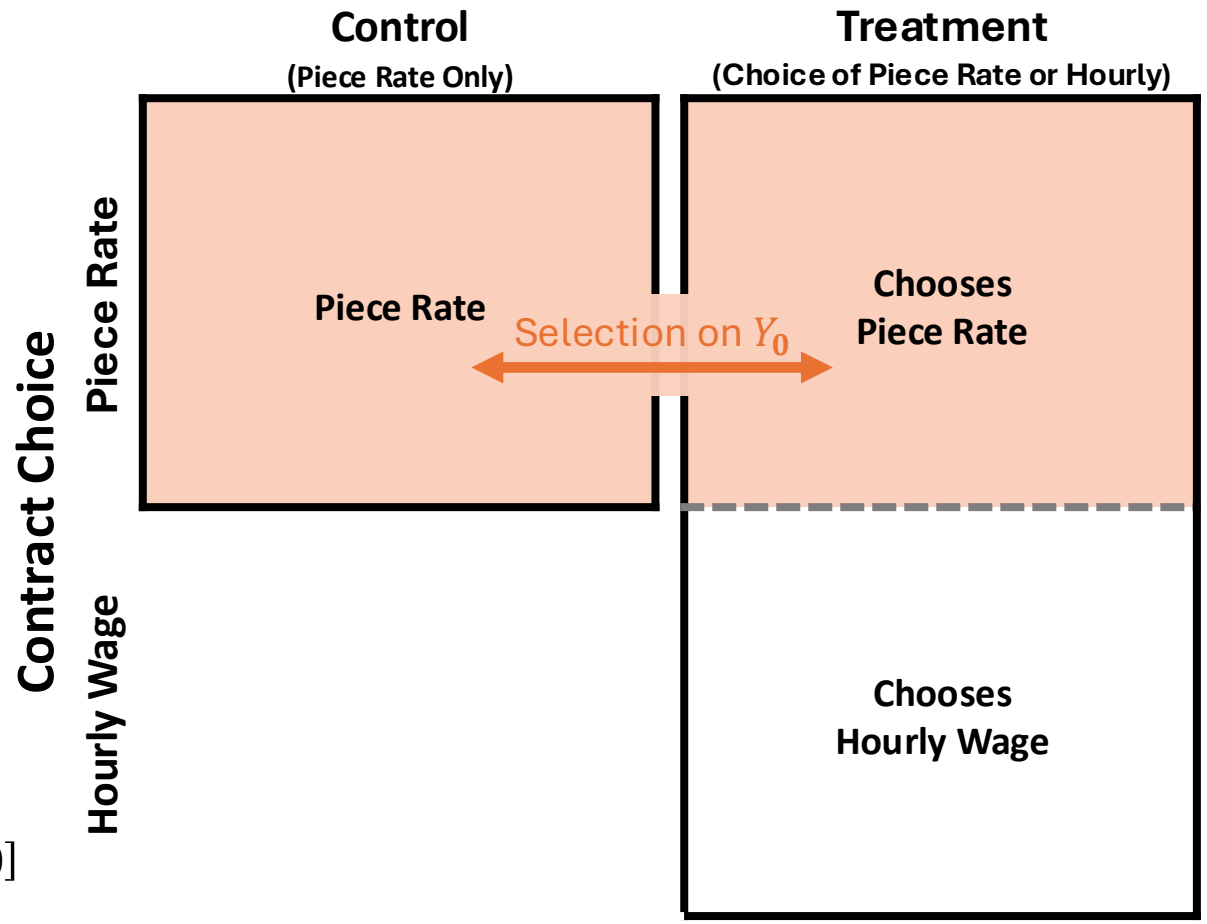
Comparing between workers facing different hourly offers identifies ***treatment effect*** of hourly wages:

$$\frac{E[Y_i|W_i=1]-E[Y_i|W_i=0]}{\Pr(H_i=1|W_i=1)} = \underbrace{E[Y_{1i}-Y_{0i}|H_i=1]}_{\text{Moral Hazard}}$$

Local Average Treatment Effect (LATE)



Experimental Design: Single Wage Offer



Comparing piece-rate workers who faced different offer sets identifies ***selection into treatment:***

$$\frac{E[Y_i|W_i=0]-E[Y_i|H_i=0,W_i=1]}{\Pr(H_i=1|W_i=1)} = \underbrace{E[Y_{0i}|H_i=1]-E[Y_{0i}|H_i=0]}_{\text{Selection on } Y_0}$$

Adverse Selection

Details

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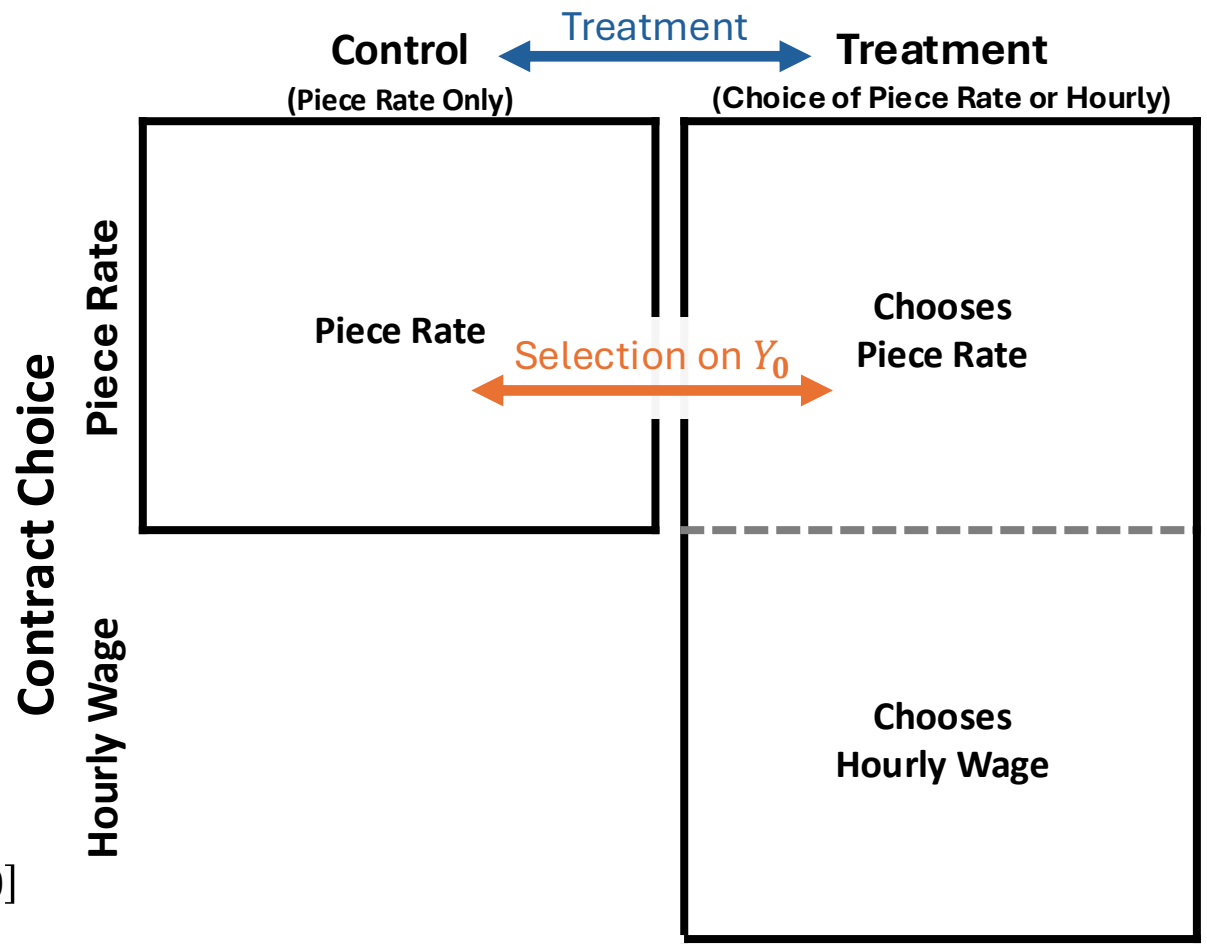
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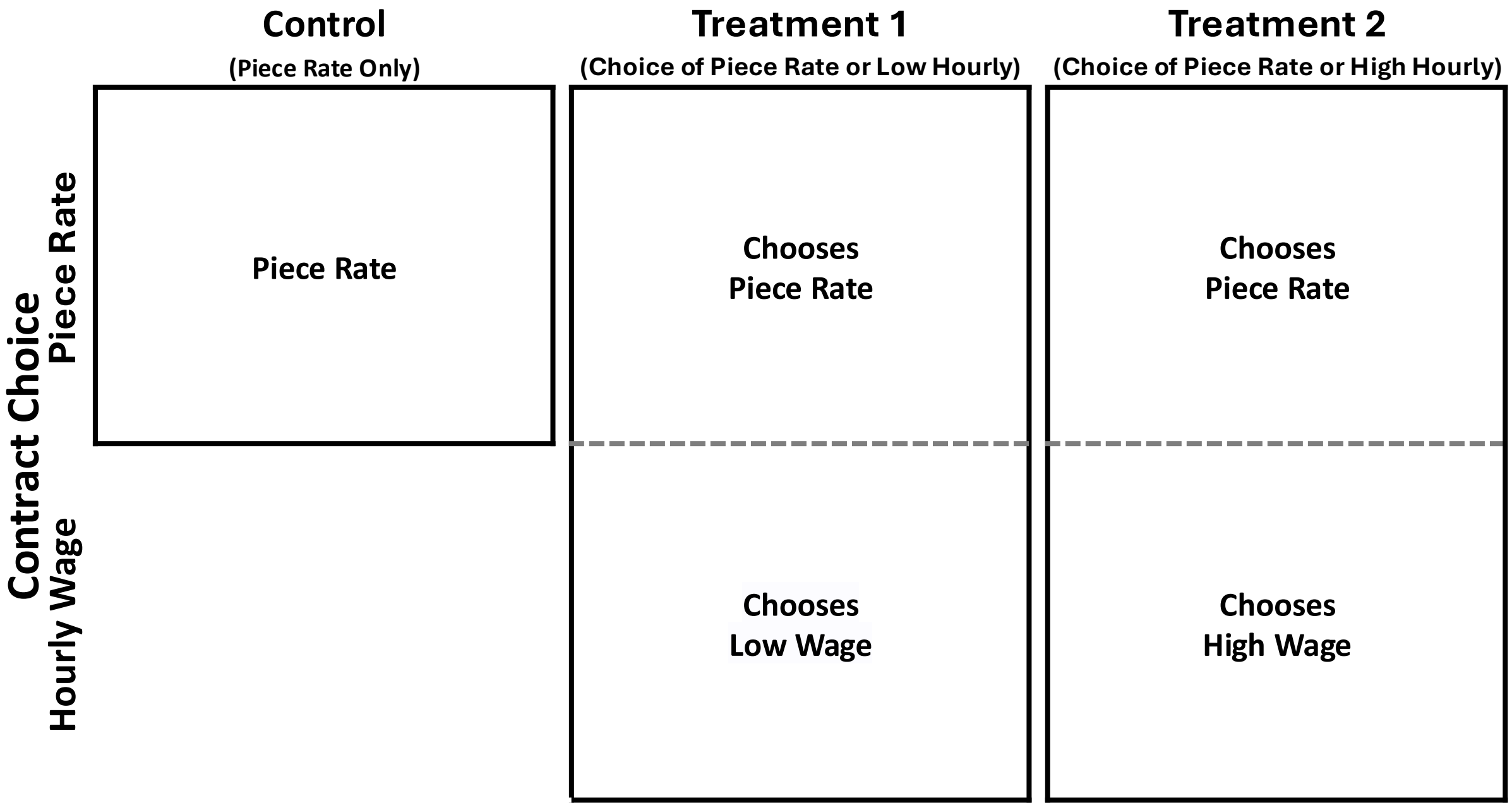
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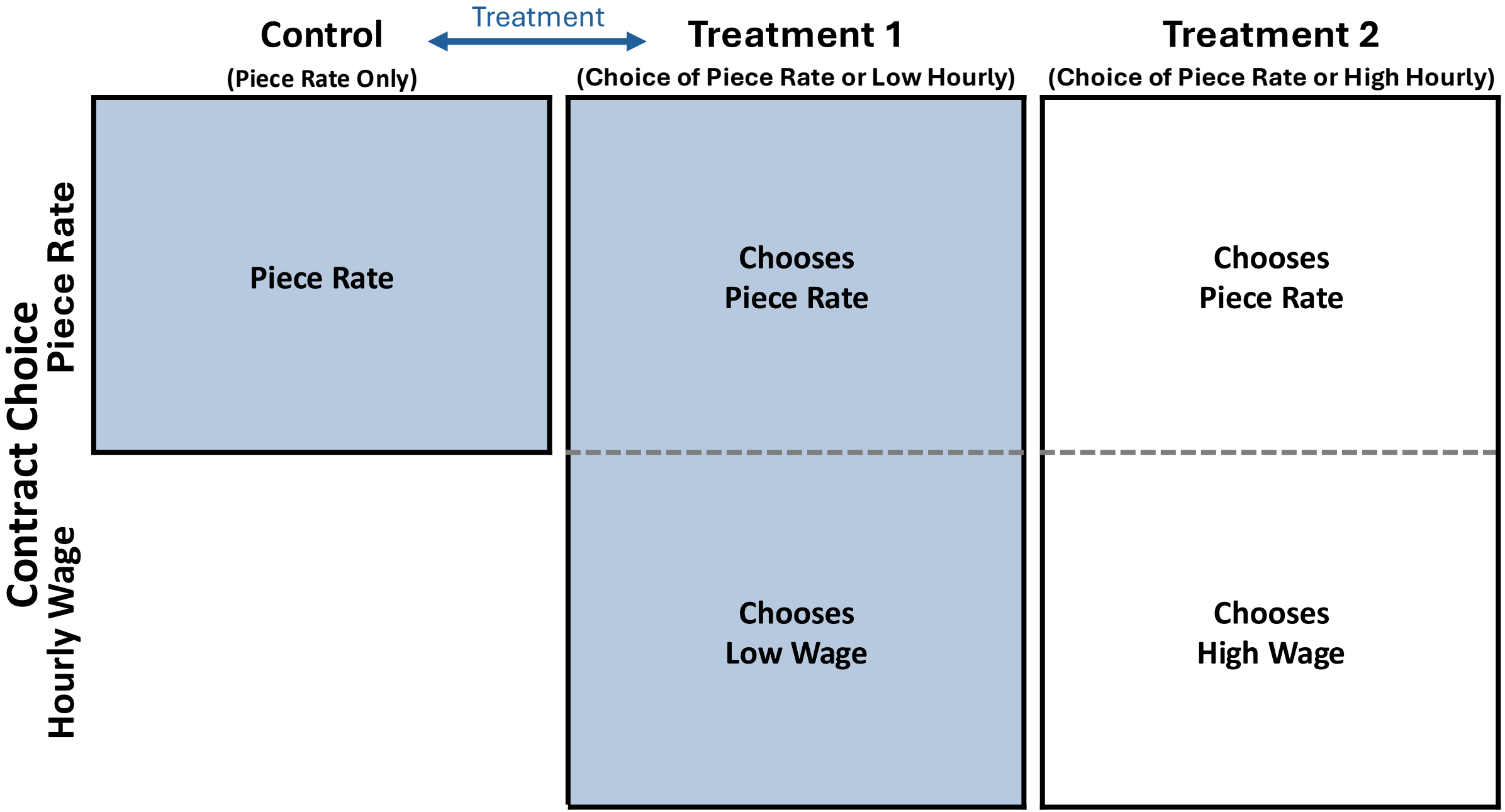
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Selection on Y_0

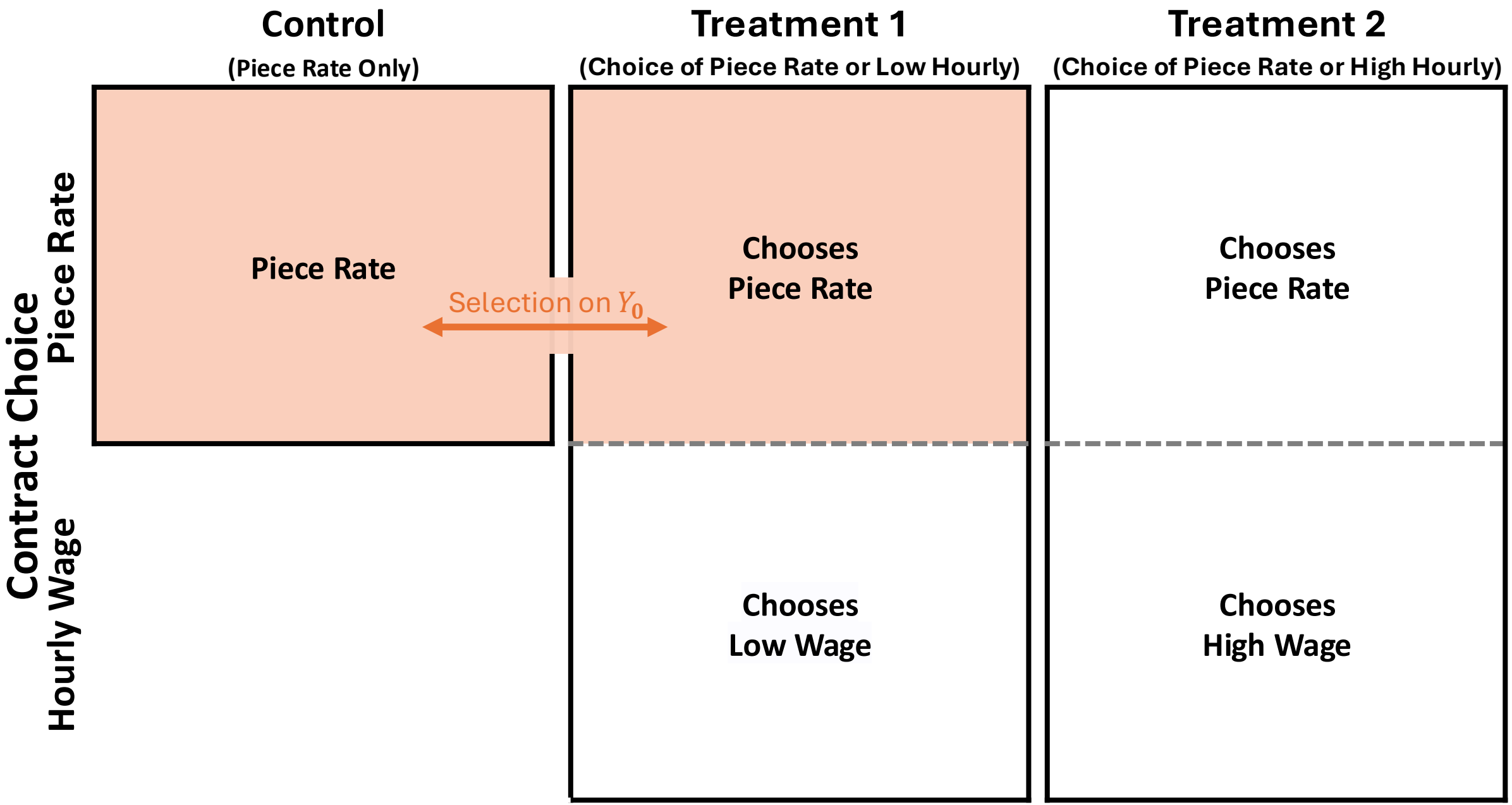




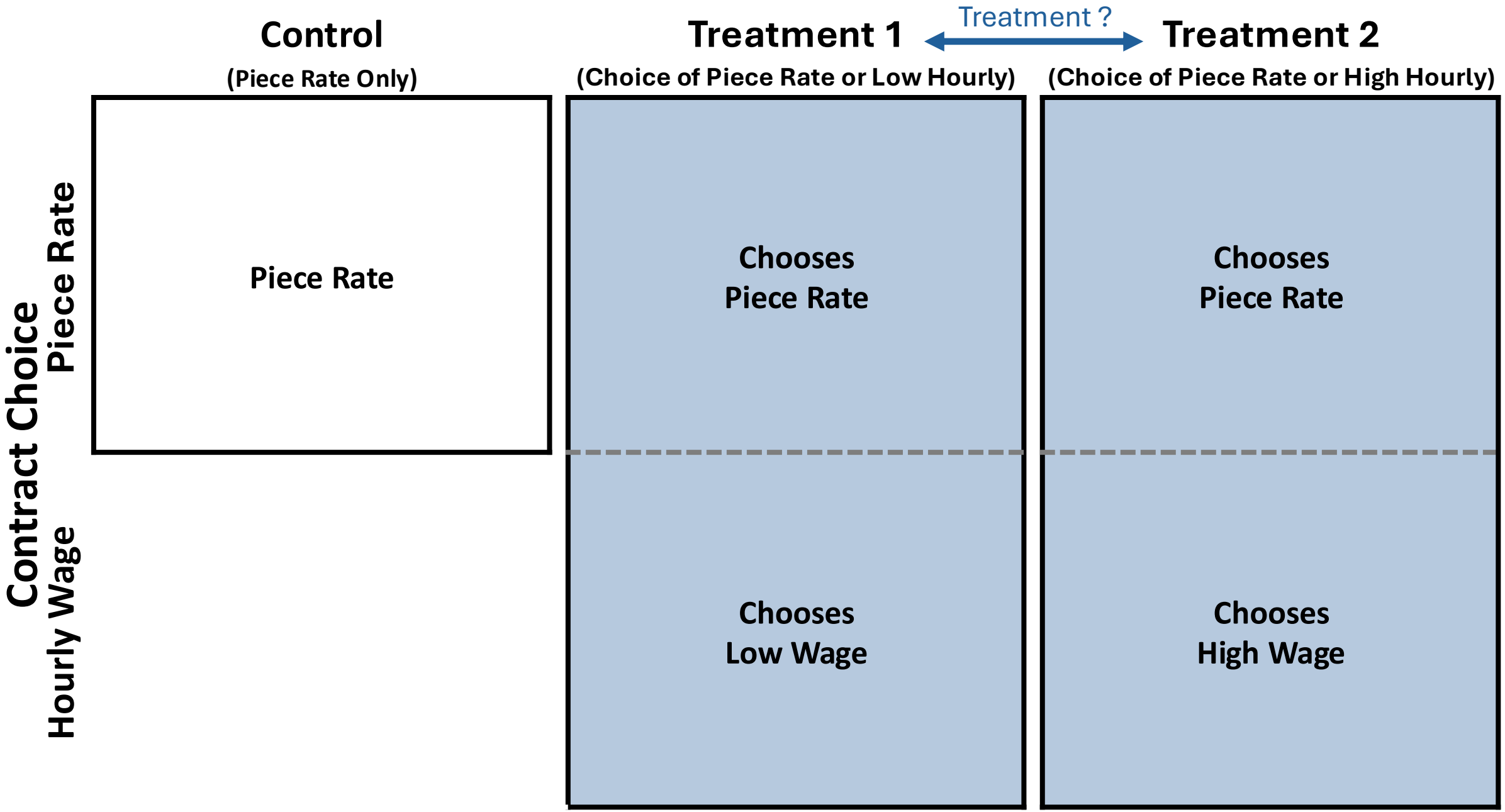
Formal Expression



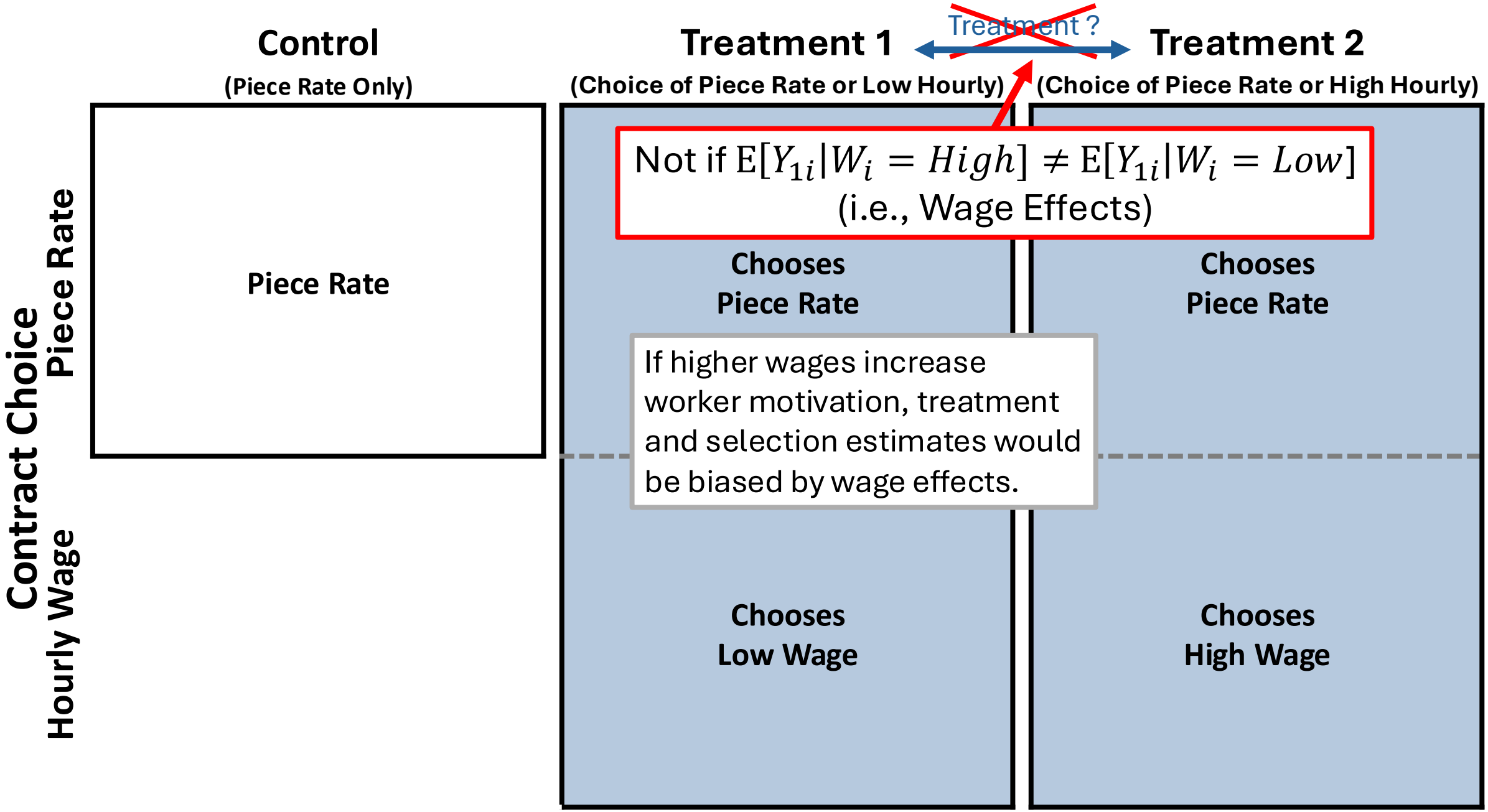
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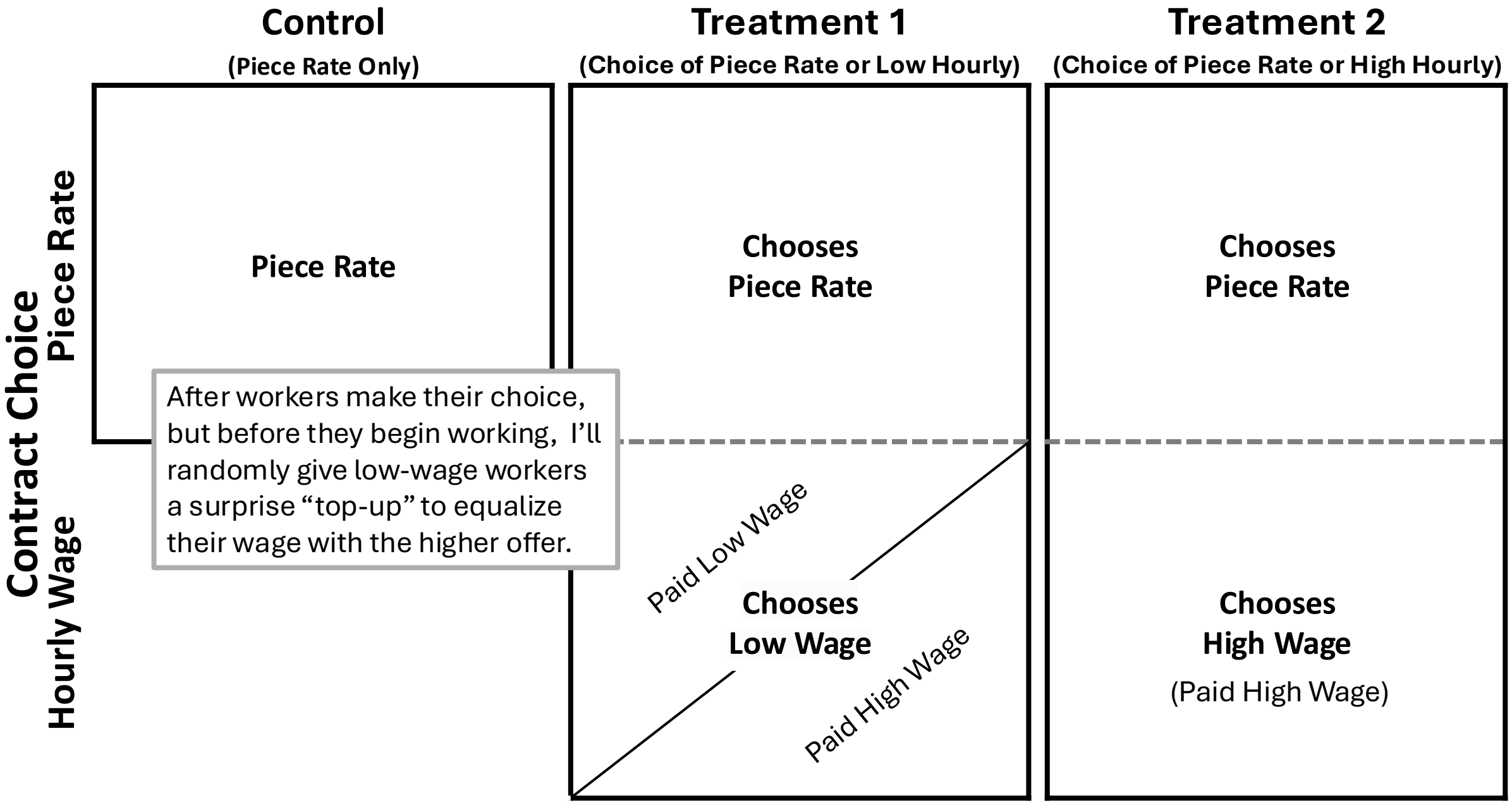
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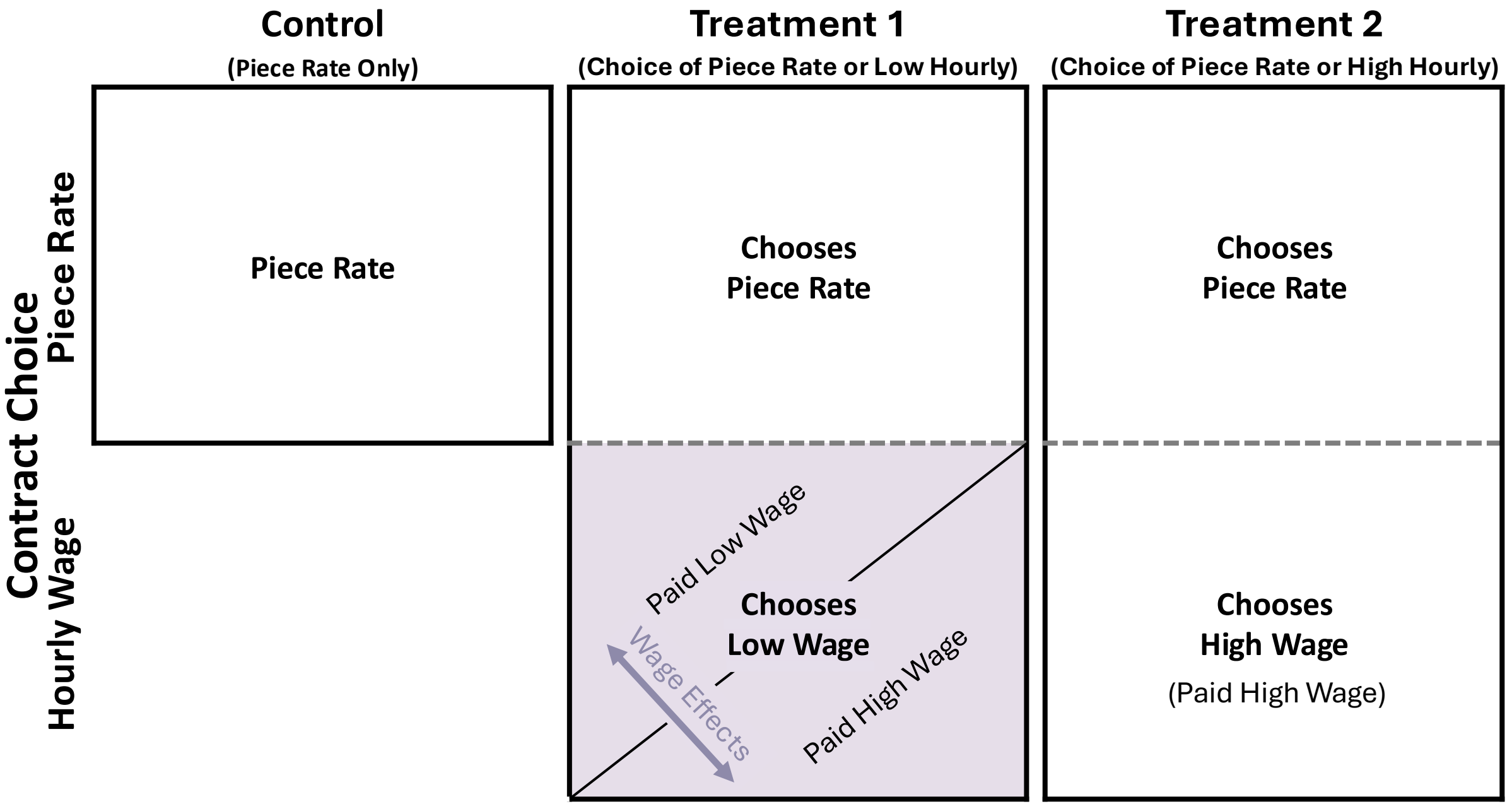
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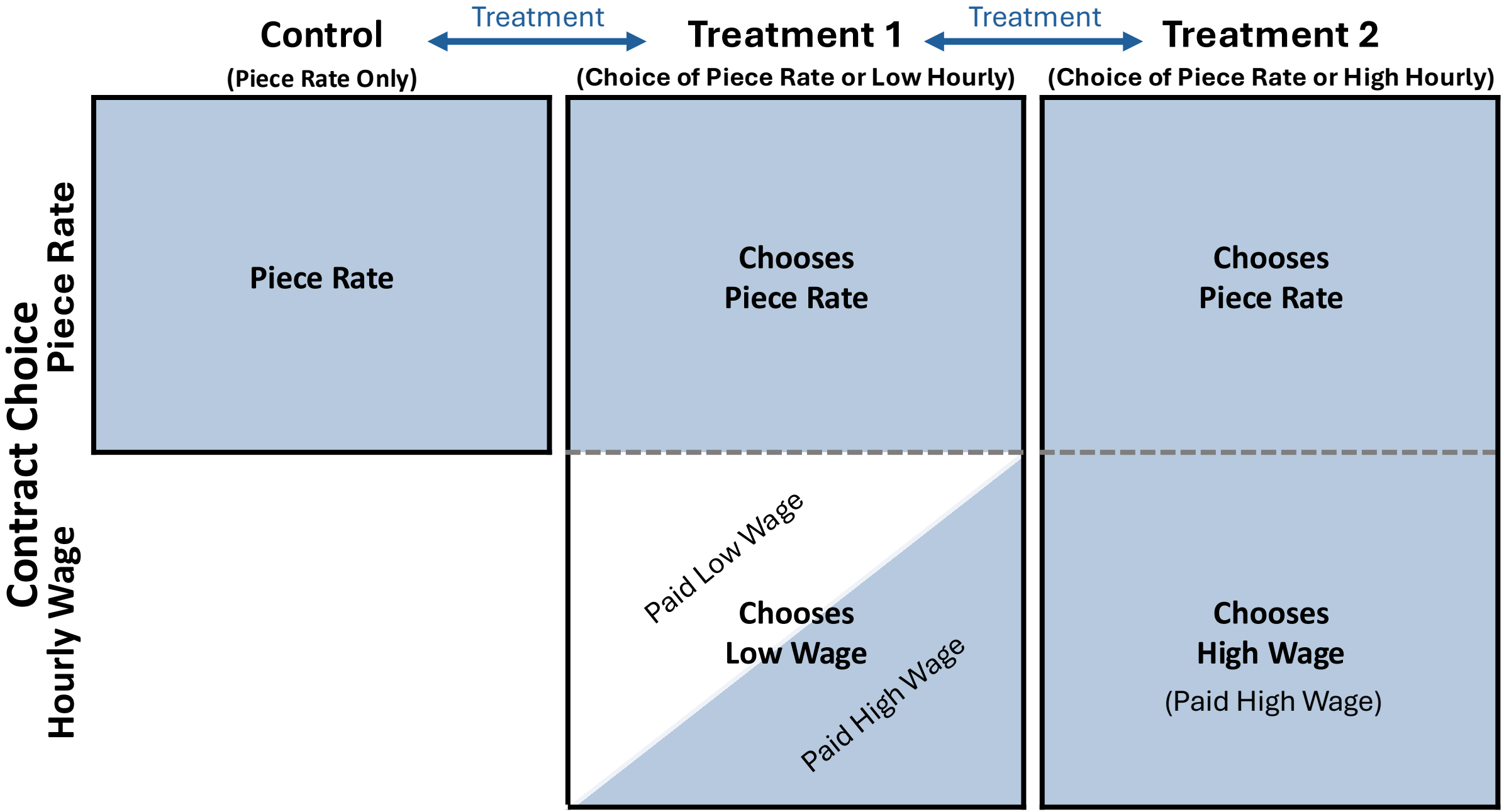
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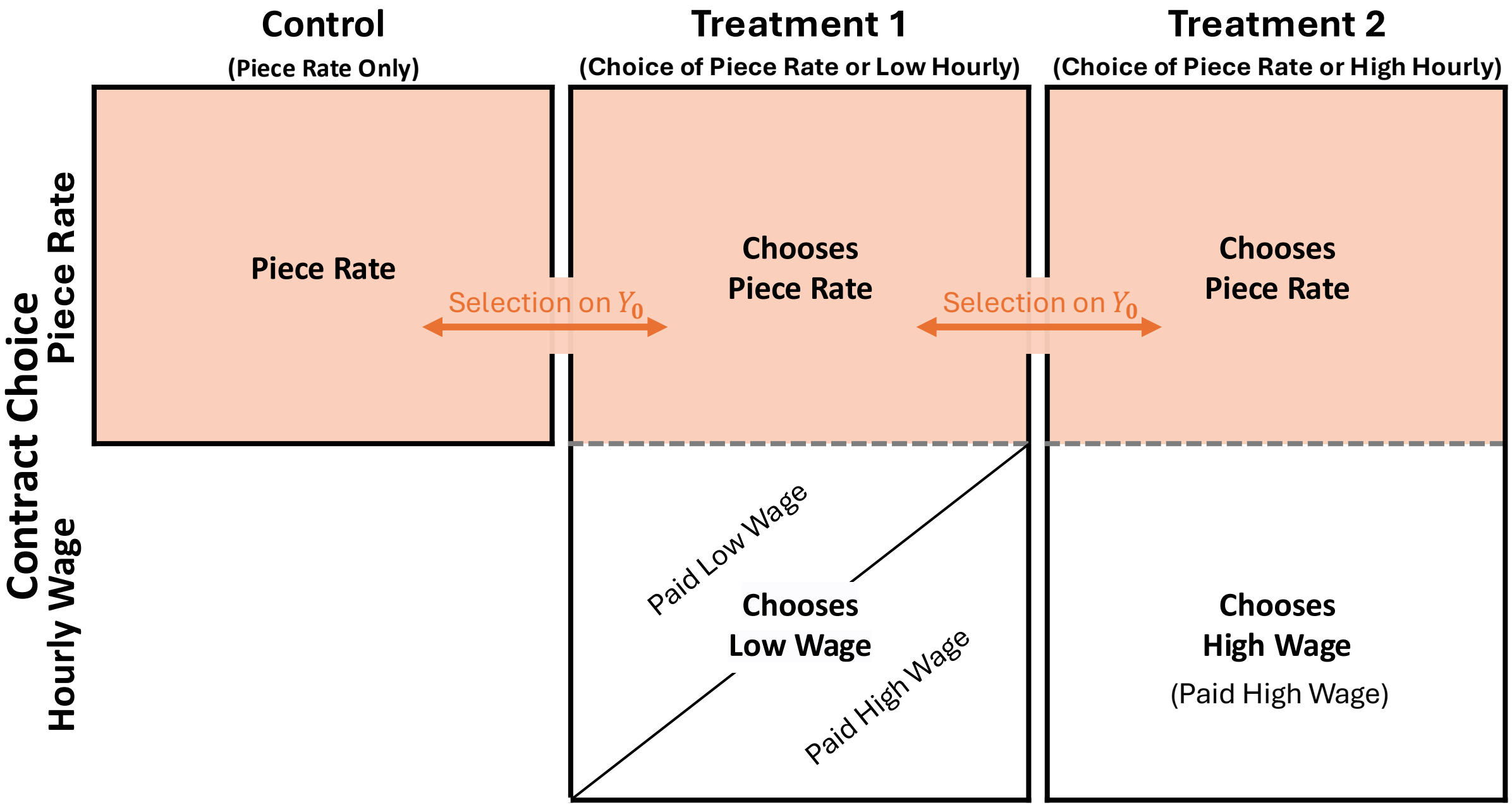
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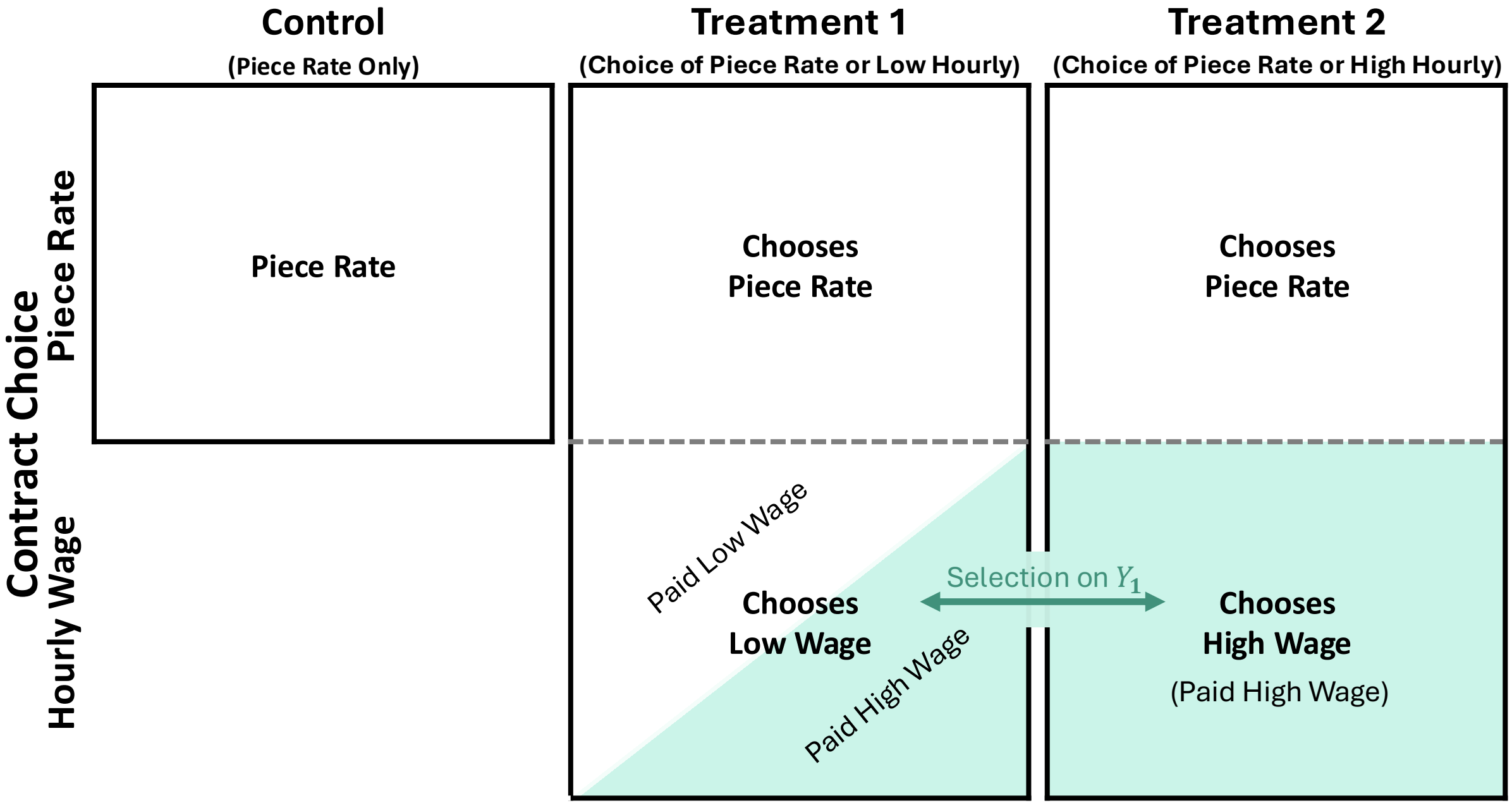
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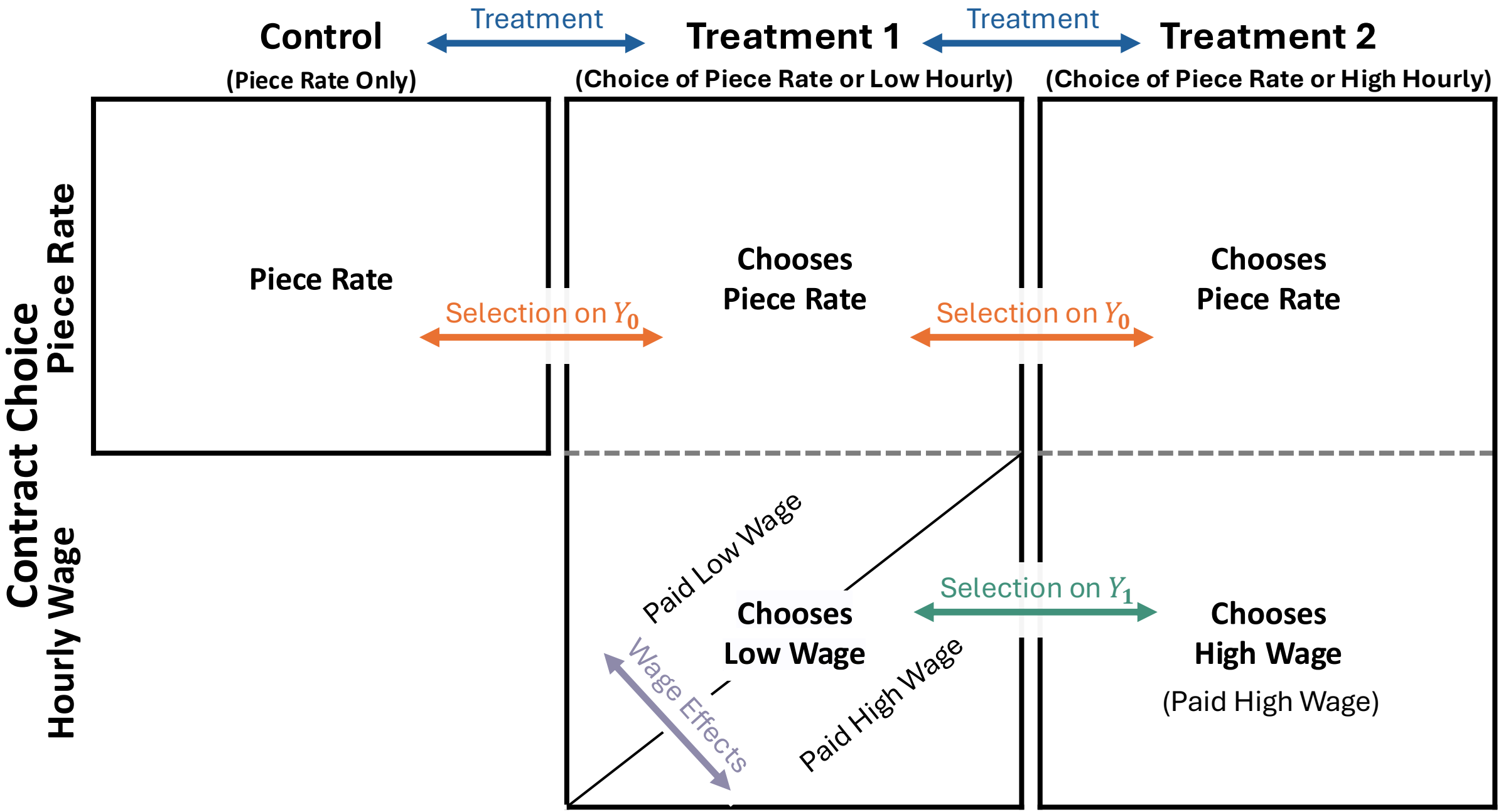
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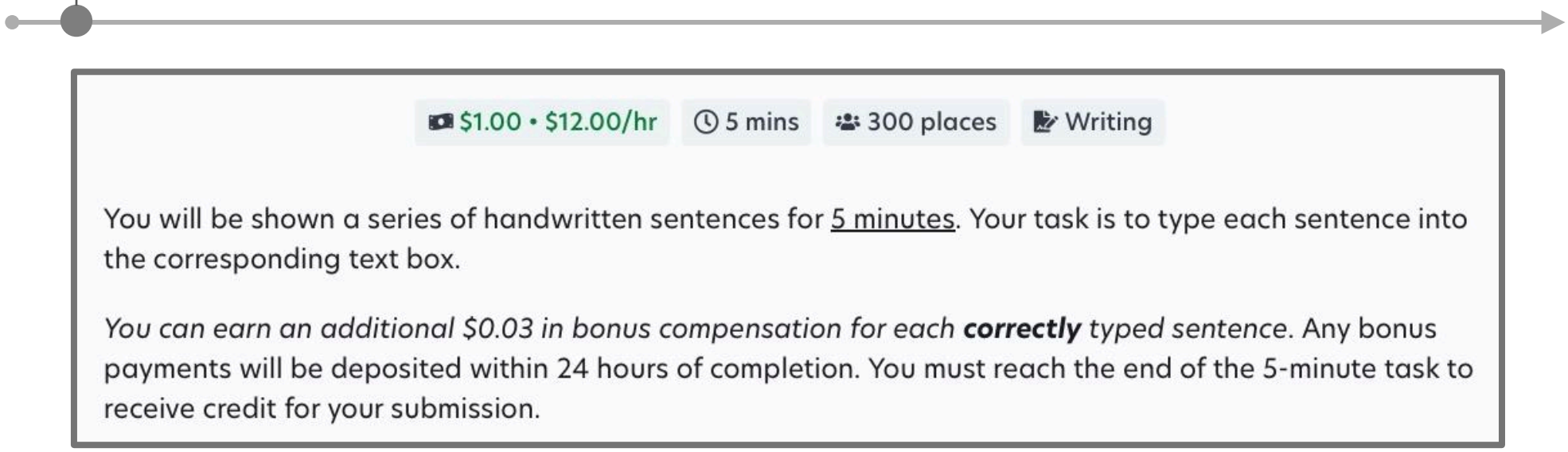
Formal Expression

Experimental Procedure

Workers Recruited

Prolific job posting advertises a $1.00 reward for a five-minute data-entry task, plus $0.03 per correct entry.

- Posting screens for
  - Approval rating ≥98%
  - Number of approved tasks ≥10
  - Located within US
- Workers are unaware of experiment
- Experiment takes place over ten waves of ≈300 workers



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Randomization #1: Wage Offers

Workers are randomly assigned to one of eighteen experimental groups, each with a different menu of bonus wage offers.

Hourly Wage Offer	Piece-Rate Offer	Number of Participants
No Hourly Offer	$0.03 per sentence	302
$1.20/hr	$0.03 per sentence	300
$1.80/hr	$0.03 per sentence	101
$2.40/hr	$0.03 per sentence	103
$3.00/hr	$0.03 per sentence	304
$3.60/hr	$0.03 per sentence	100
$4.20/hr	$0.03 per sentence	99
$4.80/hr	$0.03 per sentence	101
$5.40/hr	$0.03 per sentence	101
$6.00/hr	$0.03 per sentence	305
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$12.00/hr	$0.03 per sentence	305
$15.00/hr	$0.03 per sentence	100
$18.00/hr	$0.03 per sentence	102
$21.00/hr	$0.03 per sentence	304
	Total:	3030

Balance Test

Experimental Procedure

Workers Recruited

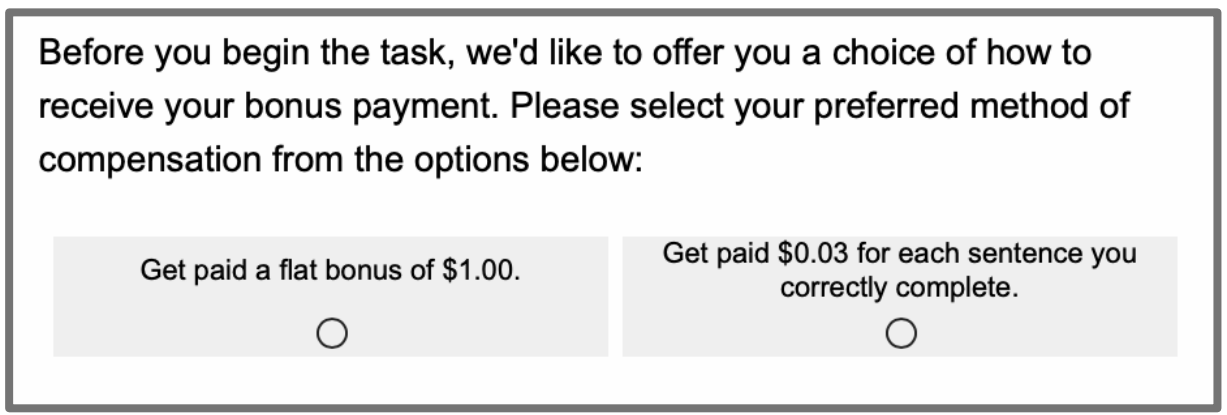
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Choice of Compensation

Treated workers choose between a $0.03 piece rate and hourly wage offers ranging from $1.20 to $21.00 per hour.

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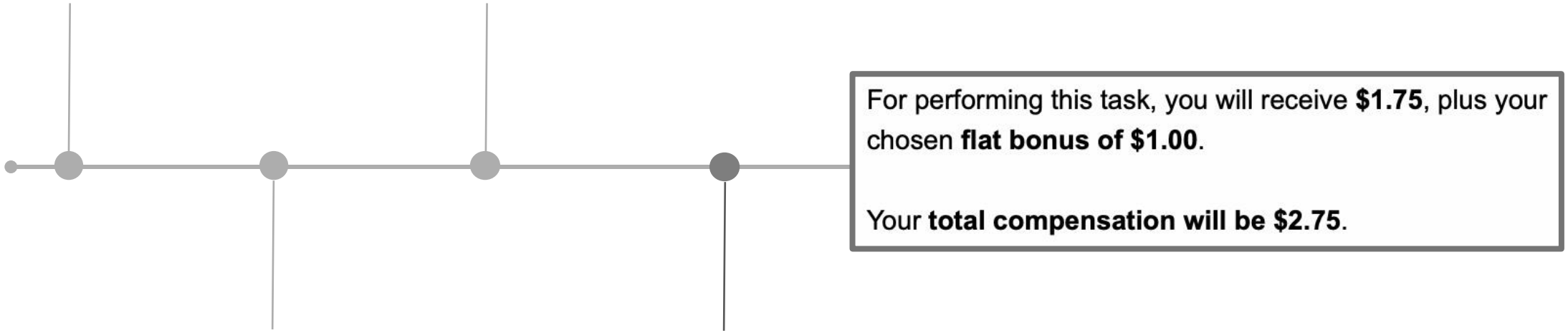
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Randomization #2: Wage Raises

Before work begins, hourly wages are raised to $21.00 per hour for a random 50 percent of workers who accepted lower-valued hourly offers.

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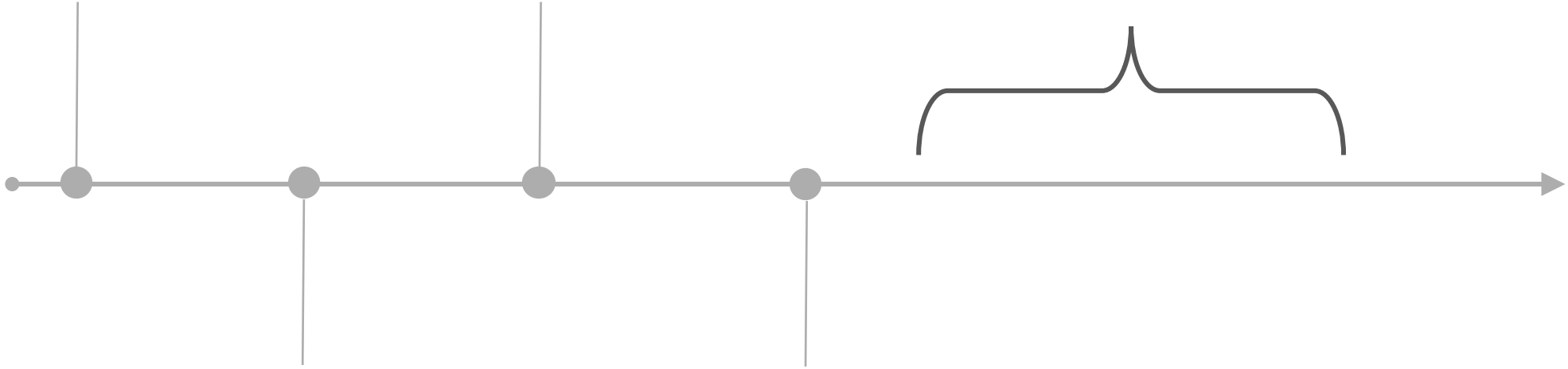
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Task Performed

Each worker has five minutes to complete as many data entries as possible. Screen displays worker's progress and remaining time.



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Randomization #2: Wage Raises

Before work begins, hourly wages are raised to $21.00 per hour for a random 50 percent of workers who accepted lower-valued hourly offers.

You will be shown a series of handwritten sentences. On each page, your task is to type the sentence into the text box below.

Here is one example of a completed sentence:



The quick brown fox jumps over the lazy dog.

Your answers should be as accurate as possible. Please be mindful of capitalization, spacing, and punctuation. When you've completed a sentence, click the "→" button to move on.

You will have 5 minutes to complete as many sentences as you can. You cannot start over, and you can only perform this task once.

→

Stills

Experimental Procedure

Workers Recruited

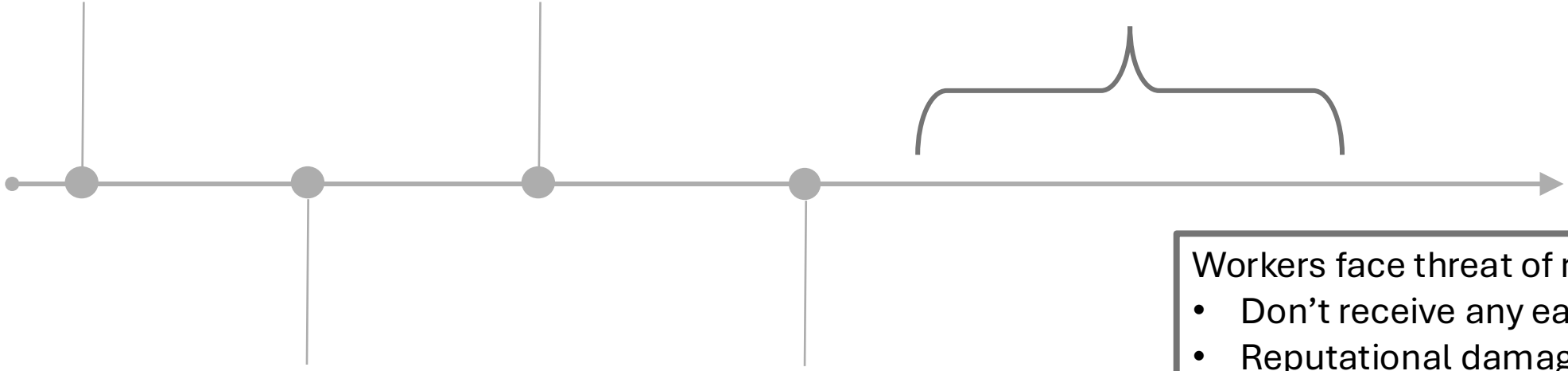
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Workers face threat of rejection ⇒

- Don't receive any earnings
- Reputational damage (approval rate ↓)

Experimental Procedure

Workers Recruited

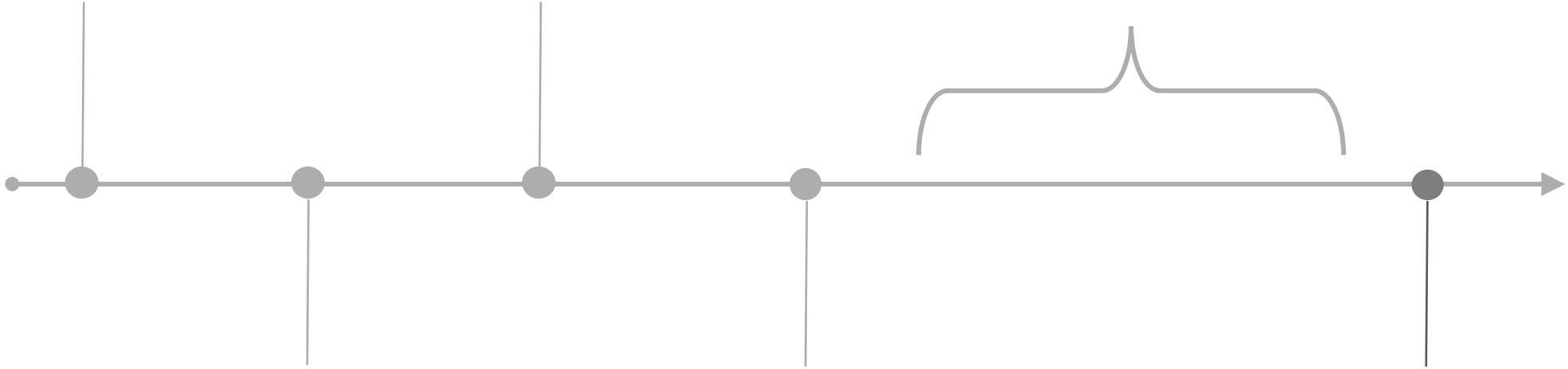
Prolific job posting advertises a $1.00 reward for a five-minute data-entry task, plus $0.03 per correct entry.

Choice of Compensation

Treated workers choose between a $0.03 piece rate and hourly wage offers ranging from $1.20 to $21.00 per hour.

Task Performed

Each worker has five minutes to complete as many data entries as possible. Screen displays worker's progress and remaining time.



Randomization #1: Wage Offers

Workers are randomly assigned to one of eighteen experimental groups, each with a different menu of bonus wage offers.

Randomization #2: Wage Raises

Before work begins, hourly wages are raised to $21.00 per hour for a random 50 percent of workers who accepted lower-valued hourly offers.

Wages and Bonuses Paid

Within 24 hours of completing the task, workers are paid the $1.00 base reward plus any bonus earnings.

Key Advantages

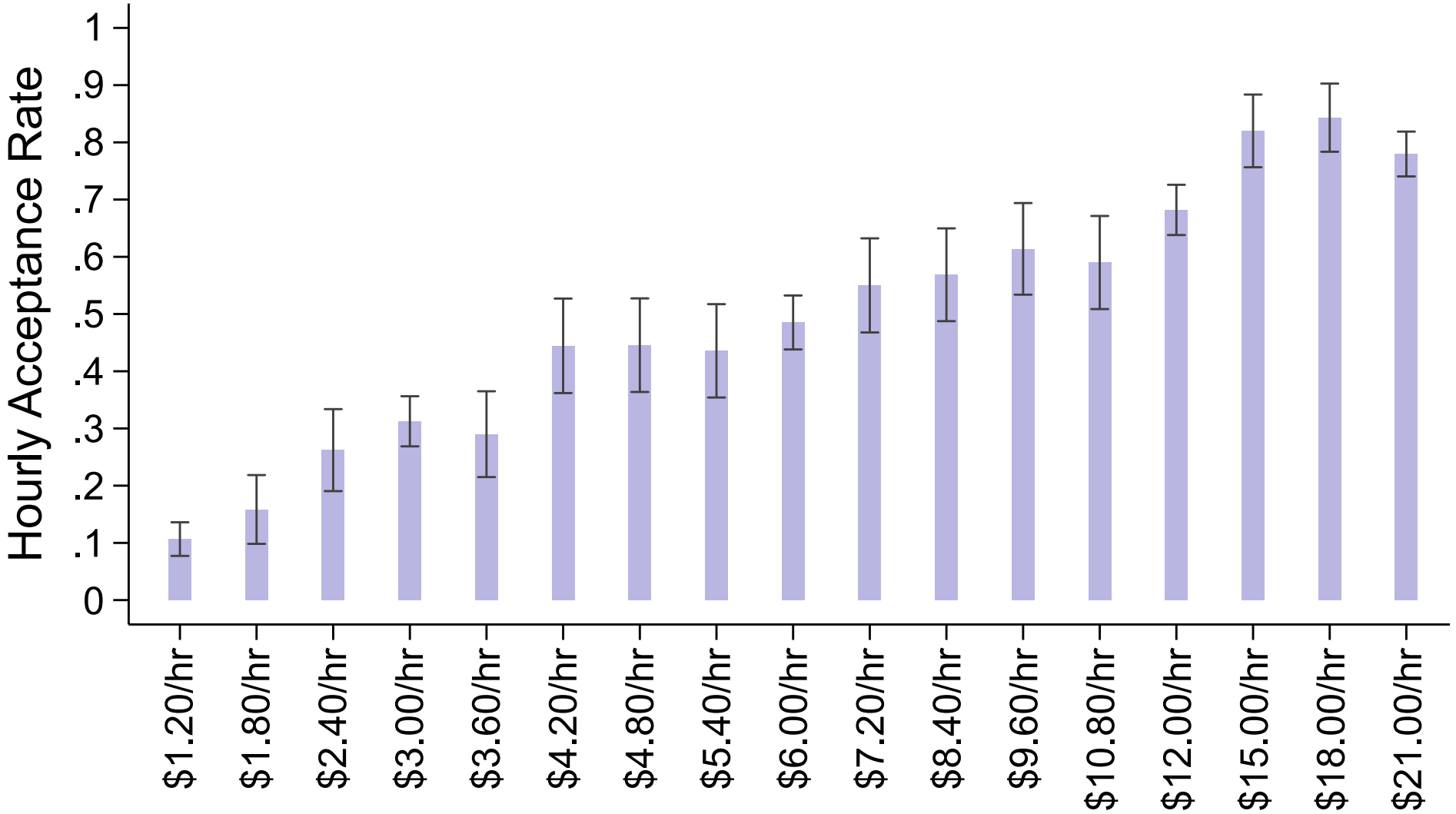
- Sample is not selected into pre-existing employer or contract
  - e.g., sample of firm's existing employees would exclude freelancers
- Control of piece-rate outside option ⇒ *market* supply curve
- Wide range of contract offers, not just those "under the lamppost"
  - i.e., options include potentially *unraveled* contracts that are not profitable
- Observe output for all workers—both *accepters* and *decliners*

  ⇒ Can estimate selection on Y_0 and Y_1
- Experimental task is designed to be as generalizable as possible
  - 66% of US workers type on the job
  - Workers don't know it's an experiment
  - Task features tend to *attenuate* welfare effects relative to other gig markets
- Highly Replicable!

Summary Statistics

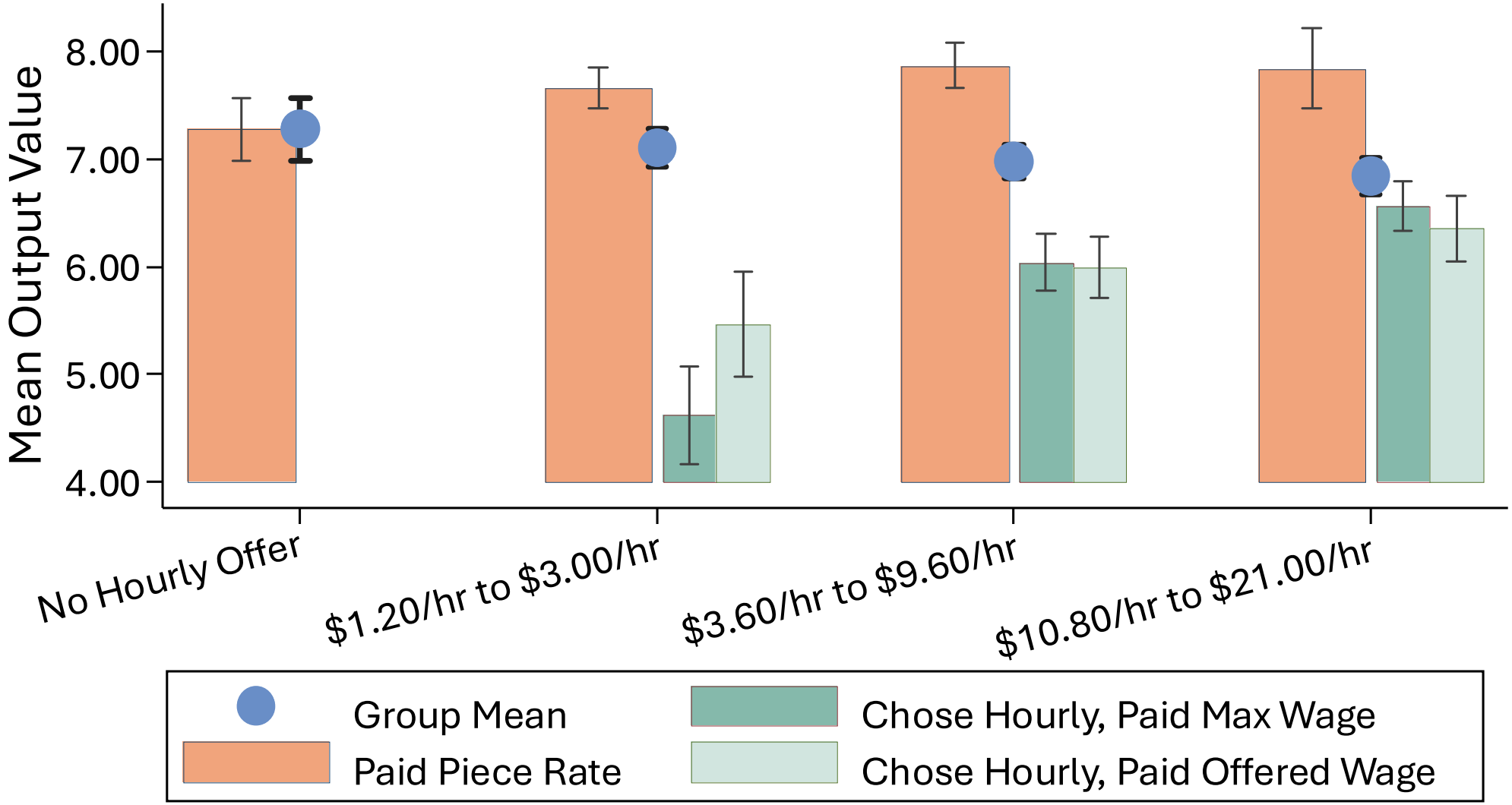
Category	Variable	Mean	SD
Panel A: Task Performance	Accepted Hourly Offer	0.486	0.500
	Completed Sentences	21.98	8.143
	Correct Sentences	19.45	8.480
	Output Value	7.003	3.053
	Finished	0.986	0.117
Panel B: Demographics & Employment	Age	37.23	12.18
	Female	0.643	0.479
	Minority	0.357	0.479
	Employed	0.685	0.465
	Student	0.187	0.390
	Number of Previous Tasks	1281.6	1746.4

Hourly Wage Take-up



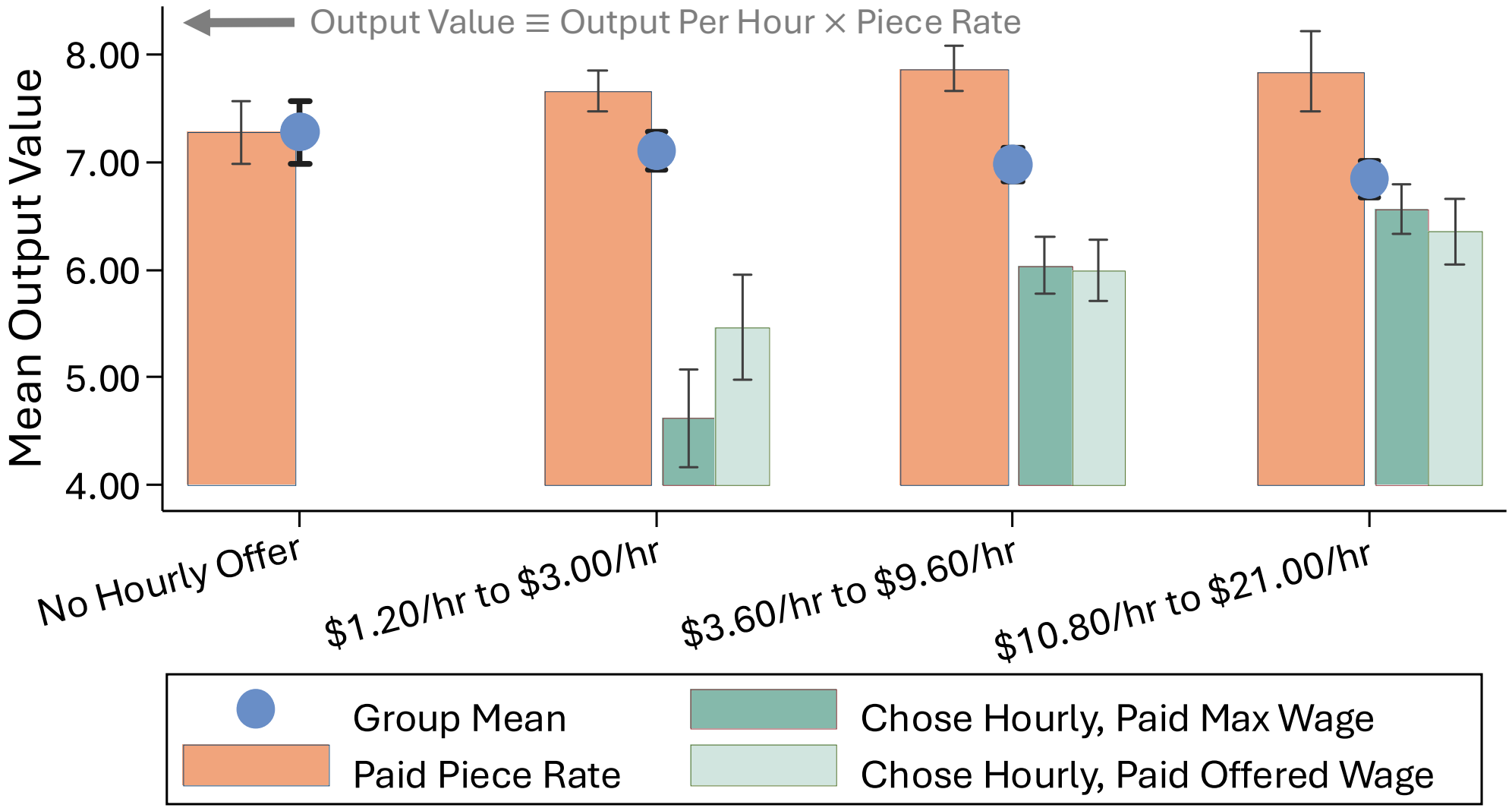
Logit Estimates

Worker Output by Offer & Contract



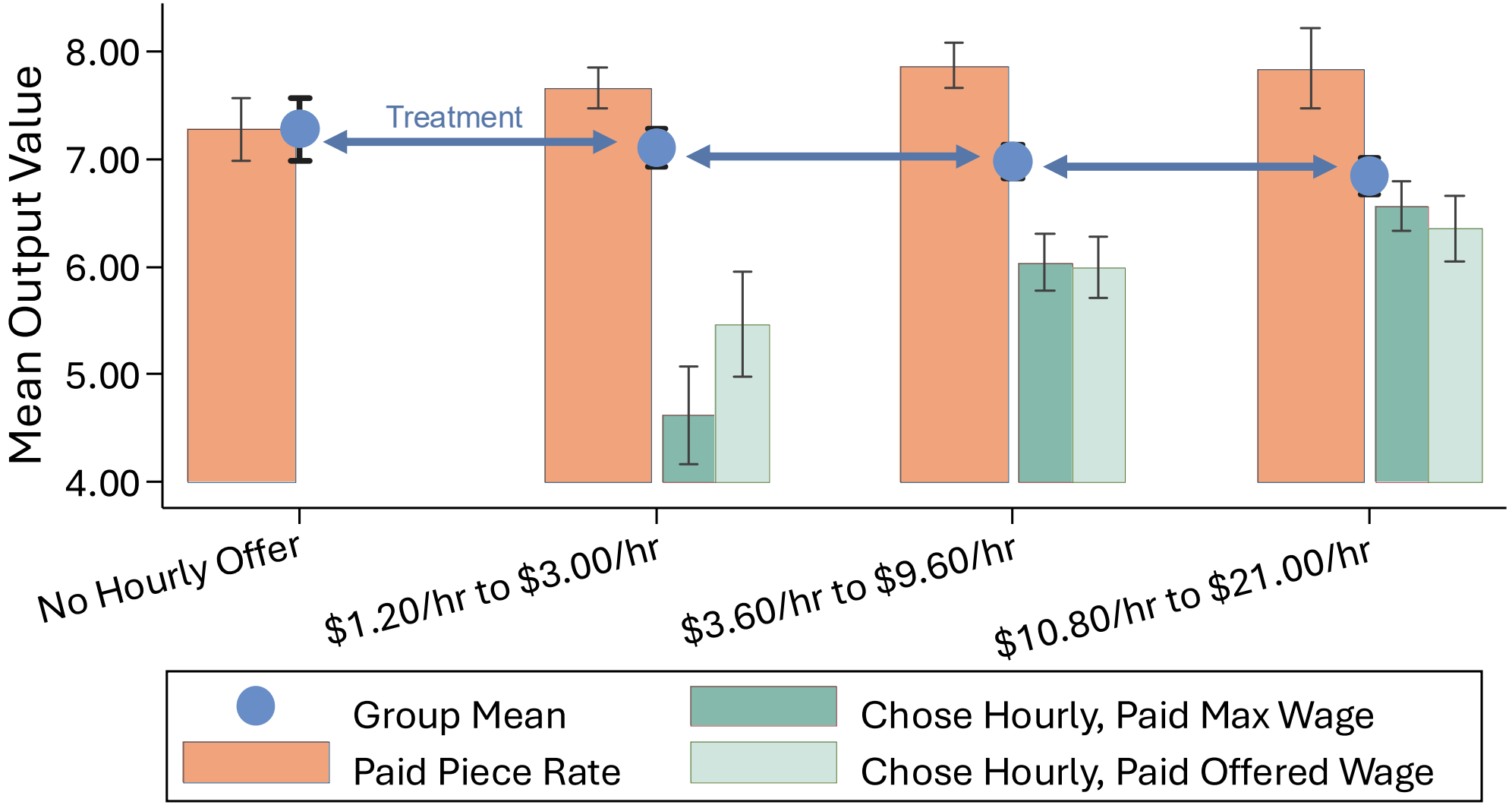
Raw Output Value

Worker Output by Offer & Contract



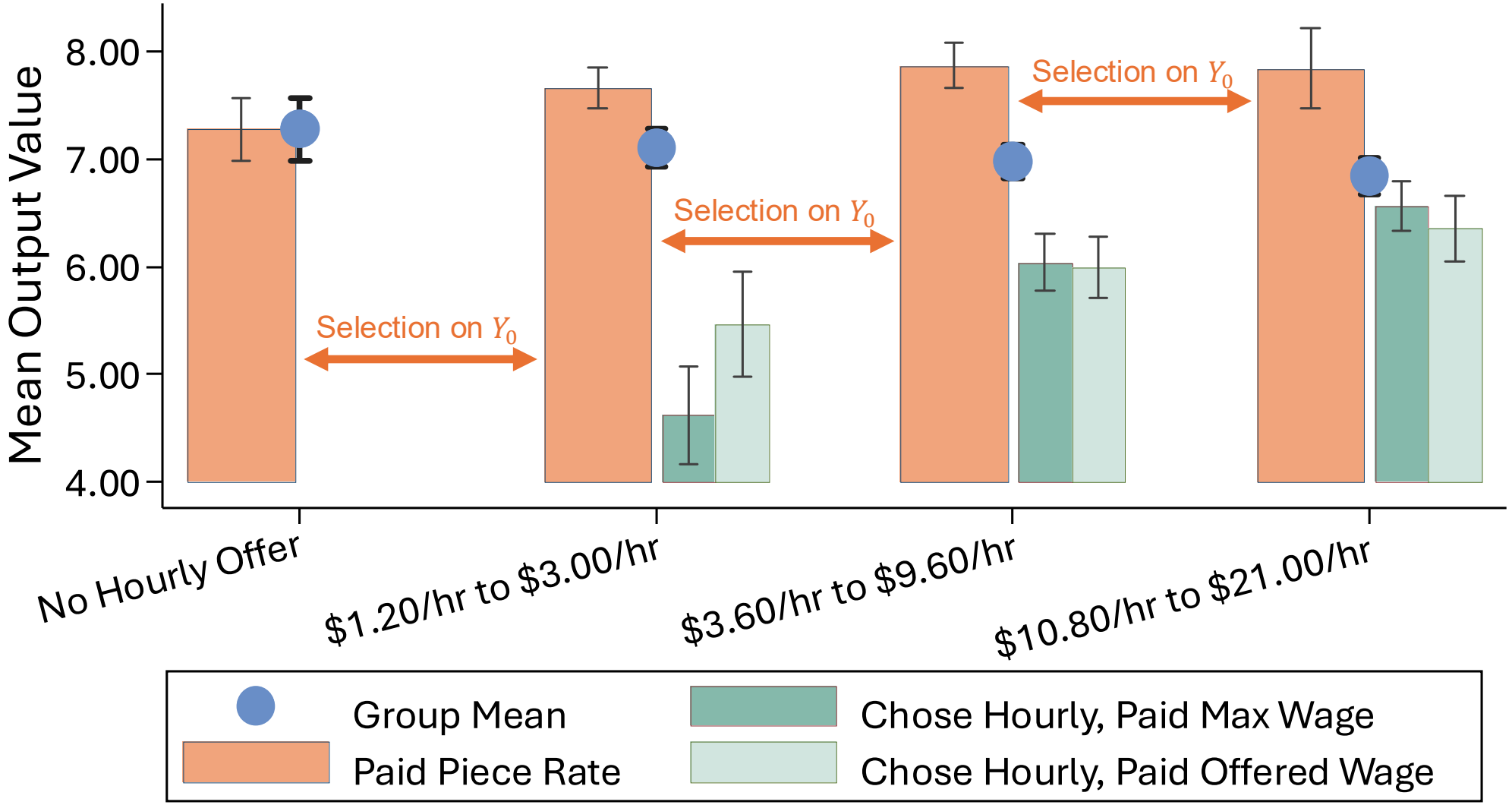
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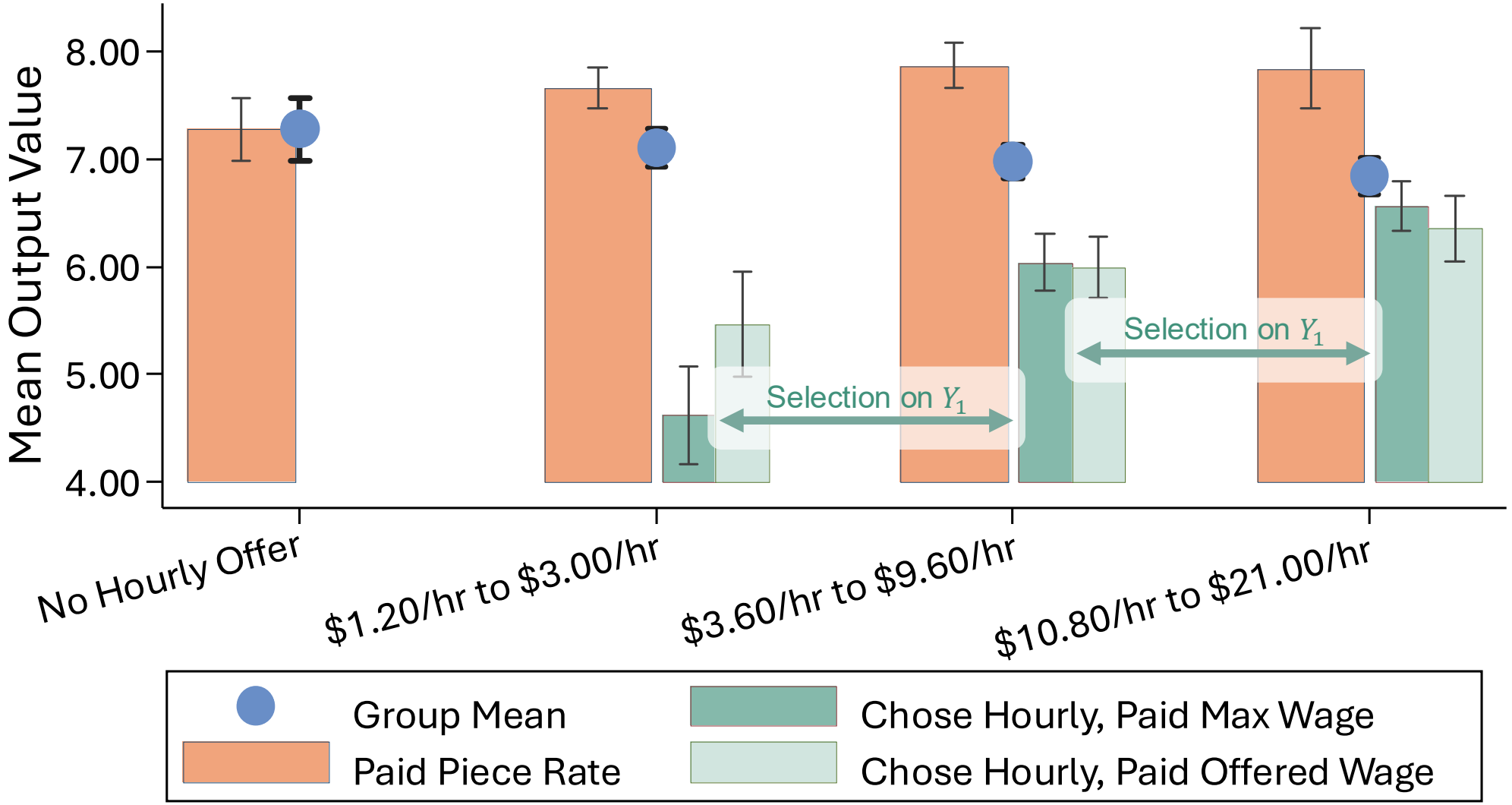
Raw Output Value

Worker Output by Offer & Contract



Raw Output Value

Worker Output by Offer & Contract



Raw Output Value

2SLS Estimates of Treatment Effects

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	-0.612^{***}	-0.606^{***}	-0.591^{***}	-0.475^{**}
	(0.215)	(0.208)	(0.207)	(0.193)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
Dep. Mean	7.00	7.00	7.00	7.00
R-squared	0.040	0.088	0.104	0.221
N	3030	3030	3030	3030

$$\tilde{Y}_i = \delta H_i + \boldsymbol{\eta} \boldsymbol{X_i} + \epsilon_i$$

$$\tilde{Y}_i \equiv Y_i - \gamma H_i \times W_i^P$$

Raw Output Value

Selection by Log Wage Offer

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	−2.664***	−2.552***	−2.513***	−2.387***
	(0.340)	(0.333)	(0.334)	(0.312)
Declined × Log Hourly Wage Offer	0.0927	0.119	0.141	0.163*
	(0.100)	(0.0970)	(0.0965)	(0.0906)
Accepted × Log Hourly Wage Offer	0.666***	0.610***	0.605***	0.555***
	(0.121)	(0.117)	(0.117)	(0.110)
Accepted × Log Effective Hourly Wage	−0.109	−0.0860	−0.0815	−0.0410
	(0.126)	(0.123)	(0.123)	(0.116)
Control Group (No Hourly Offer)	−0.361	−0.323	−0.263	−0.263
	(0.234)	(0.226)	(0.226)	(0.209)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
Dep. Mean	7.00	7.00	7.00	7.00
R-squared	0.075	0.121	0.137	0.254
N	3030	3030	3030	3030

$$Y_i = \alpha H_i + \beta_0(1 - H_i) \times W_i + \beta_1 H_i \times W_i + \gamma H_i \times W_i^P + \boldsymbol{\xi} \boldsymbol{X_i} + \epsilon_i$$

Marginal Selection Raw Output Value

Selection by Log Wage Offer

As wage offer increases...

output of **decliners** ↑

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
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Marginal Selection Raw Output Value

Selection by Log Wage Offer

As wage offer increases...

output of **accepters** ↑

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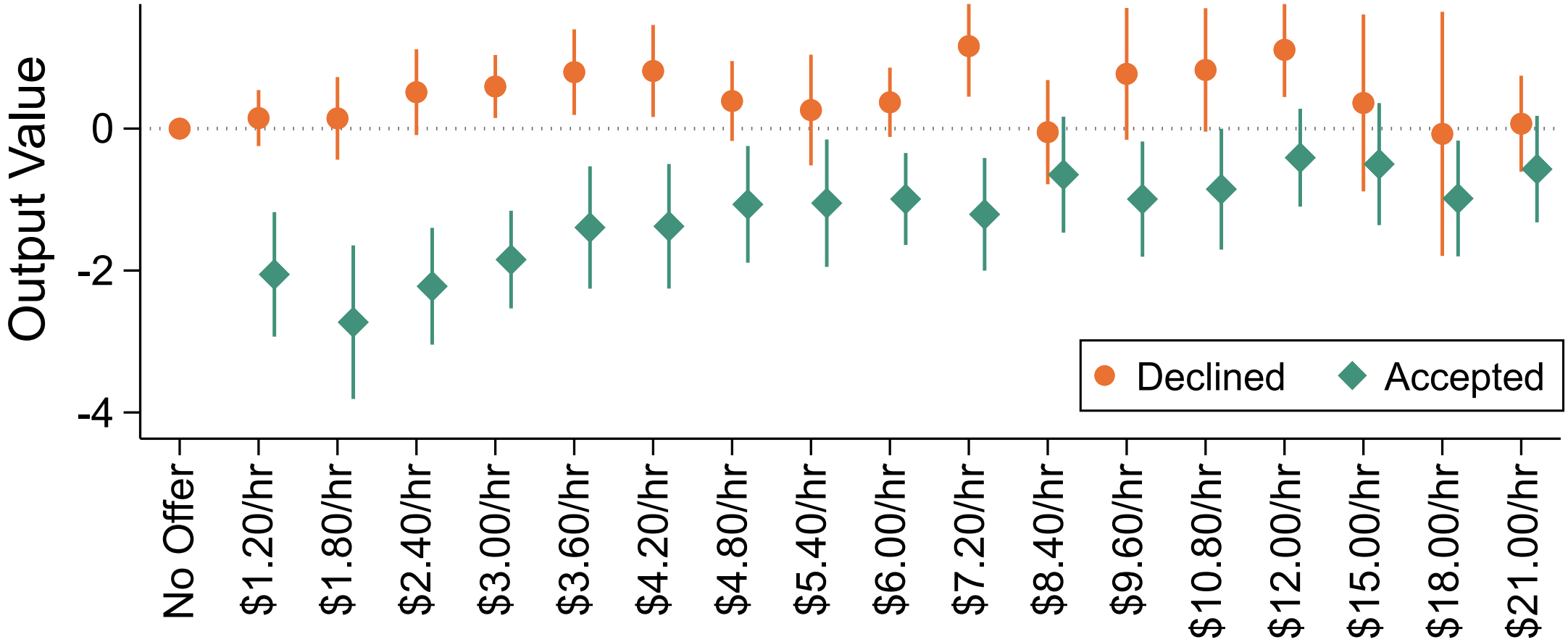
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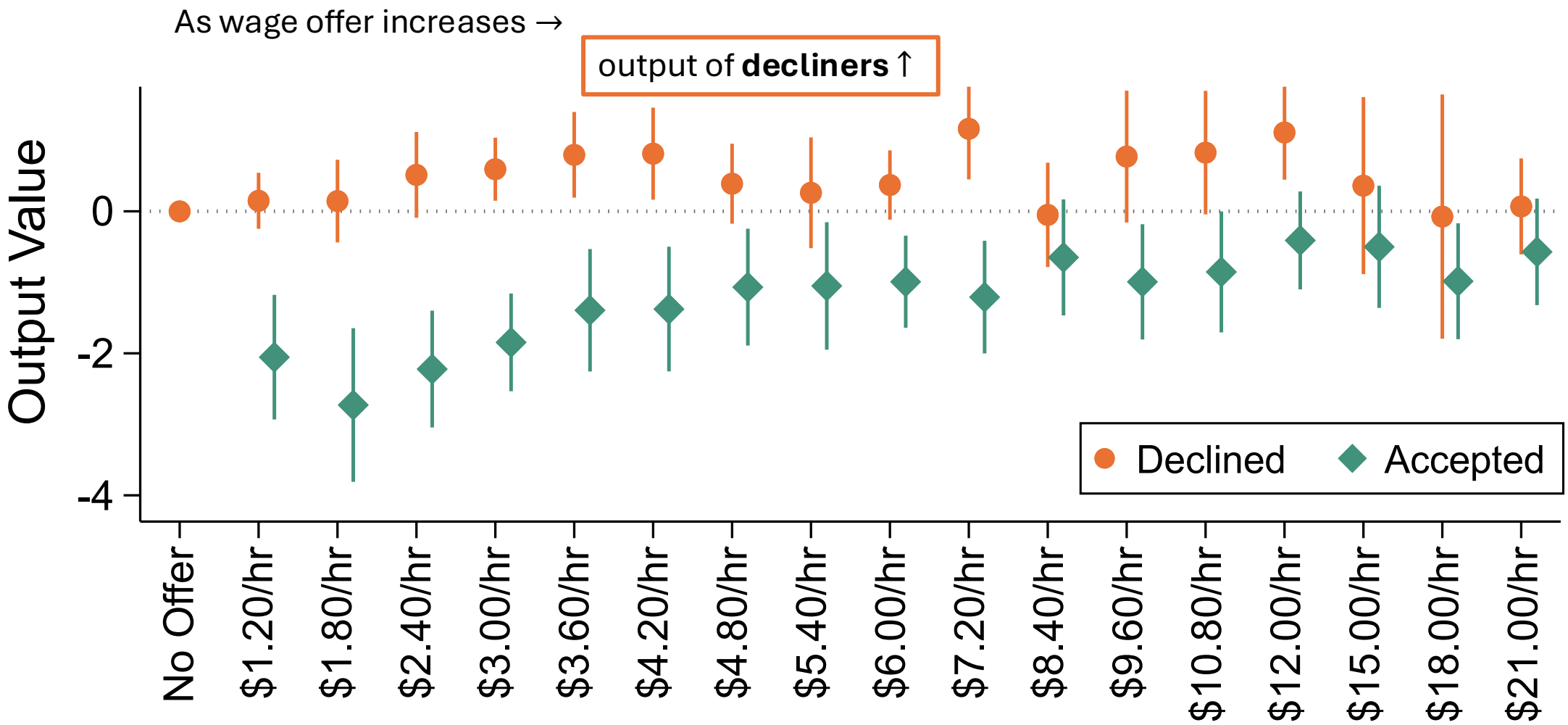
Selection by Log Wage Offer



$$Y_i = \sum_{w \in \mathbb{W}} [(\beta_0^w (1 - H_i) + \beta_1^w H_i) \times \mathbf{1}\{W_i = w\}] + \gamma H_i \times W_i^P + \boldsymbol{\xi} \boldsymbol{X_i} + \epsilon_i$$

Raw Output Value

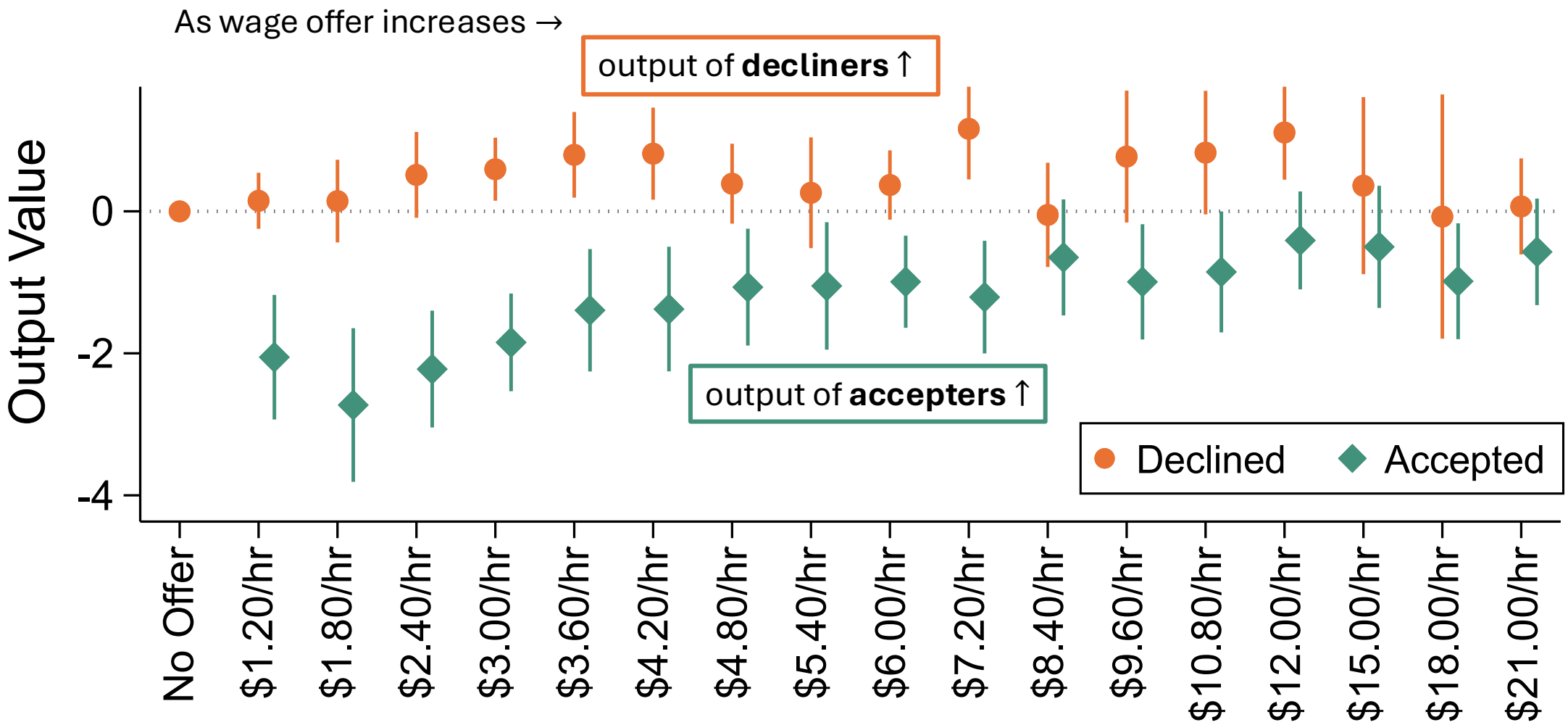
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Raw Output Value

Summary so far…

Let $\overline{w}_i \equiv$ **hourly reservation wage** (the lowest hourly wage i would accept)
Estimates of *average* treatment and selection into fixed wage, $w \Rightarrow$

Evidence of Moral Hazard

Hourly wages reduce average output among workers who accept.

$$E[Y_{1i} - Y_{0i} | \overline{w}_i \leq w] < 0$$

Treatment Effect

Evidence of Adverse Selection

Low-productivity workers are more likely to accept low hourly wage offers.

$$\frac{\partial}{\partial w} E[Y_{0i} | \overline{w}_i > w] > 0$$

Selection on Y_0

$$\frac{\partial}{\partial w} E[Y_{1i} | \overline{w}_i \leq w] > 0$$

Selection on Y_1

What are the implications for… Market equilibrium? Social welfare? Public policy?

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Selection on Y_1

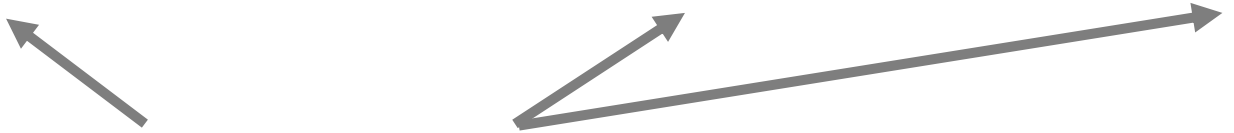
What are the implications for… Market equilibrium? Social welfare? Public policy?

Model shows how answers depend on the *marginal* worker ($\overline{w}_i = w$)

$$E[Y_{1i} - Y_{0i} | \overline{w}_i = w]$$

$$E[Y_{0i} | \overline{w}_i = w]$$

$$E[Y_{1i} | \overline{w}_i = w]$$



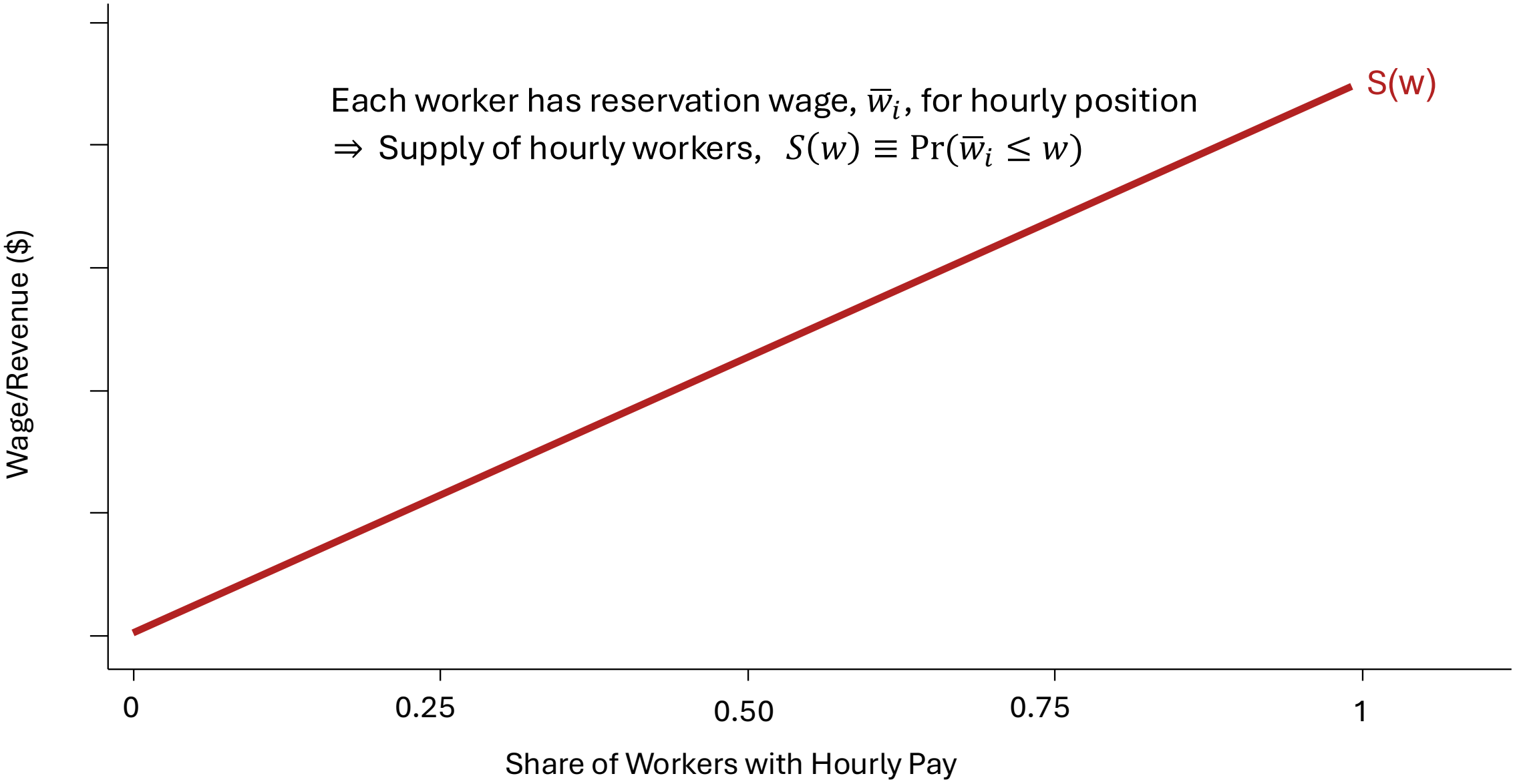
Identified with Marginal Treatment Effects (MTE)

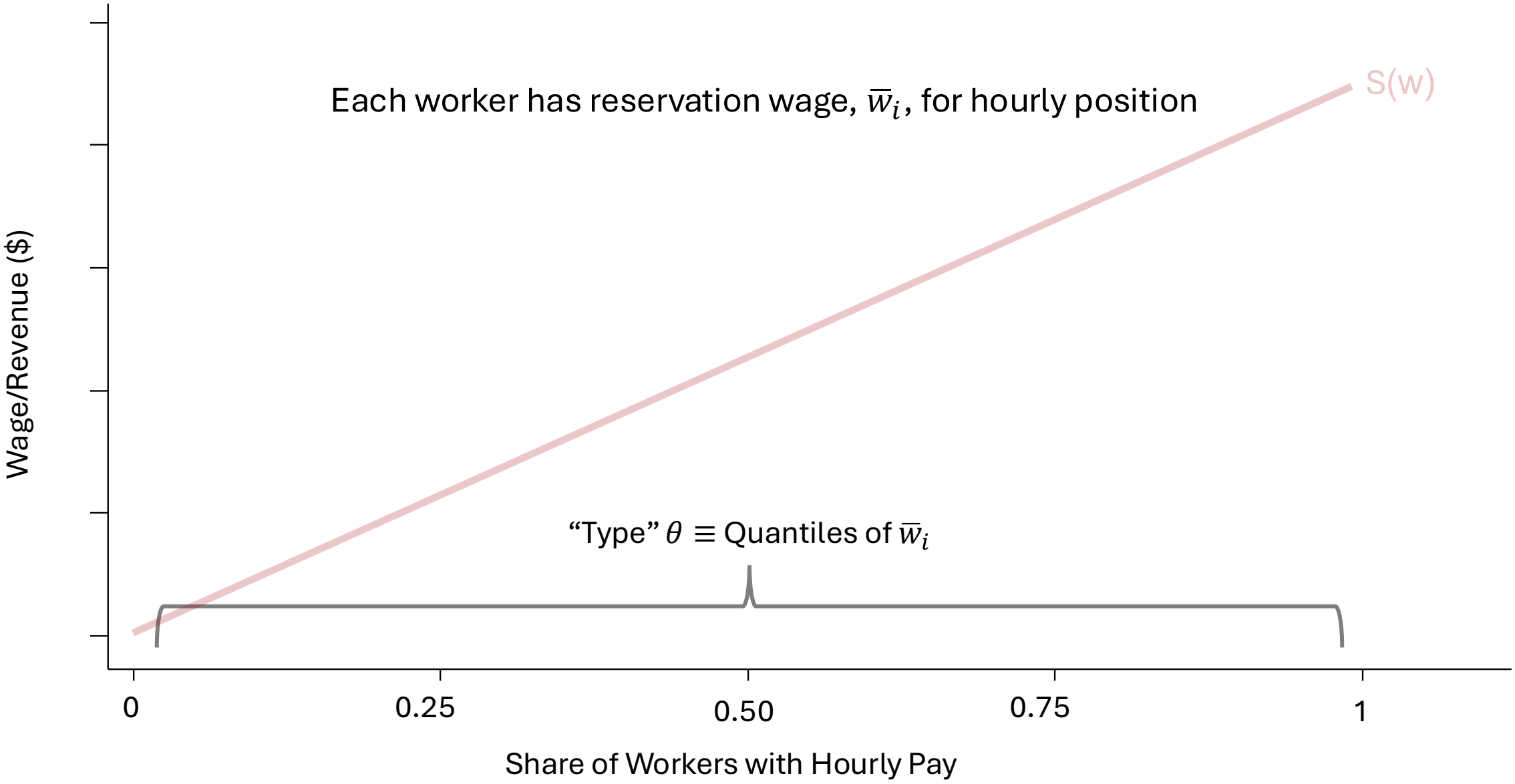
Outline

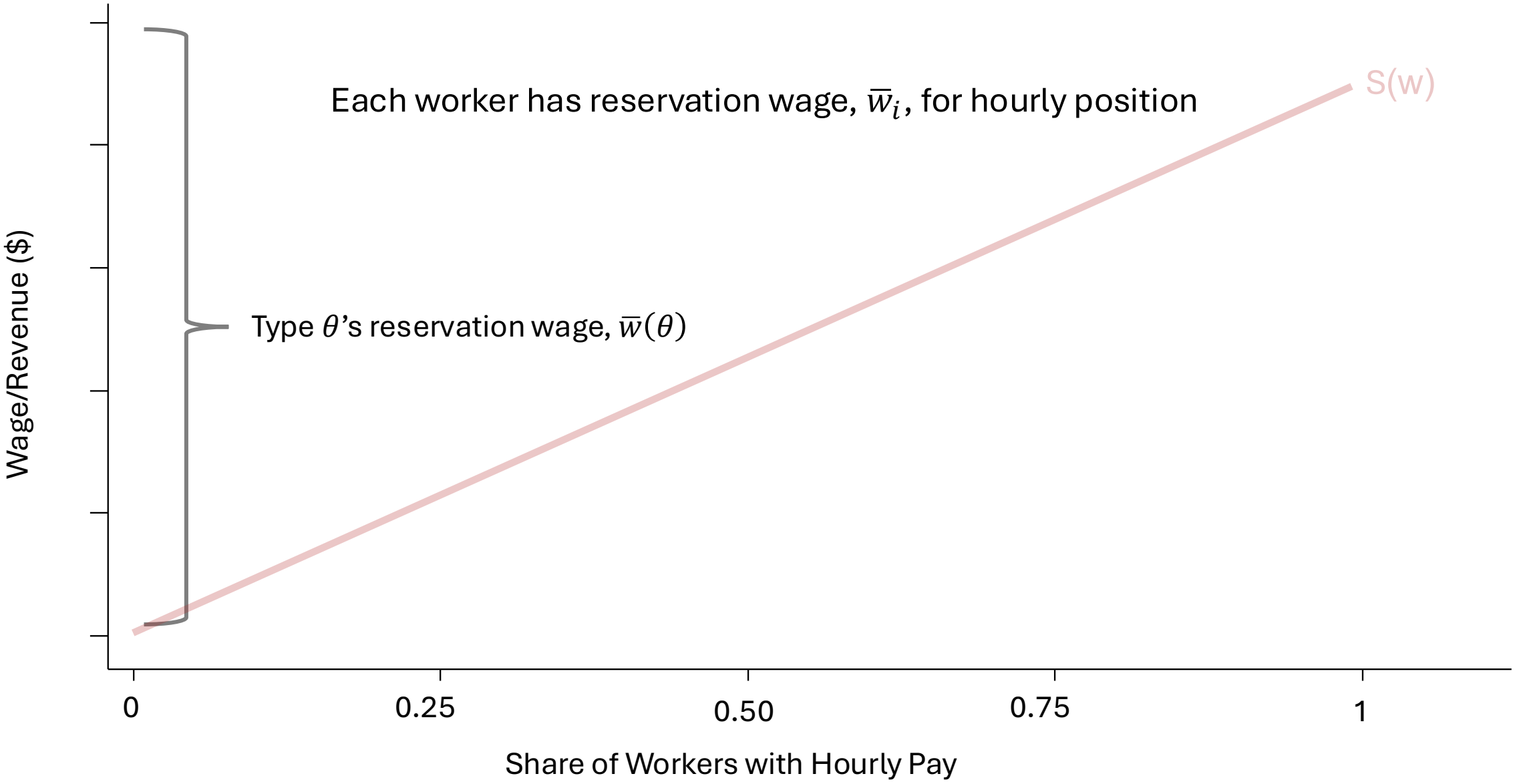
1. Experimental Design and Results
2. Model of Asymmetric Information
3. Estimates of Marginal Value and Welfare Loss
4. Optimal Fixed-Wage Subsidy and Piece-Rate Tax
5. Extensions

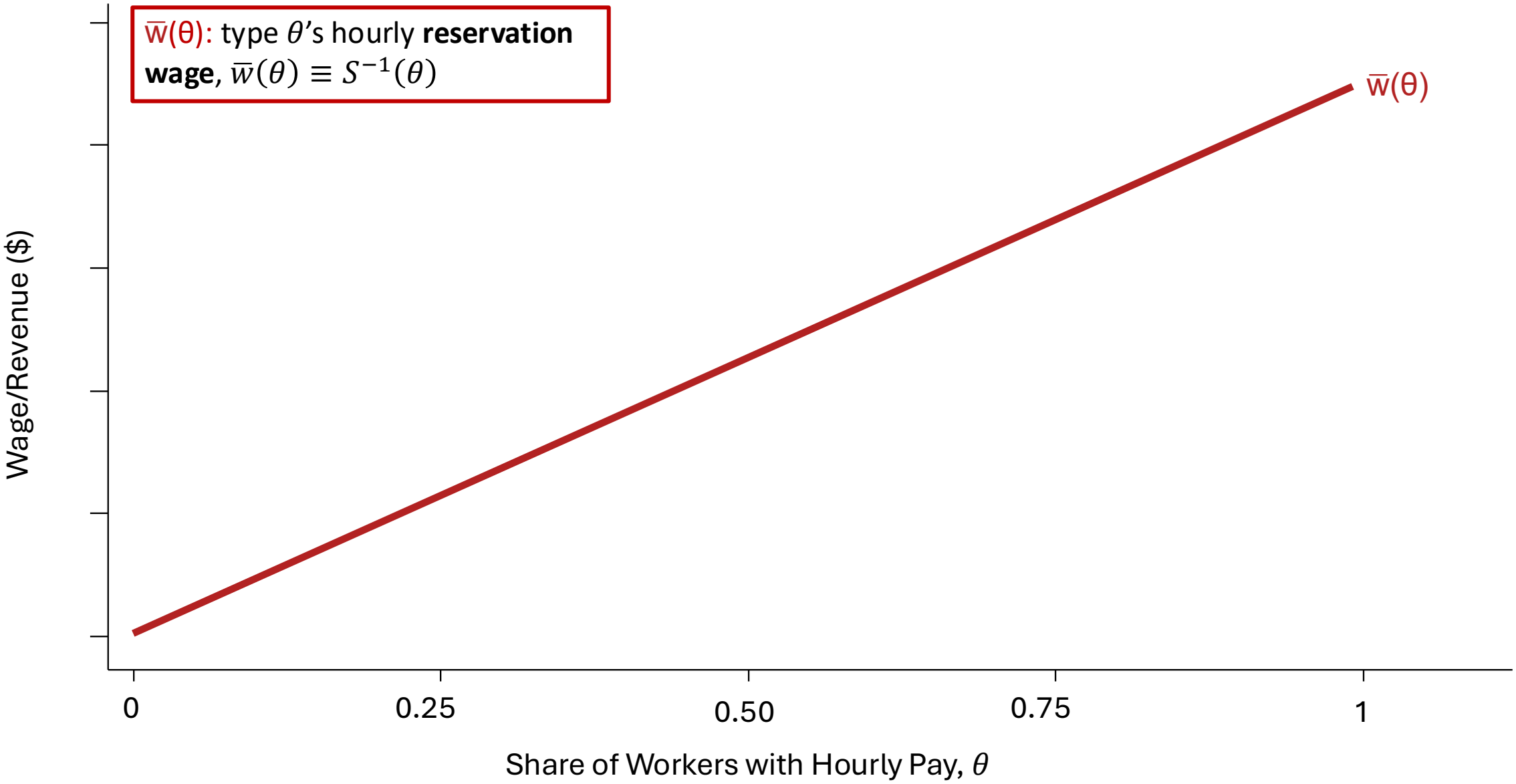
Model

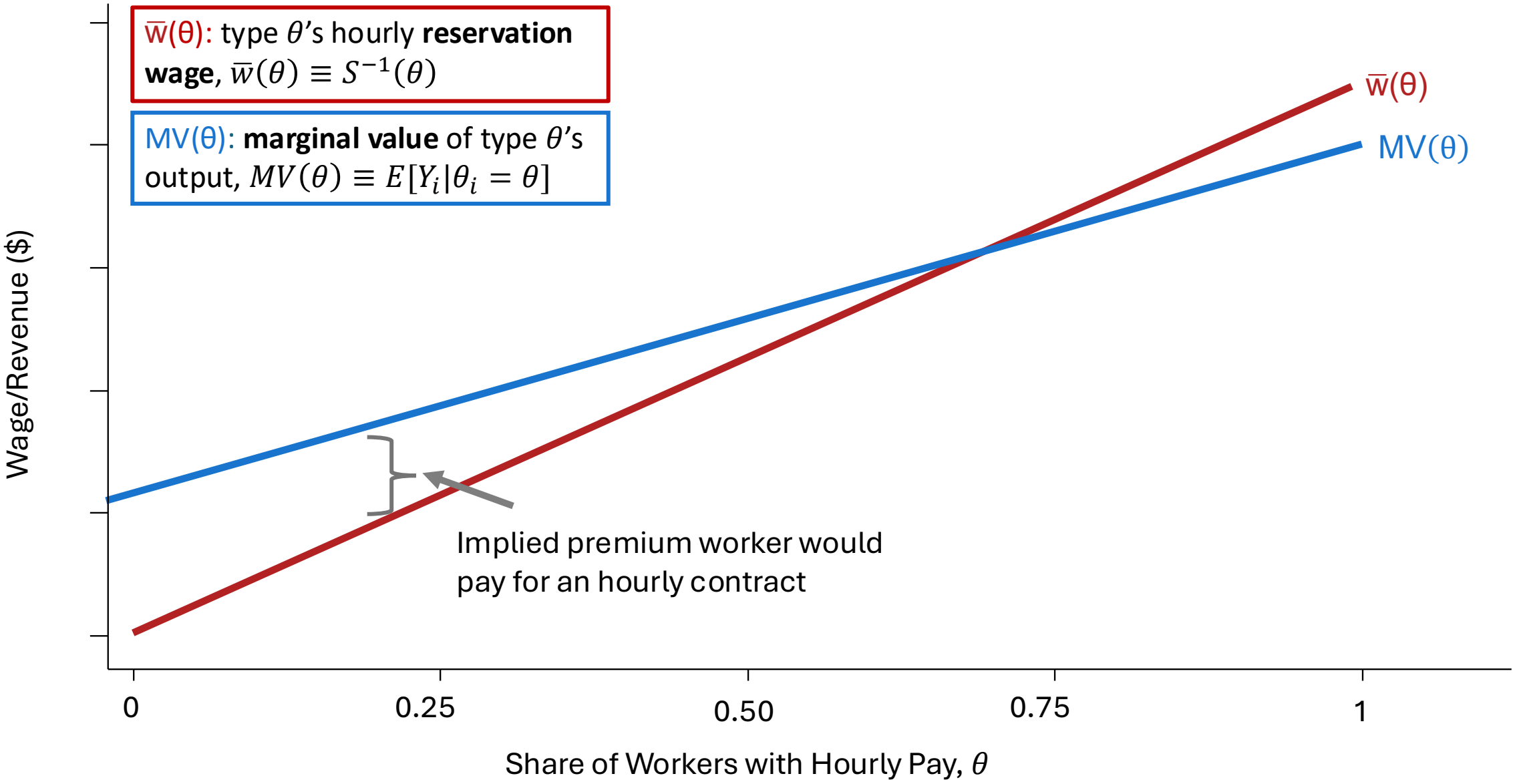
- Population of observationally equivalent (pre-screened) workers
- Each worker produces hourly output $q_i = f(\zeta_i, e_i, \nu_i)$
  - ζ_i: (unobserved) worker characteristics
  - e_i: individual effort, potentially influenced by the contract (moral hazard)
  - ν_i: random noise
- Value of worker i's output is $Y_i \equiv pq_i$, where $p \equiv$ price per unit q
- Firms have two options:
  1. Buy output at per-unit market price p (e.g., piece rate, freelance hire)
  2. Offer worker *ex-ante* hourly wage, w, in exchange for claim on realized q_i

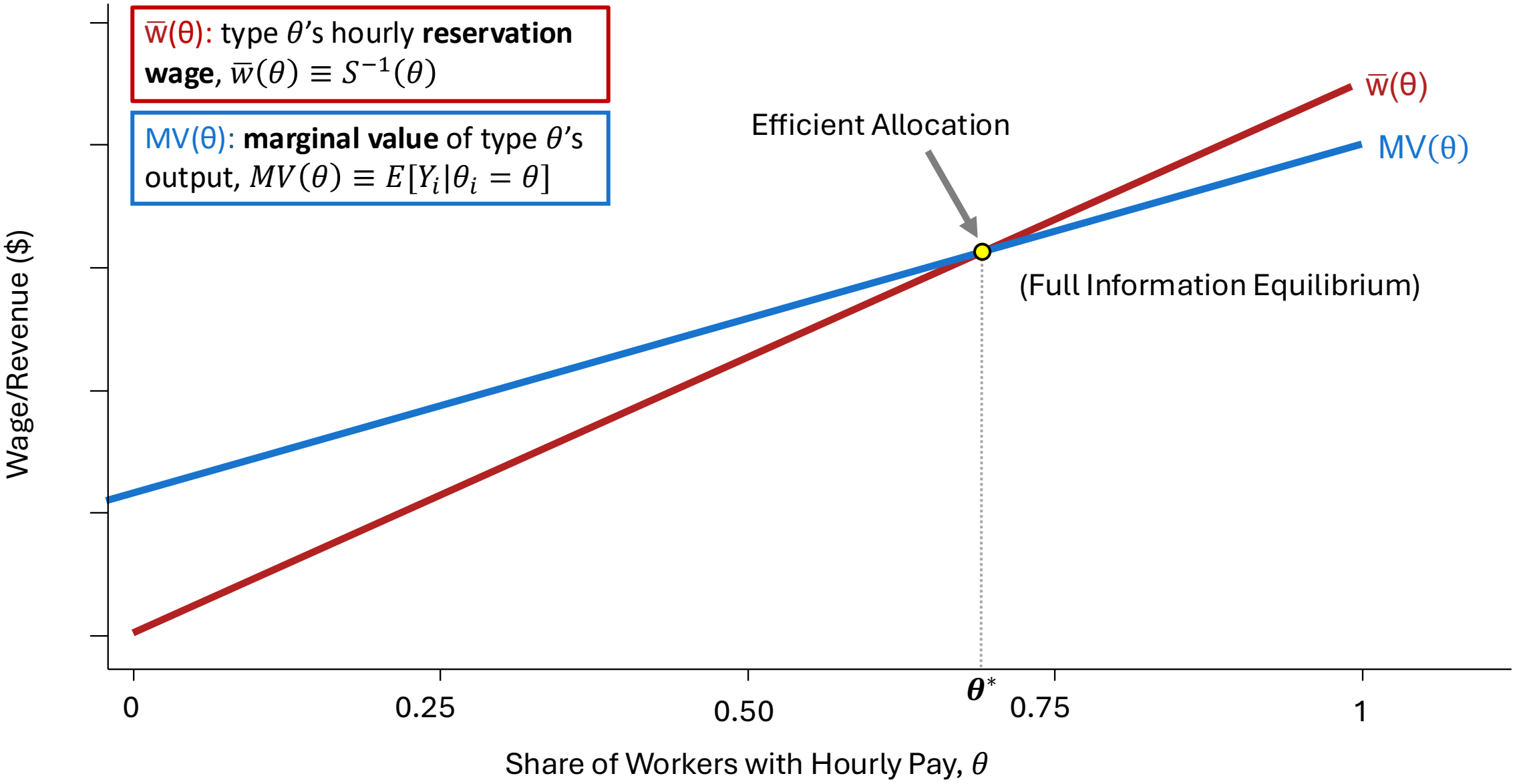

Each worker has reservation wage, $\overline{w}_i$, for hourly position
⇒ Supply of hourly workers, $S(w) \equiv \Pr(\overline{w}_i \leq w)$
S(w)
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay

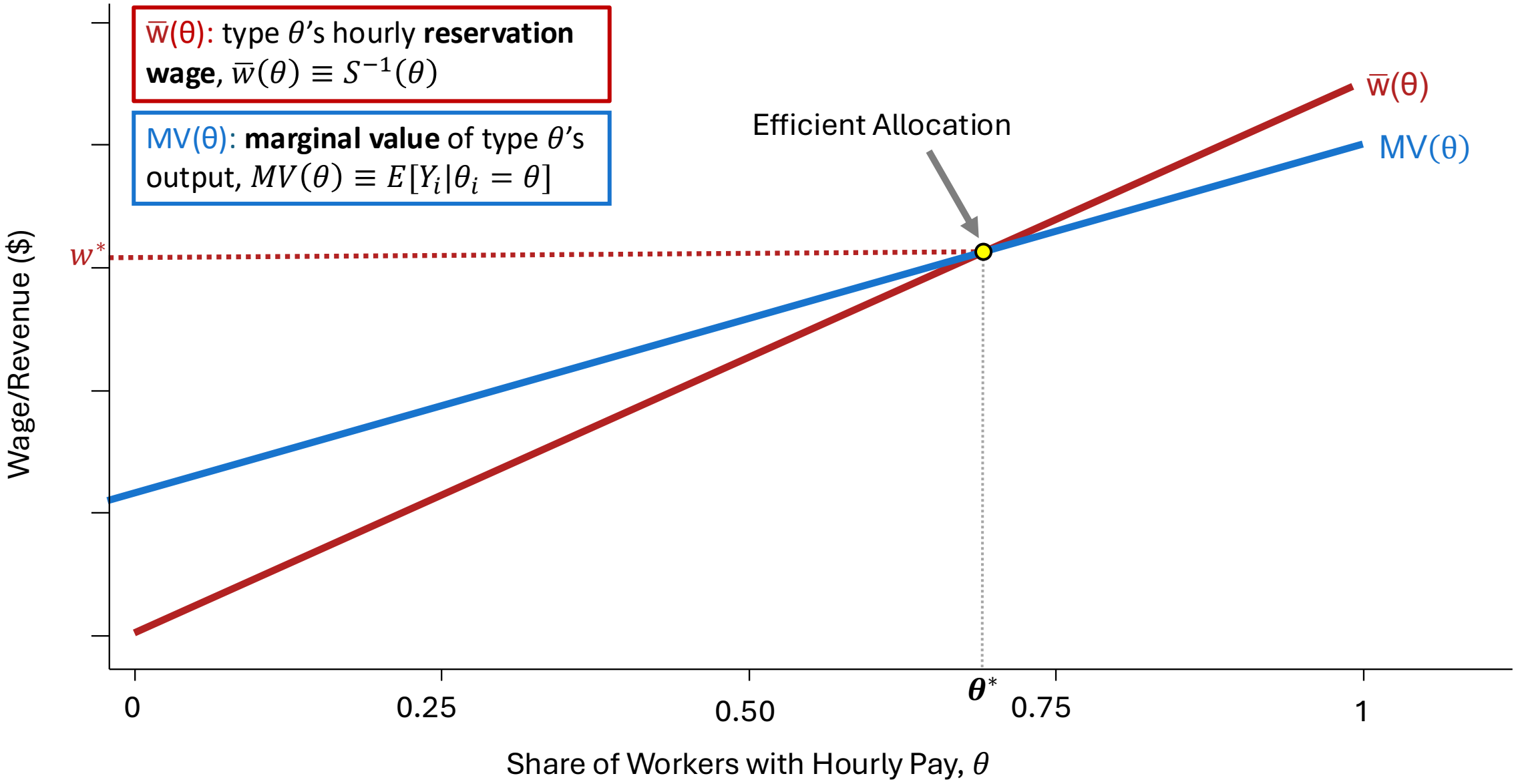

Each worker has reservation wage, $\overline{w}_i$, for hourly position
S(w)
"Type" $\theta \equiv$ Quantiles of $\overline{w}_i$
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay

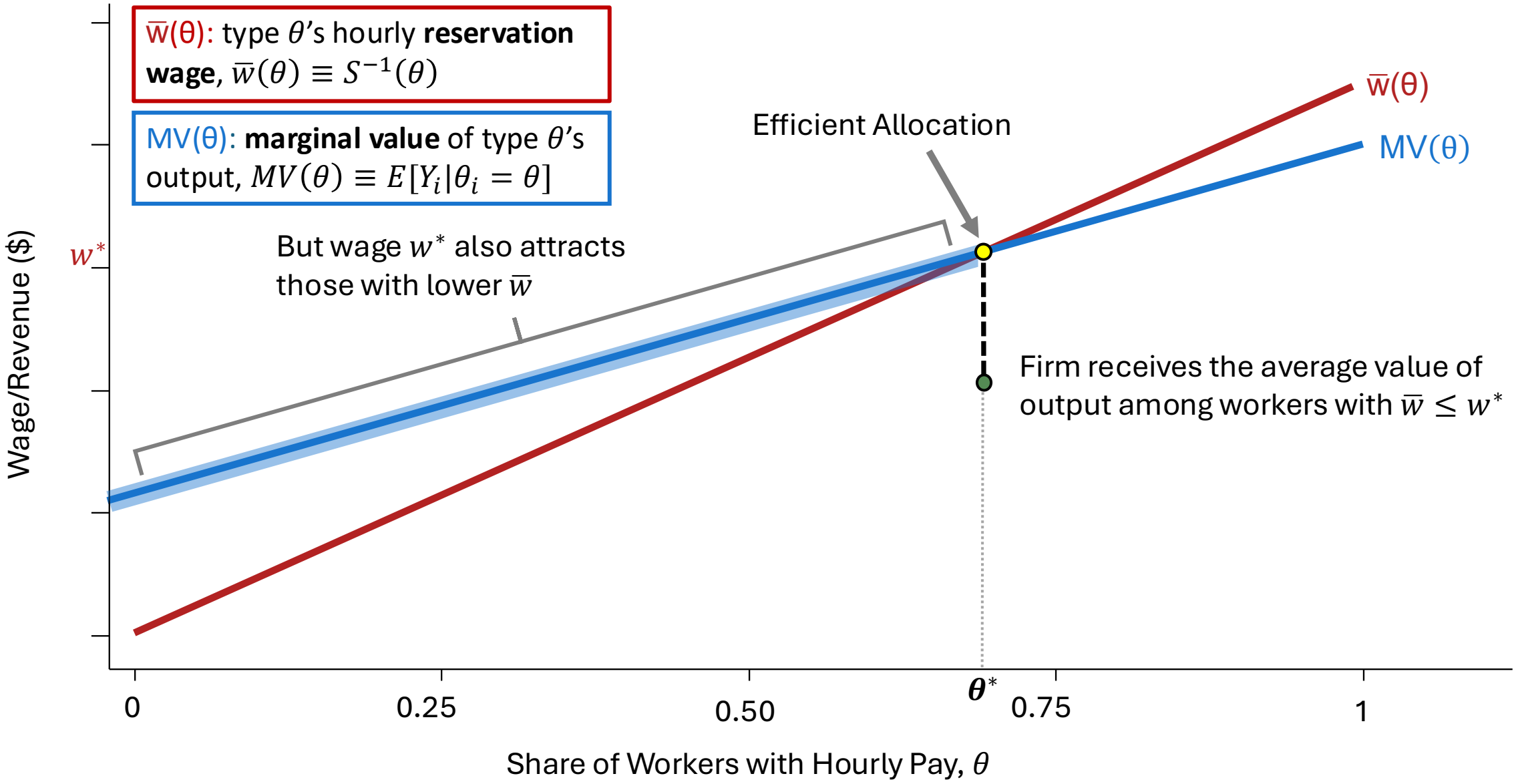

Each worker has reservation wage, $\overline{w}_i$, for hourly position
S(w)
Type θ's reservation wage, $\overline{w}(\theta)$
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay

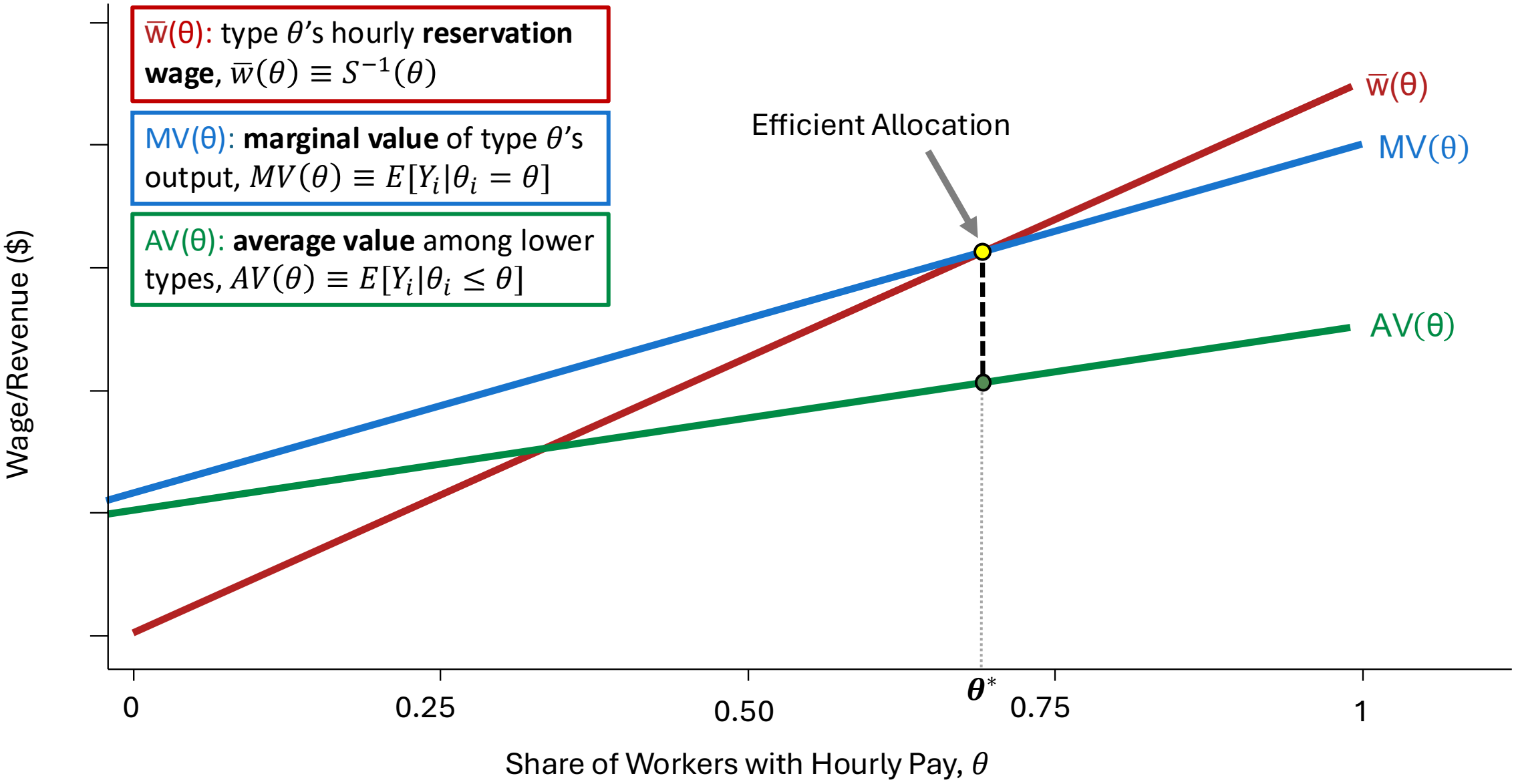

w̄(θ): type θ's hourly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
w̄(θ)
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ

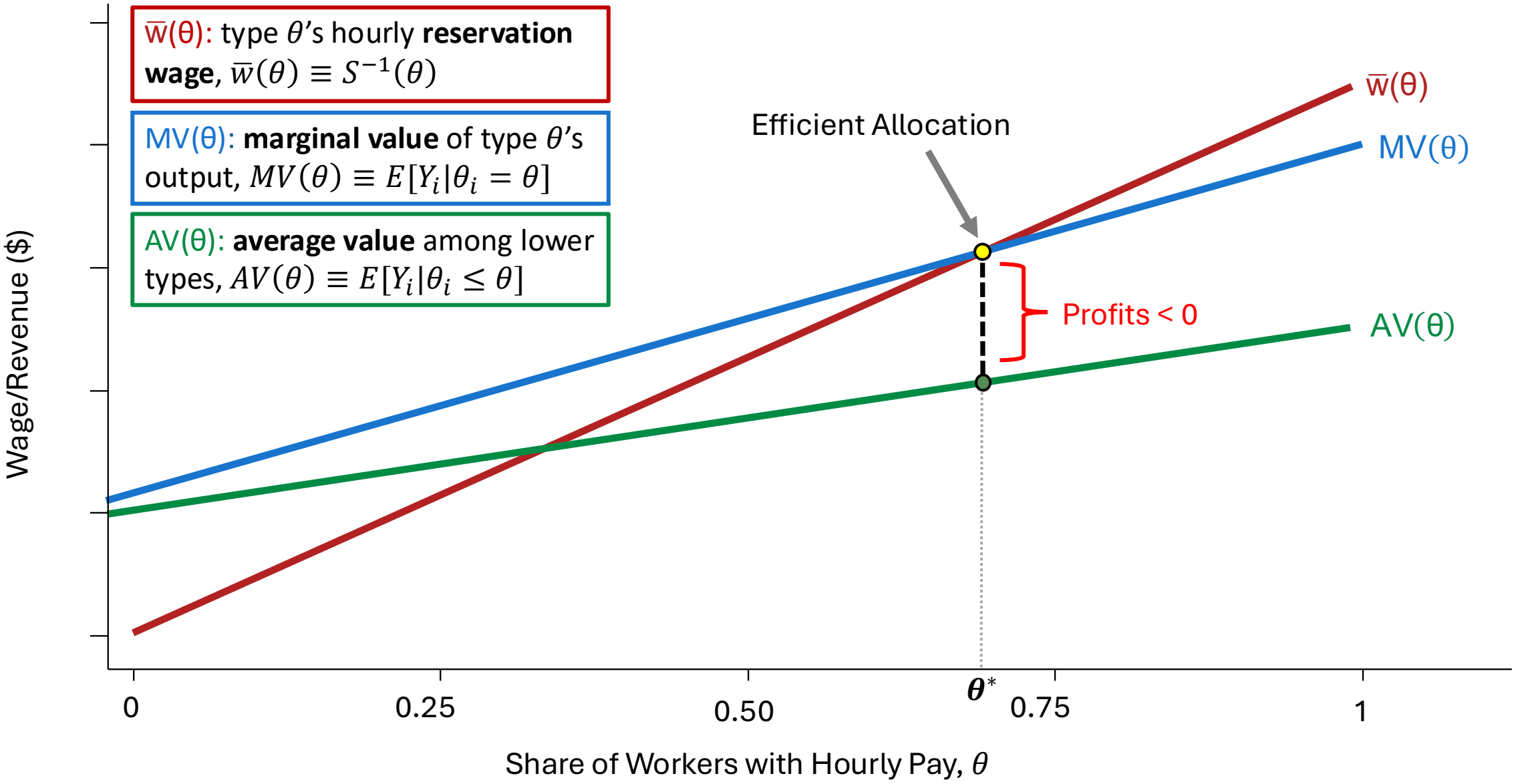

w̄(θ): type θ's hourly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
MV(θ): **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$
w̄(θ)
MV(θ)
Implied premium worker would pay for an hourly contract
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ

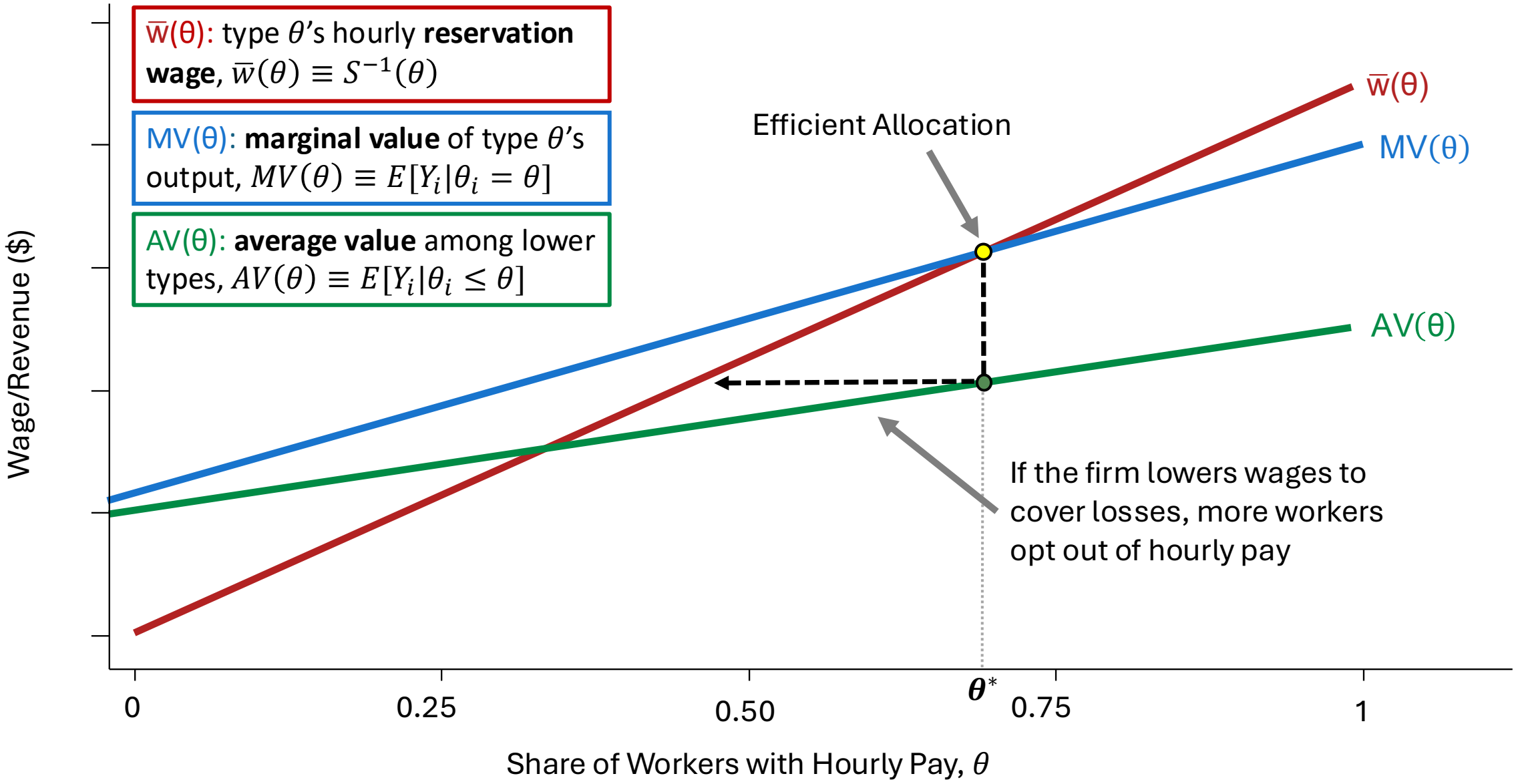

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w̄(θ)
MV(θ)
Efficient Allocation
(Full Information Equilibrium)
Wage/Revenue ($)
0
0.25
0.50
θ^*
0.75
1
Share of Workers with Hourly Pay, θ

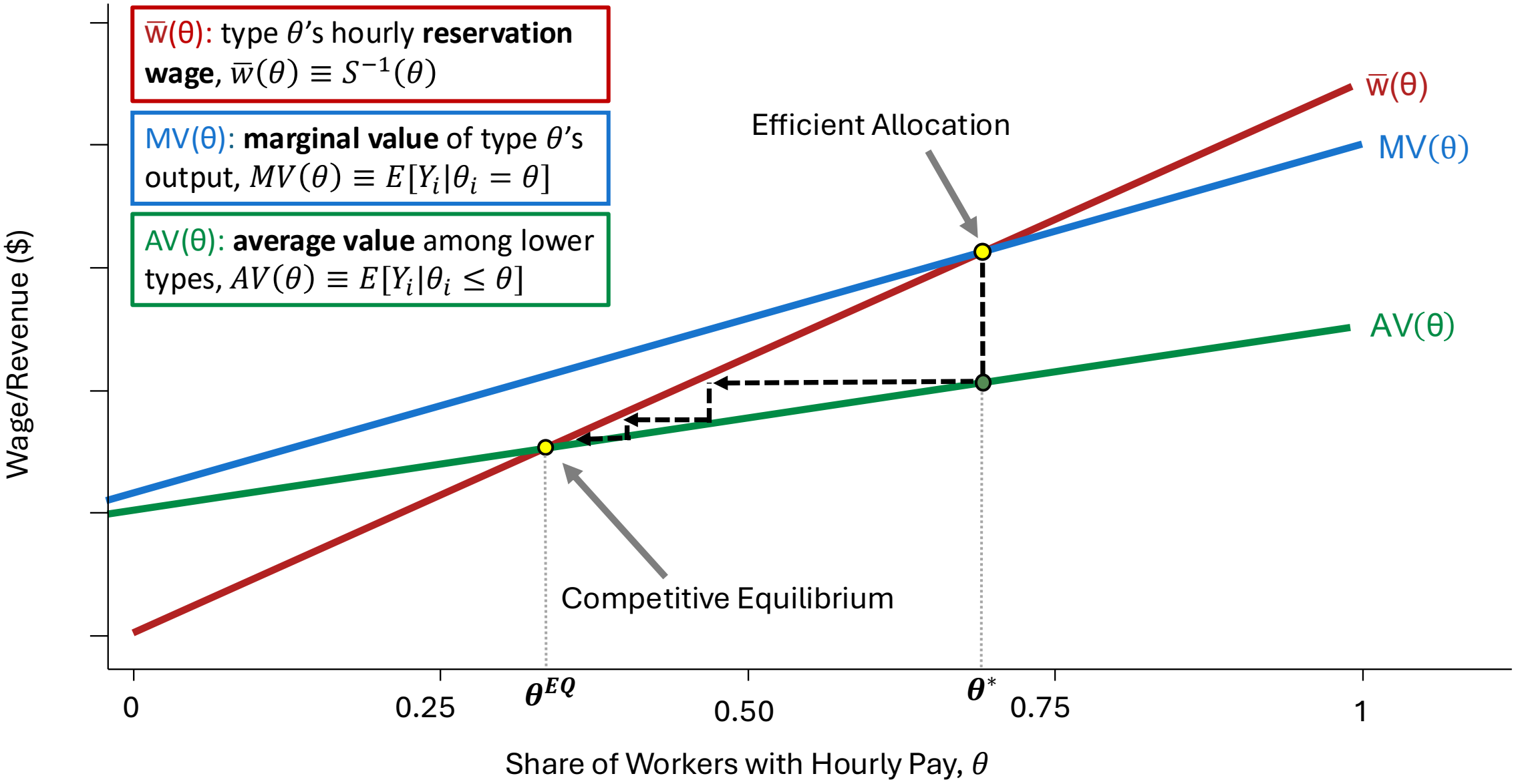

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w̄(θ)
MV(θ)
Efficient Allocation
w^*
Wage/Revenue ($)
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Share of Workers with Hourly Pay, θ

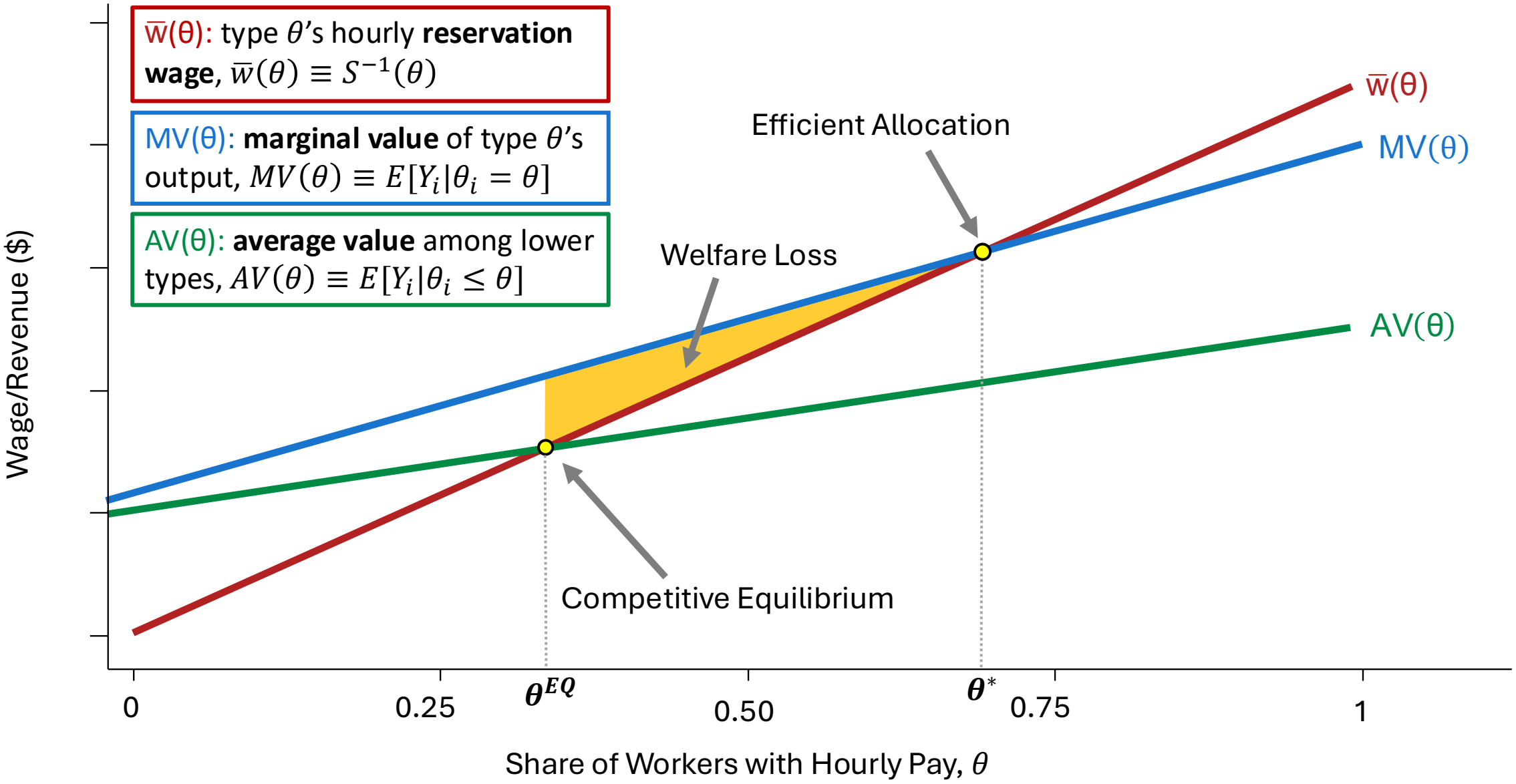

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w̄(θ)
MV(θ)
Efficient Allocation
But wage w^* also attracts those with lower $\bar{w}$
w^*
Firm receives the average value of output among workers with $\bar{w} \leq w^*$
Wage/Revenue ($)
0
0.25
0.50
θ^*
0.75
1
Share of Workers with Hourly Pay, θ


w̄(θ): type θ's hourly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
MV(θ): **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$
AV(θ): **average value** among lower types, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$
w̄(θ)
MV(θ)
AV(θ)
Efficient Allocation
Wage/Revenue ($)
0
0.25
0.50
θ^*
0.75
1
Share of Workers with Hourly Pay, θ


w̄(θ): type θ's hourly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
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w̄(θ)
MV(θ)
AV(θ)
Efficient Allocation
Profits < 0
Wage/Revenue ($)
0
0.25
0.50
θ^*
0.75
1
Share of Workers with Hourly Pay, θ


w̄(θ): type θ's hourly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
MV(θ): **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$
AV(θ): **average value** among lower types, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$
w̄(θ)
MV(θ)
AV(θ)
Efficient Allocation
If the firm lowers wages to cover losses, more workers opt out of hourly pay
Wage/Revenue ($)
0
0.25
0.50
θ^*
0.75
1
Share of Workers with Hourly Pay, θ


w̄(θ): type θ's hourly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
MV(θ): **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$
AV(θ): **average value** among lower types, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$
w̄(θ)
MV(θ)
AV(θ)
Efficient Allocation
Competitive Equilibrium
Wage/Revenue ($)
0
0.25
θ^{EQ}
0.50
θ^*
0.75
1
Share of Workers with Hourly Pay, θ


w̄(θ): type θ's hourly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
MV(θ): **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$
AV(θ): **average value** among lower types, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$
w̄(θ)
MV(θ)
AV(θ)
Efficient Allocation
Welfare Loss
Competitive Equilibrium
Wage/Revenue ($)
0
0.25
θ^{EQ}
0.50
θ^*
0.75
1
Share of Workers with Hourly Pay, θ

Incorporating Moral Hazard

- Where does moral hazard fit in? Consider potential outputs

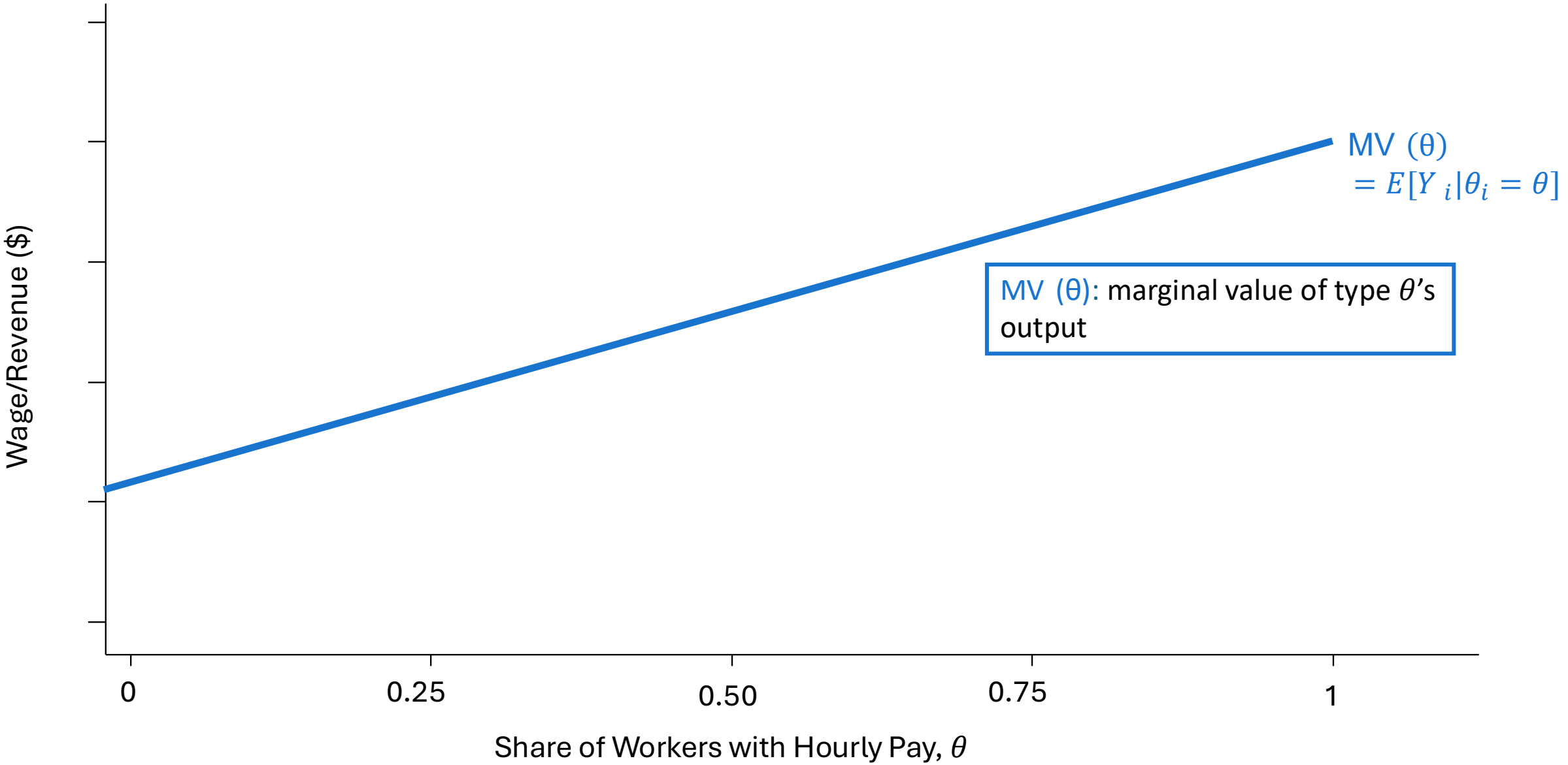
  Y_{1i}: Potential output value under hourly wage
  Y_{0i}: Potential output value under piece rate
  $\Big\}\ Y_{1i} = Y_{0i} + MH_i$

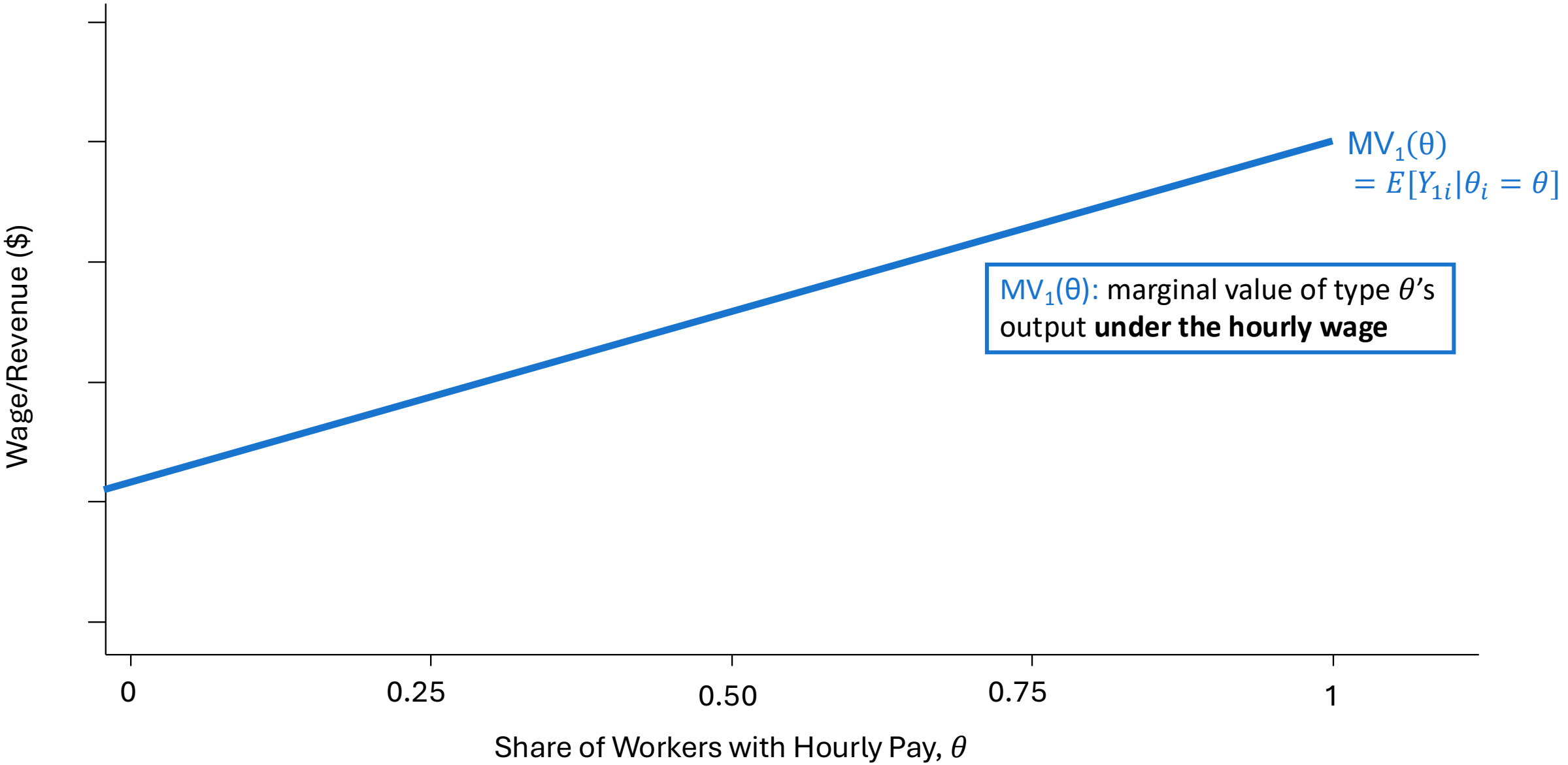
- Firms care about Y_{1i}, but they don't care about Y_{0i}
  - Piece-rate workers sell their output at a constant price per unit, so their productivity has no effect on firm profits
- $MV(\theta)$ & $AV(\theta)$ are defined *conditional* on accepting the hourly wage

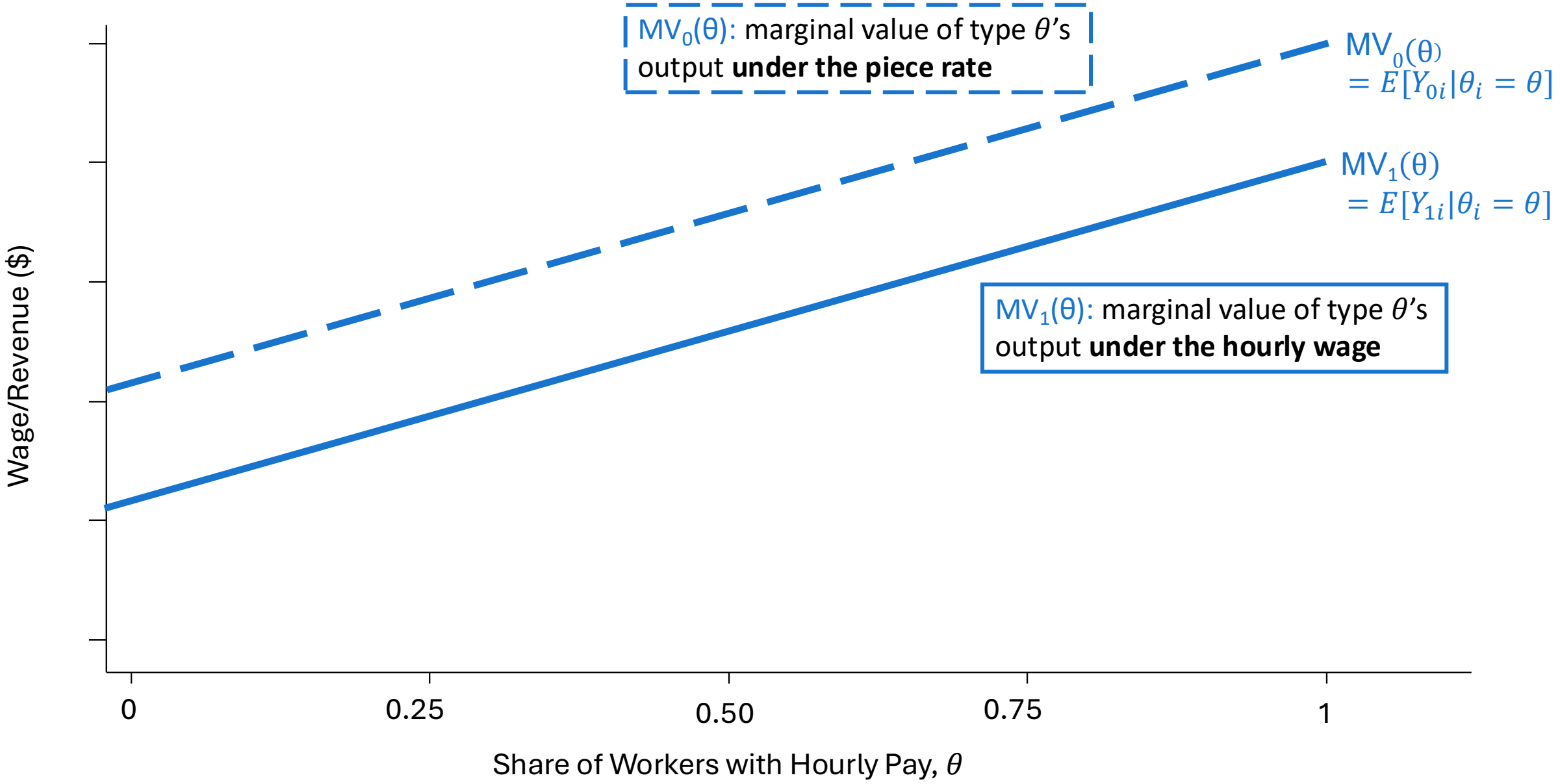
$$MV(\theta) \equiv E[Y_i|\theta_i = \theta] = E[Y_{1i}|\theta_i = \theta]$$
$$AV(\theta) \equiv E[Y_i|\theta_i \leq \theta] = E[Y_{1i}|\theta_i \leq \theta]$$

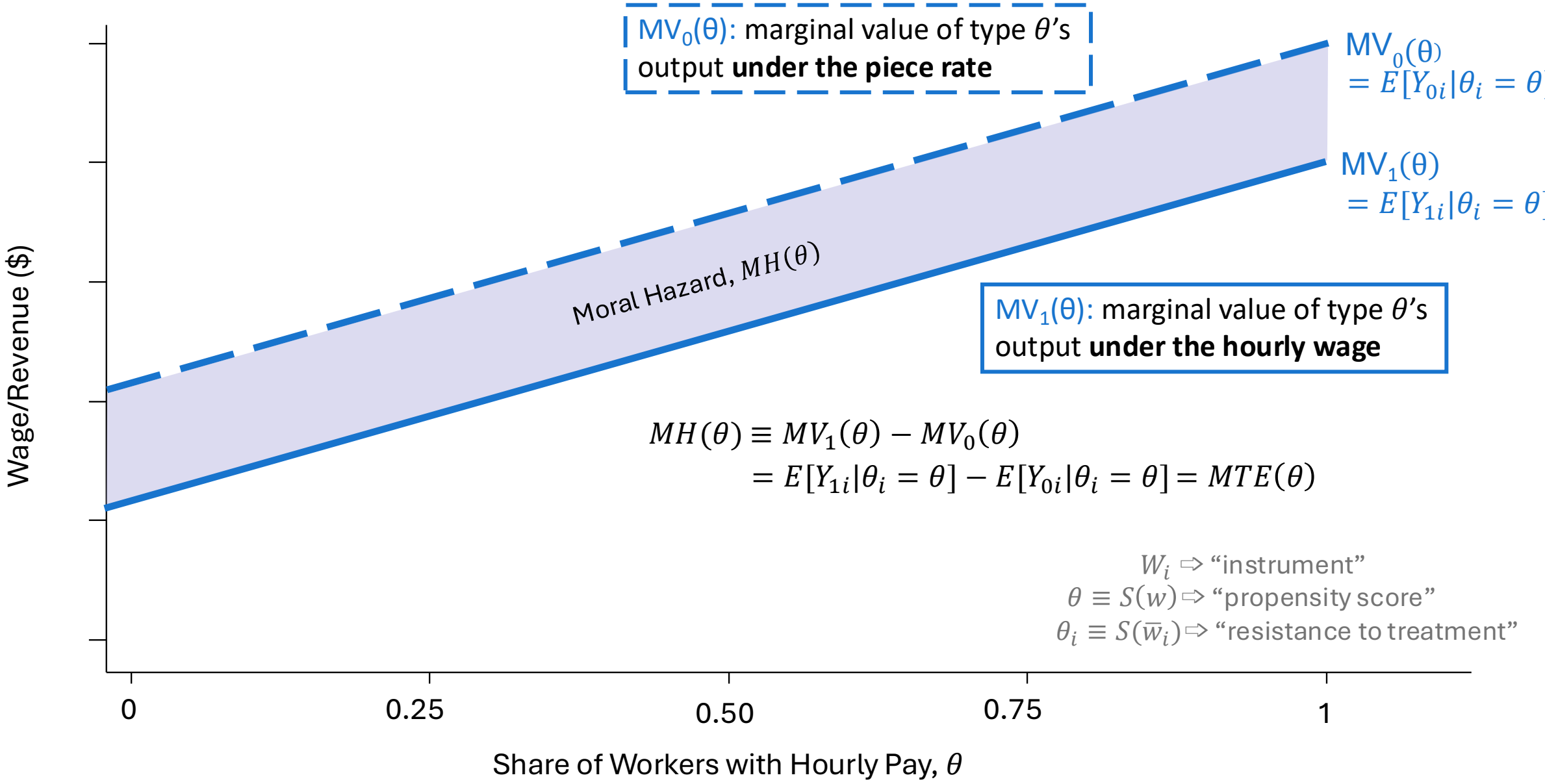
  ⇒ Equilibrium is inclusive of workers' moral hazard response to the hourly wage

- Still want to separately identify moral hazard
  - Firms might mitigate MH with "partial insurance" (e.g., sales commission, tips)
  - MH needed to measure welfare impact of policies promoting fixed-wages
    - Trade-off between insurance value and fiscal externality (MH ⇒ tax revenue ↓)


MV (θ)
$= E[Y_i|\theta_i = \theta]$
MV (θ): marginal value of type θ's output
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ

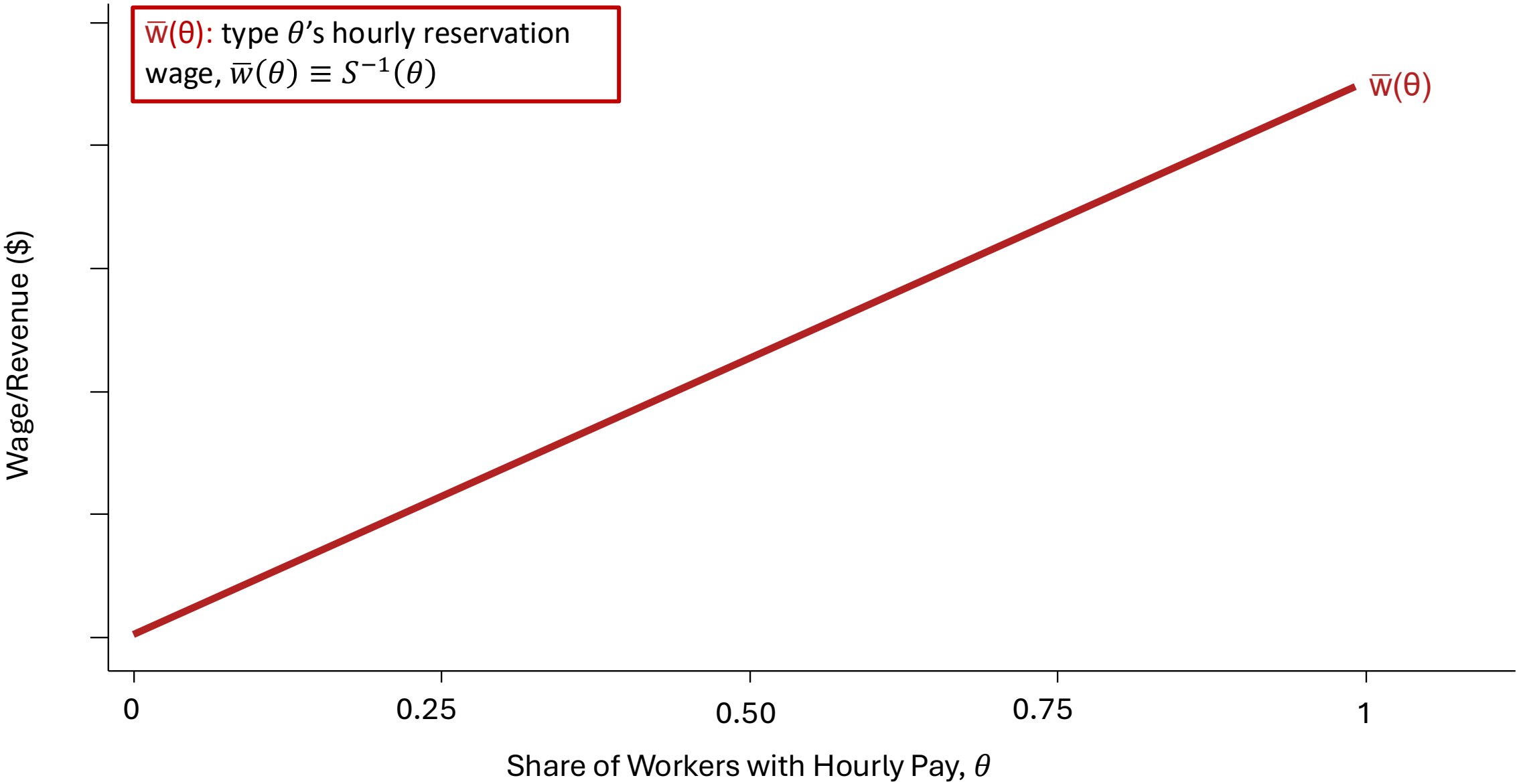

Wage/Revenue ($)
$MV_1(\theta) = E[Y_{1i}|\theta_i = \theta]$
$MV_1(\theta)$: marginal value of type θ's output **under the hourly wage**
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ

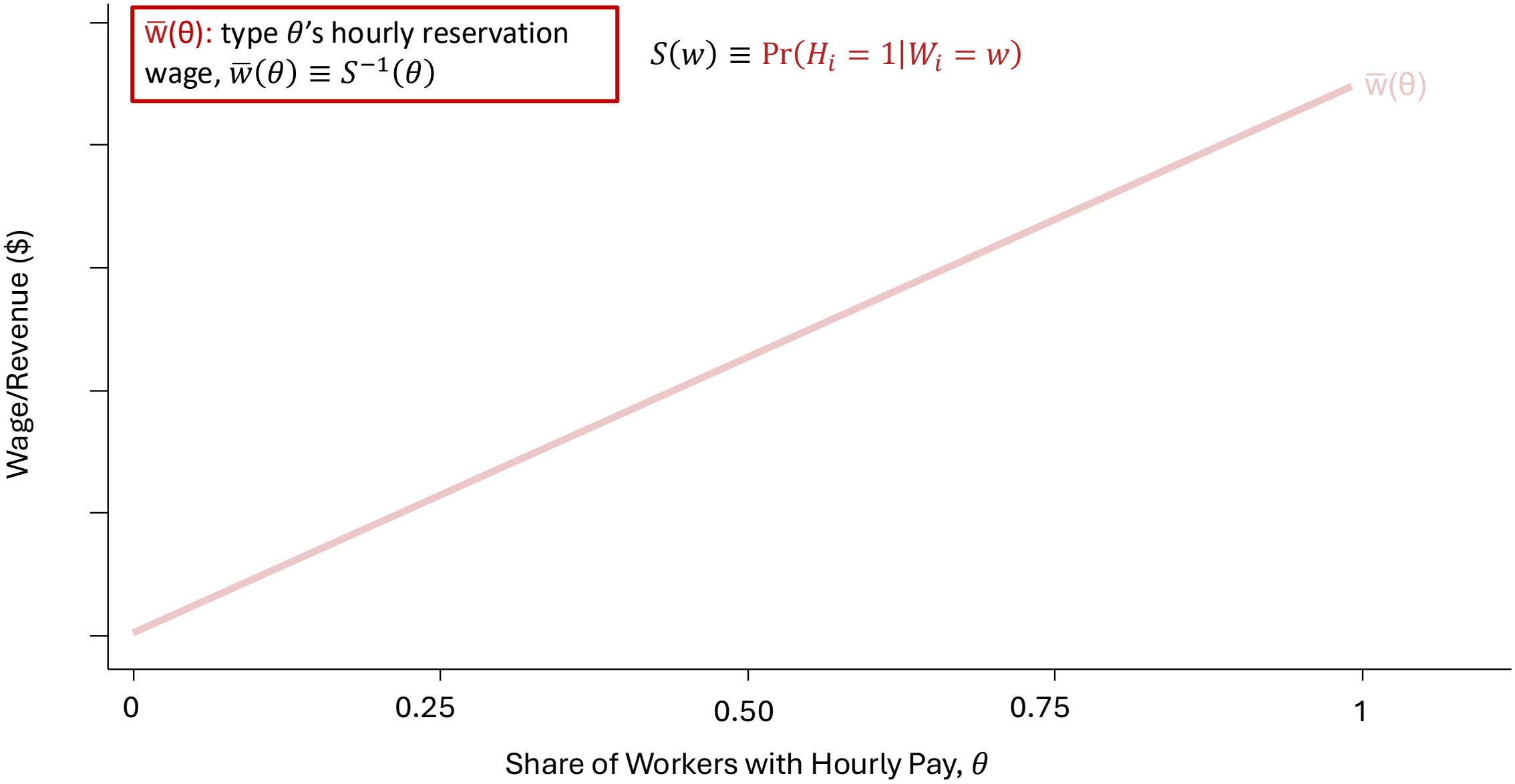

MV0(θ): marginal value of type θ's output under the piece rate
MV0(θ) = E[Y0i|θi = θ]
MV1(θ) = E[Y1i|θi = θ]
MV1(θ): marginal value of type θ's output under the hourly wage
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ

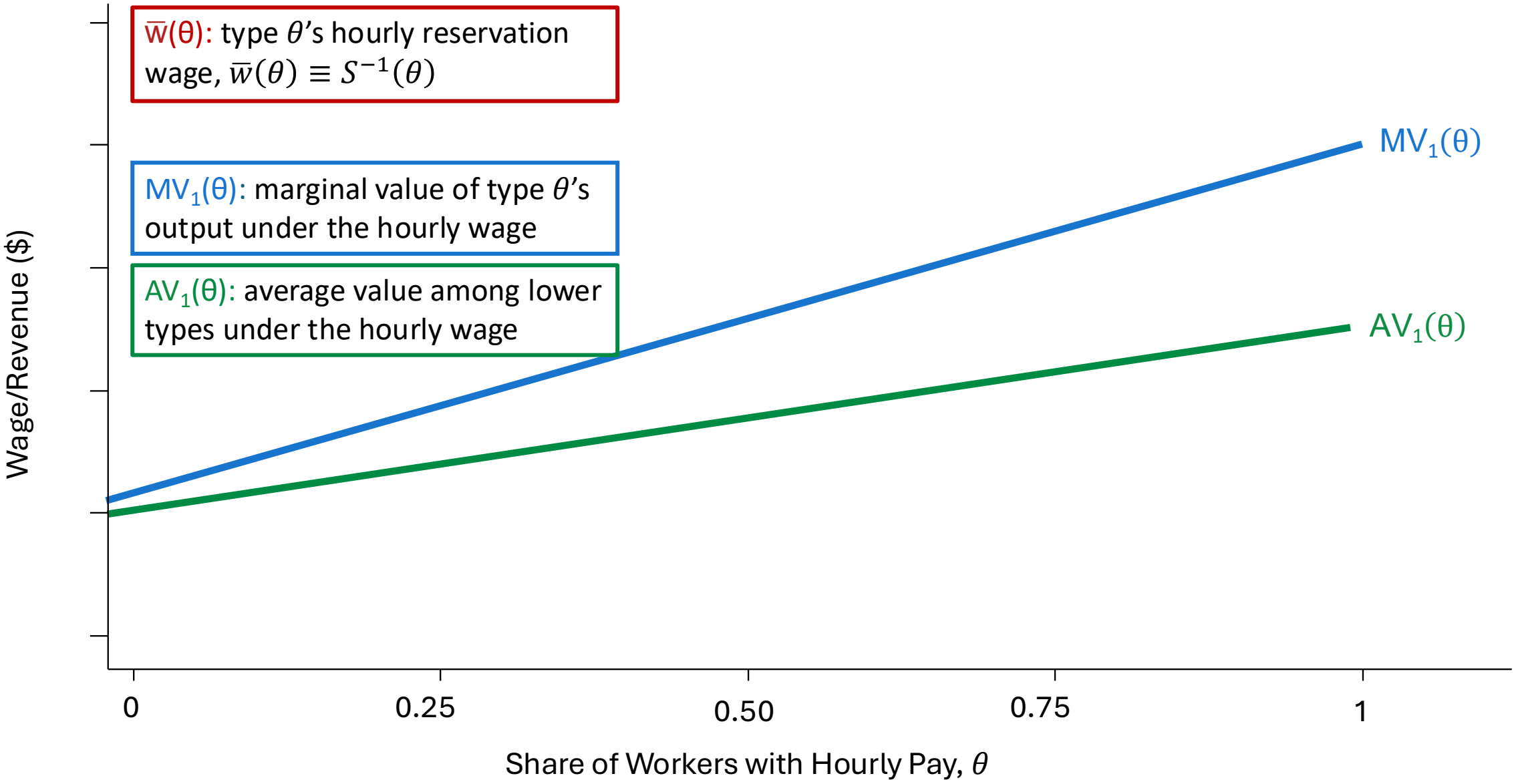

MV$_0$(θ): marginal value of type θ's output **under the piece rate**
MV$_0$(θ) $= E[Y_{0i}|\theta_i = \theta]$
MV$_1$(θ) $= E[Y_{1i}|\theta_i = \theta]$
Moral Hazard, MH(θ)
MV$_1$(θ): marginal value of type θ's output **under the hourly wage**
$MH(\theta) \equiv MV_1(\theta) - MV_0(\theta)$
$= E[Y_{1i}|\theta_i = \theta] - E[Y_{0i}|\theta_i = \theta] = MTE(\theta)$
W_i ⇨ "instrument"
$\theta \equiv S(w)$ ⇨ "propensity score"
$\theta_i \equiv S(\overline{w}_i)$ ⇨ "resistance to treatment"
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ

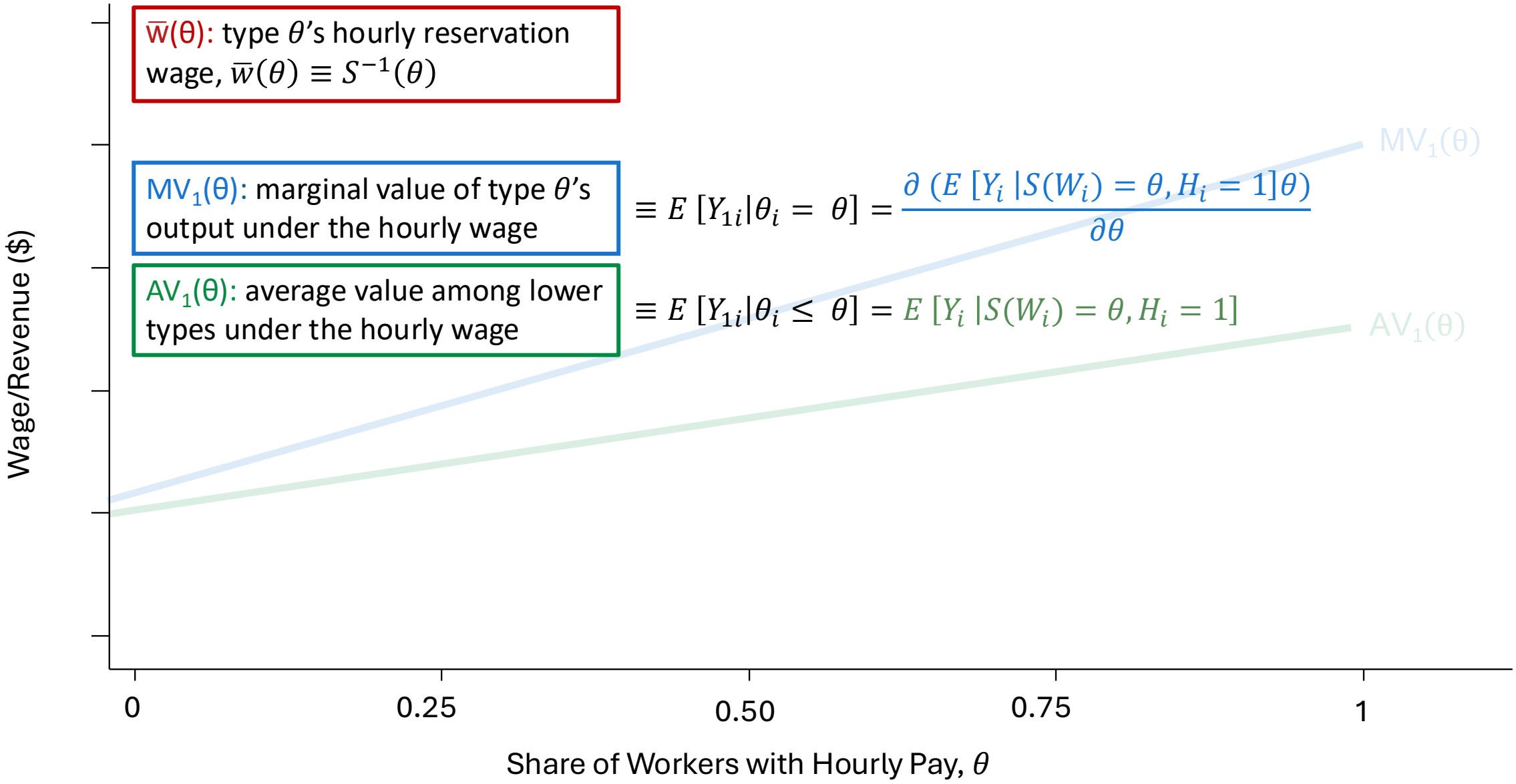
Outline

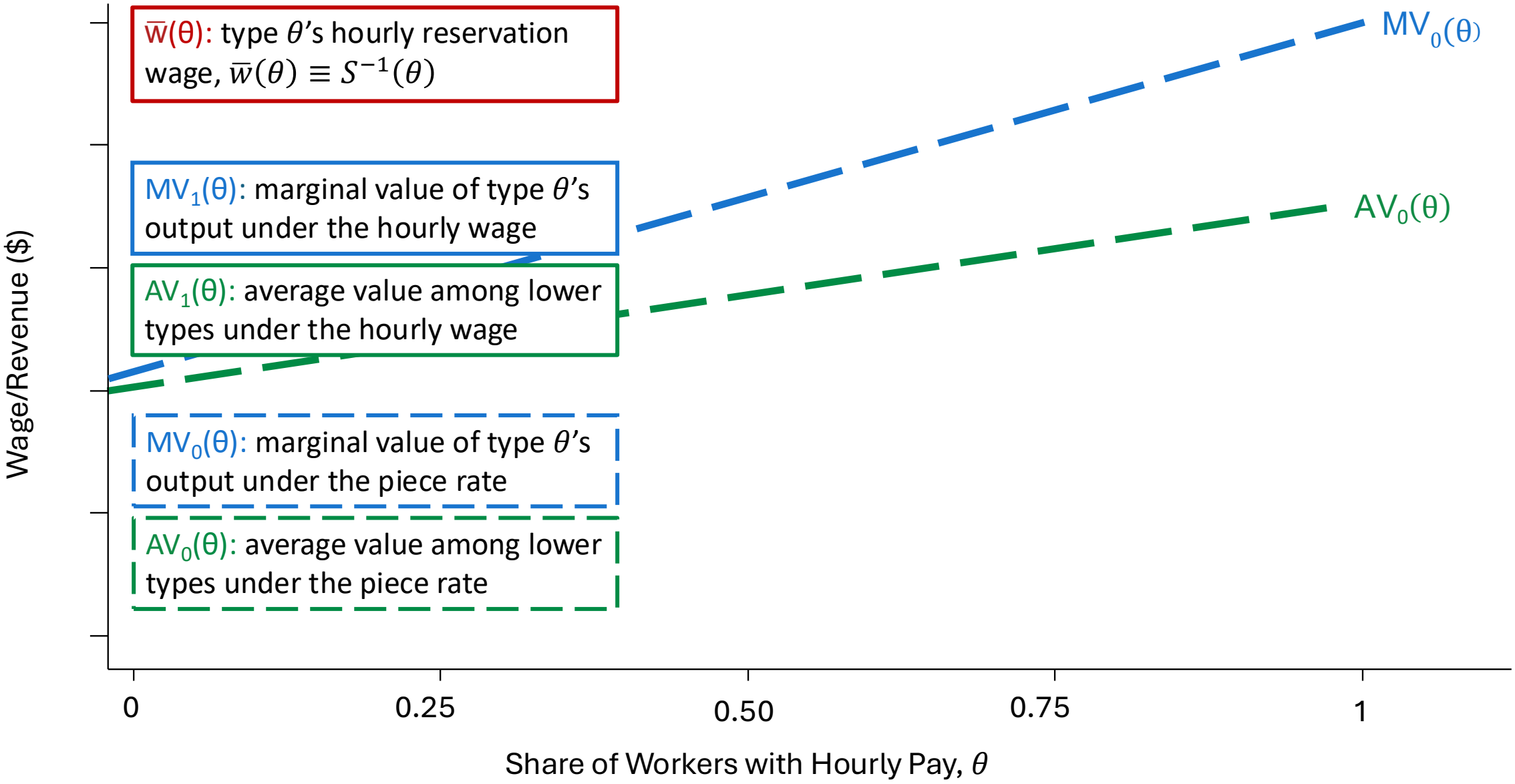
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w̄(θ)
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ

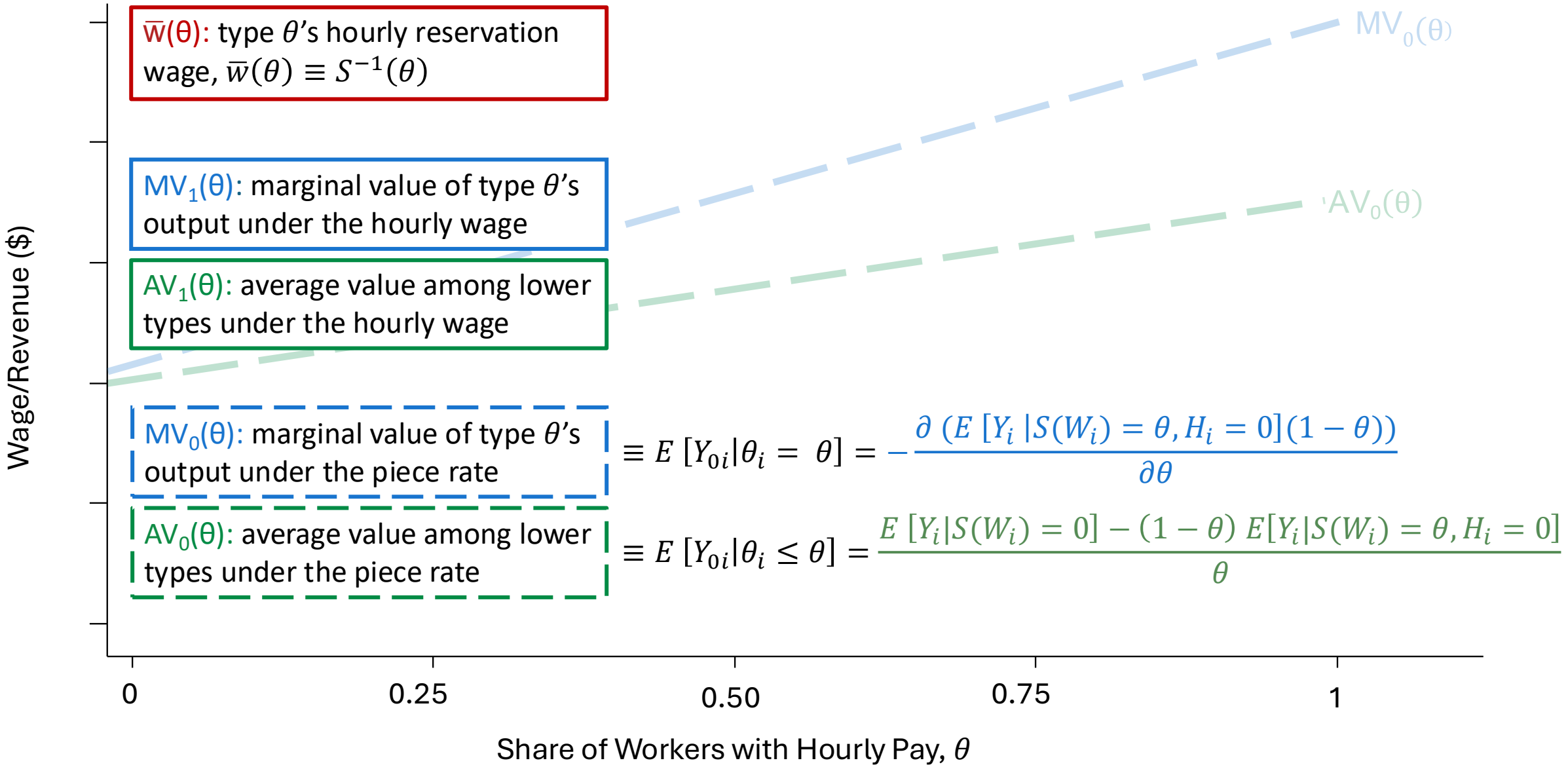

w̄(θ): type θ's hourly reservation wage, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
$S(w) \equiv \Pr(H_i = 1 | W_i = w)$
w̄(θ)
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ


$\bar{w}(\theta)$: type θ's hourly reservation wage, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
$MV_1(\theta)$: marginal value of type θ's output under the hourly wage
$AV_1(\theta)$: average value among lower types under the hourly wage
$MV_1(\theta)$
$AV_1(\theta)$
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ


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$AV_1(\theta)$: average value among lower types under the hourly wage
$\equiv E[Y_{1i}|\theta_i \leq \theta] = E[Y_i|S(W_i) = \theta, H_i = 1]$
$MV_1(\theta)$
$AV_1(\theta)$
Wage/Revenue ($)
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$MV_0(\theta)$: marginal value of type θ's output under the piece rate
$AV_0(\theta)$: average value among lower types under the piece rate
$MV_0(\theta)$
$AV_0(\theta)$
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ

AV Details


$\bar{w}(\theta)$: type θ's hourly reservation wage, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
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$\equiv E[Y_{0i}|\theta_i = \theta] = -\frac{\partial (E[Y_i|S(W_i) = \theta, H_i = 0](1-\theta))}{\partial \theta}$
$\equiv E[Y_{0i}|\theta_i \leq \theta] = \frac{E[Y_i|S(W_i) = 0] - (1-\theta)\,E[Y_i|S(W_i) = \theta, H_i = 0]}{\theta}$
$MV_0(\theta)$
$AV_0(\theta)$
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ

AV Details

$\bar{w}(\theta)$: type θ's hourly reservation wage, $\bar{w}(\theta) \equiv S^{-1}(\theta)$

$$S(w) \equiv \Pr(H_i = 1 | W_i = w)$$

$MV_1(\theta)$: marginal value of type θ's output under the hourly wage

$$\equiv E\left[Y_{1i} | \theta_i = \theta\right] = \frac{\partial \left(E\left[Y_i | S(W_i) = \theta, H_i = 1\right]\theta\right)}{\partial \theta}$$

$AV_1(\theta)$: average value among lower types under the hourly wage

$$\equiv E\left[Y_{1i} | \theta_i \leq \theta\right] = E\left[Y_i | S(W_i) = \theta, H_i = 1\right]$$

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$AV_0(\theta)$: average value among lower types under the piece rate

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Estimation

(Carneiro et al. 2011)

1. Estimate $\Pr(H_i = 1|W_i = w)$
   (logit regression)

$$S(w) \equiv \Pr(H_i = 1|W_i = w)$$

$$MV_1(\theta) \equiv E\left[Y_{1i}|\theta_i = \theta\right] = \frac{\partial\left(E\left[Y_i|S(W_i) = \theta, H_i = 1\right]\theta\right)}{\partial\theta}$$

$$AV_1(\theta) \equiv E\left[Y_{1i}|\theta_i \leq \theta\right] = E\left[Y_i|S(W_i) = \theta, H_i = 1\right]$$

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Estimation

(Carneiro et al. 2011)

$$S(w) \equiv \Pr(H_i = 1 | W_i = w)$$

1. Estimate $\Pr(H_i = 1 | W_i = w)$ (logit regression)

2. Separately estimate

$$E[Y_i | S(W_i) = \theta, H_i = 1]$$

&

$$E[Y_i | S(W_i) = \theta, H_i = 0]$$

(local polynomial regression)

$$MV_1(\theta) \equiv E[Y_{1i} | \theta_i = \theta] = \frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 1]\theta)}{\partial \theta}$$

$$AV_1(\theta) \equiv E[Y_{1i} | \theta_i \leq \theta] = E[Y_i | S(W_i) = \theta, H_i = 1]$$

$$MV_0(\theta) \equiv E[Y_{0i} | \theta_i = \theta] = -\frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 0](1-\theta))}{\partial \theta}$$

$$AV_0(\theta) \equiv E[Y_{0i} | \theta_i \leq \theta] = \frac{E[Y_i | S(W_i) = 0] - (1-\theta) E[Y_i | S(W_i) = \theta, H_i = 0]}{\theta}$$

Estimation

(Carneiro et al. 2011)

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2. Separately estimate

$$E[Y_i | S(W_i) = \theta, H_i = 1]$$

&

$$E[Y_i | S(W_i) = \theta, H_i = 0]$$

(local polynomial regression)

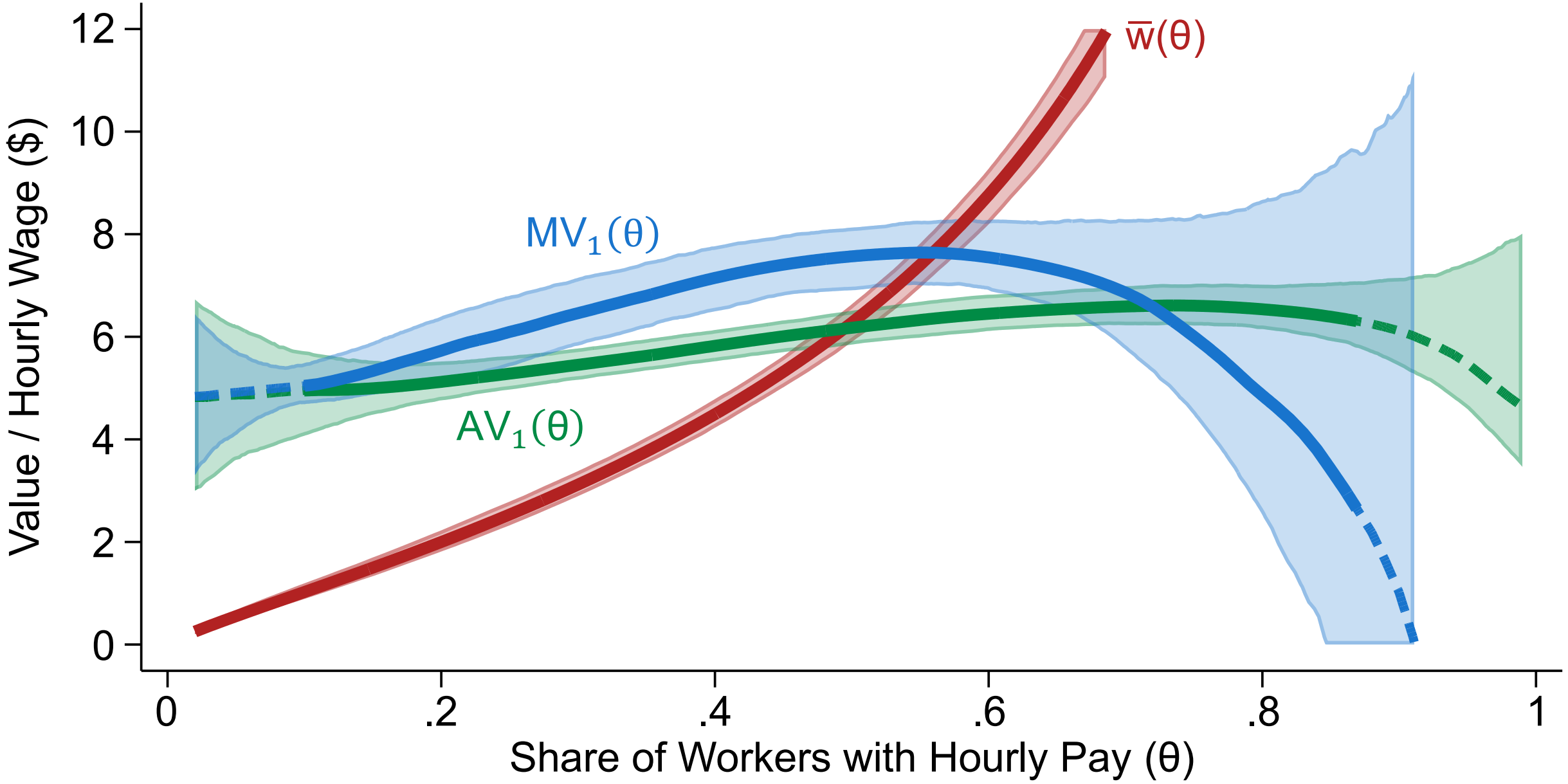
$$AV_1(\theta) \equiv E\left[Y_{1i} | \theta_i \leq \theta\right] = E\left[Y_i | S(W_i) = \theta, H_i = 1\right]$$

$$MV_0(\theta) \equiv E\left[Y_{0i} | \theta_i = \theta\right] = -\frac{\partial \left(E\left[Y_i | S(W_i) = \theta, H_i = 0\right](1-\theta)\right)}{\partial \theta}$$

3. Differentiate with respect to θ

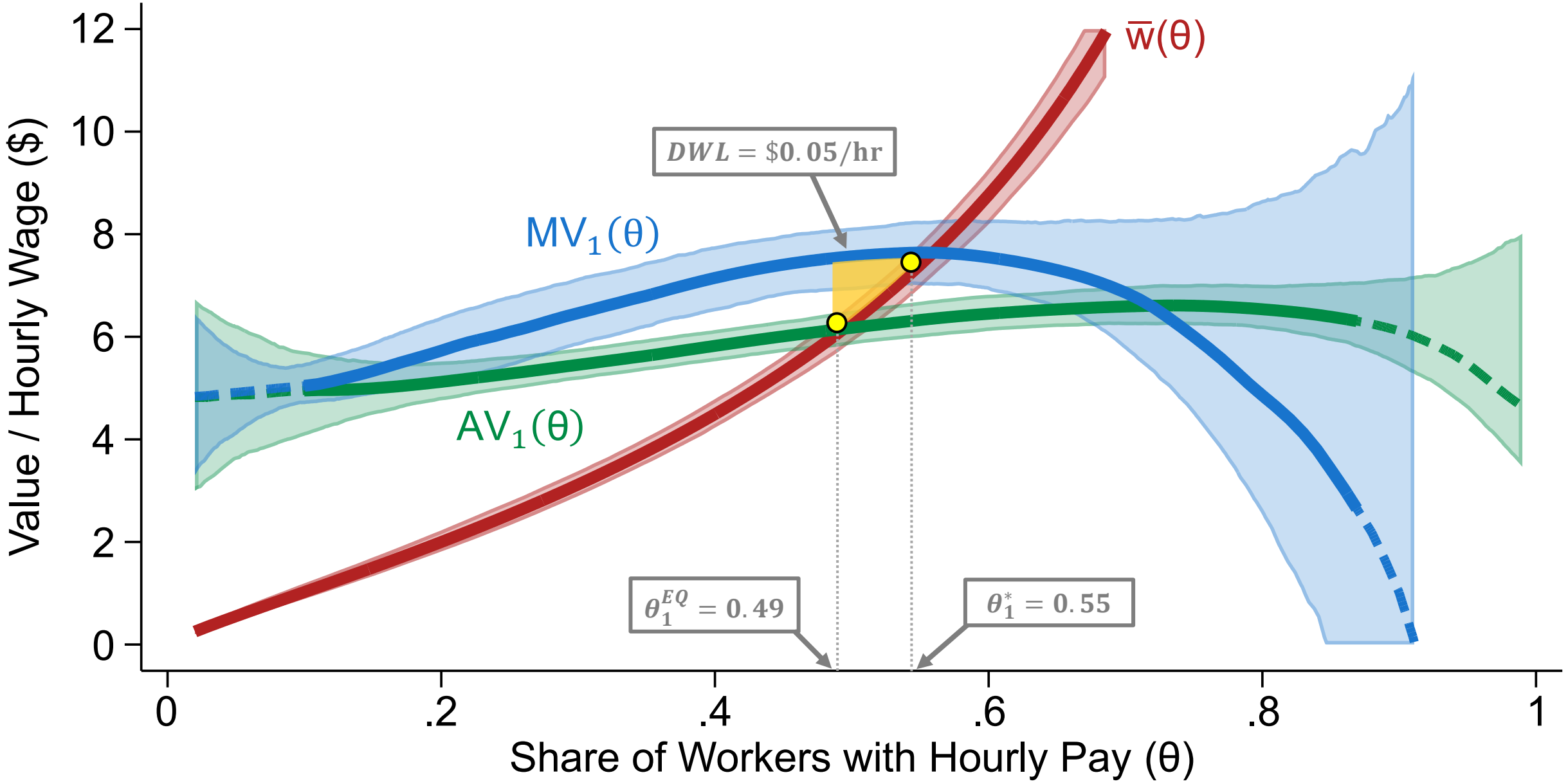
$$AV_0(\theta) \equiv E\left[Y_{0i} | \theta_i \leq \theta\right] = \frac{E\left[Y_i | S(W_i) = 0\right] - (1-\theta)\, E[Y_i | S(W_i) = \theta, H_i = 0]}{\theta}$$

Equilibrium with Adverse Selection and Moral Hazard



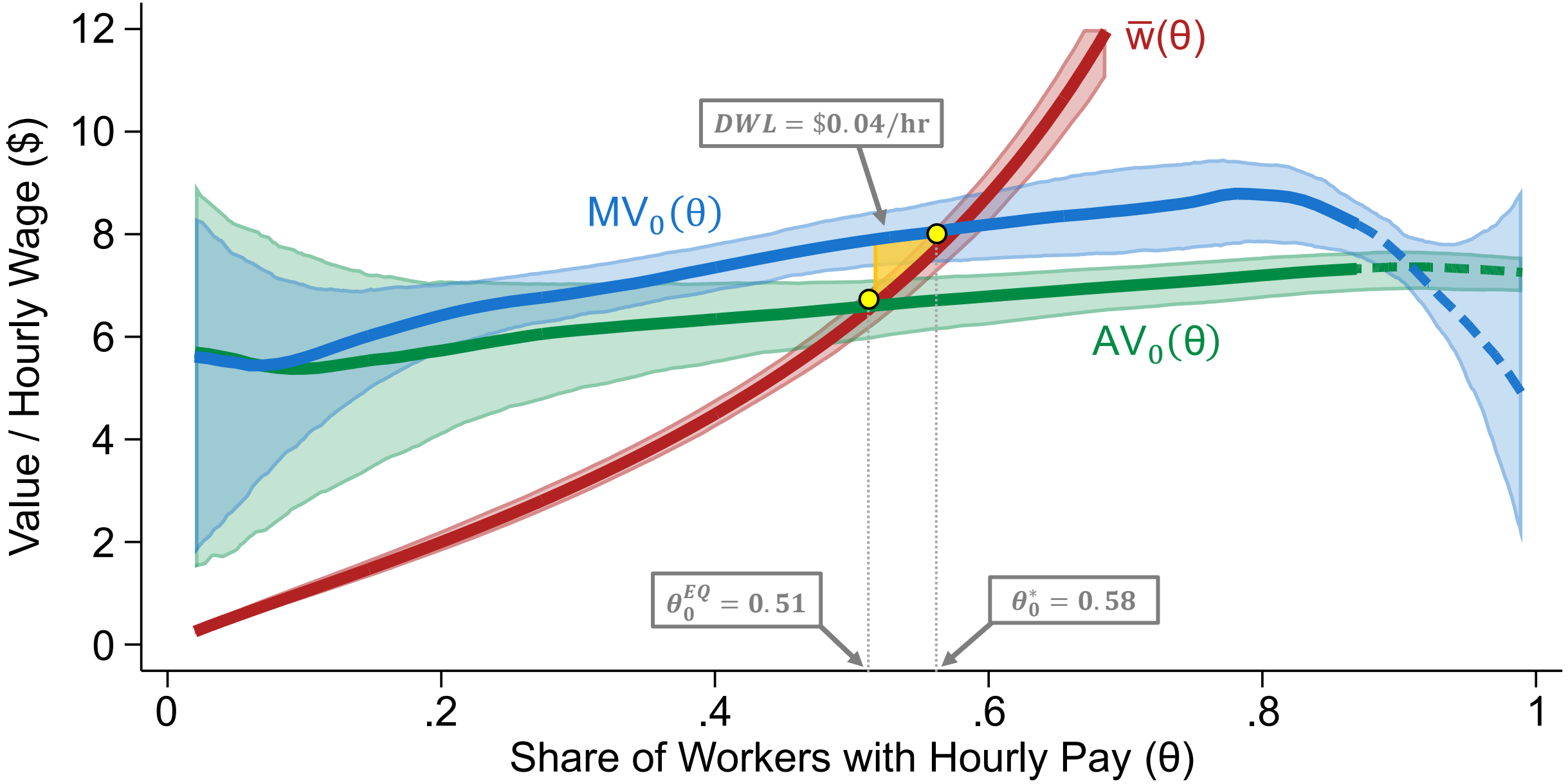
NP First Stage Raw Output Value

Equilibrium with Adverse Selection and Moral Hazard



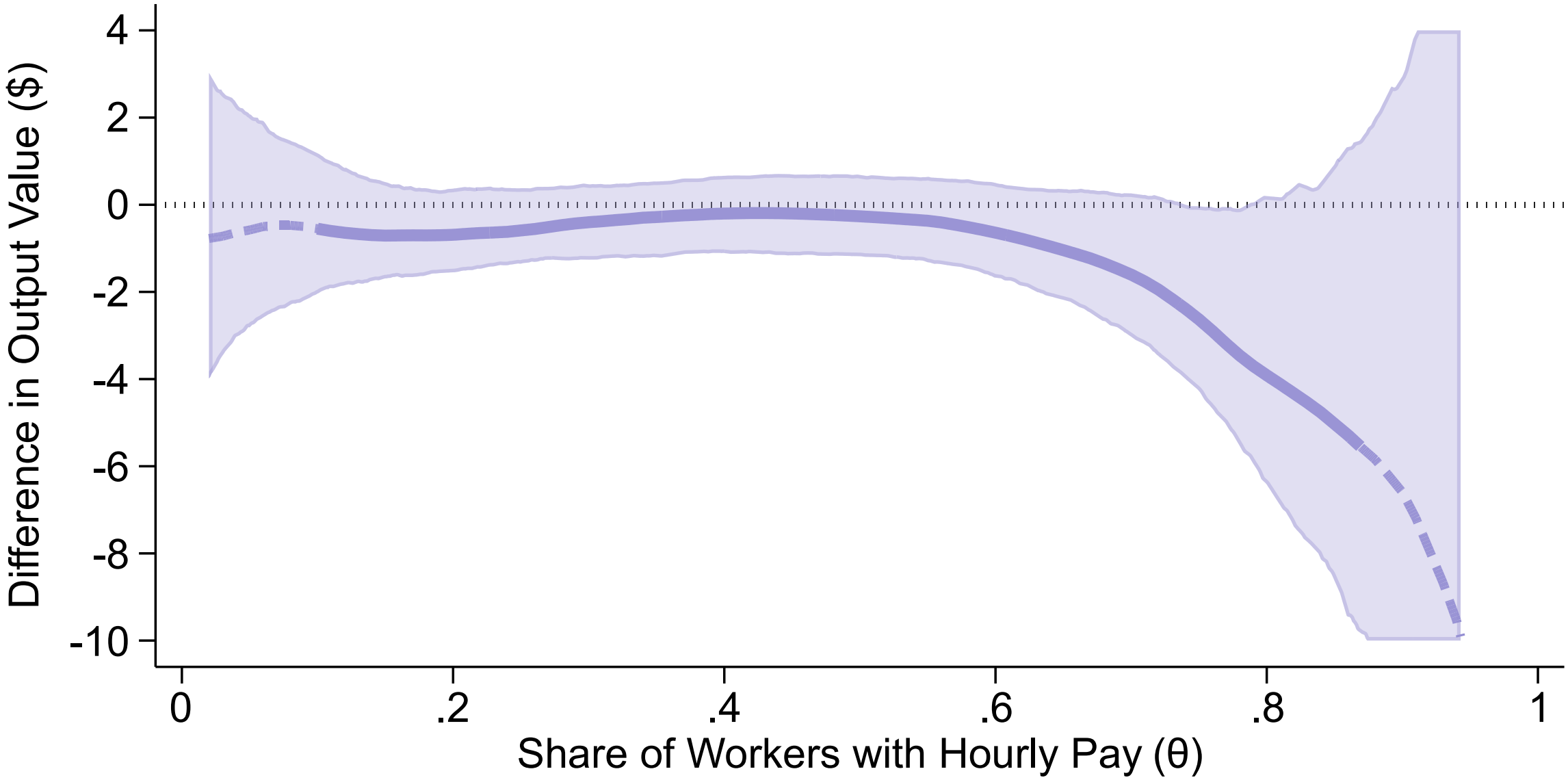
NP First Stage Raw Output Value

Equilibrium with Adverse Selection and No Moral Hazard

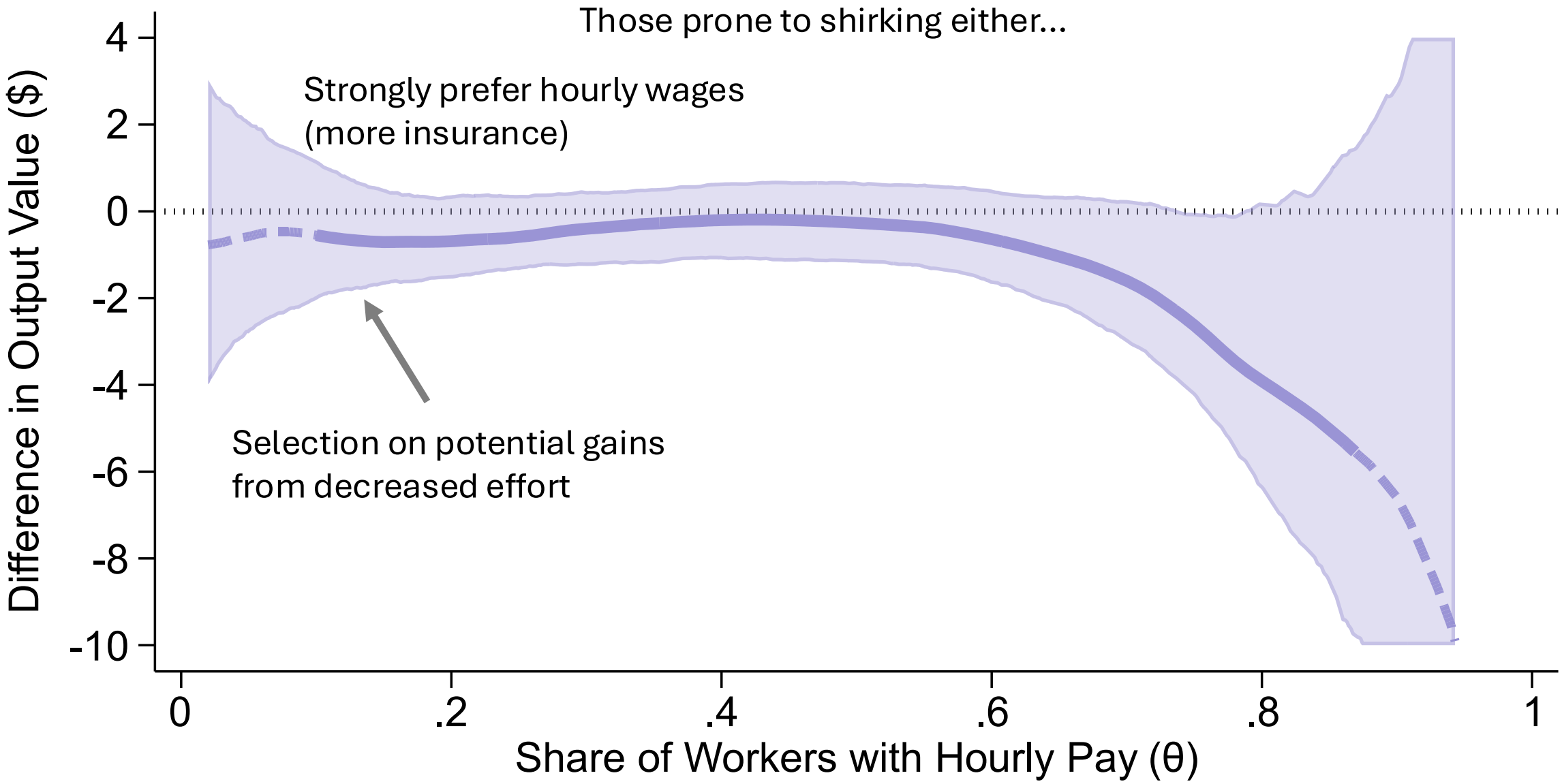


NP First Stage Raw Output Value

Moral Hazard: Estimates of Marginal Treatment Effects

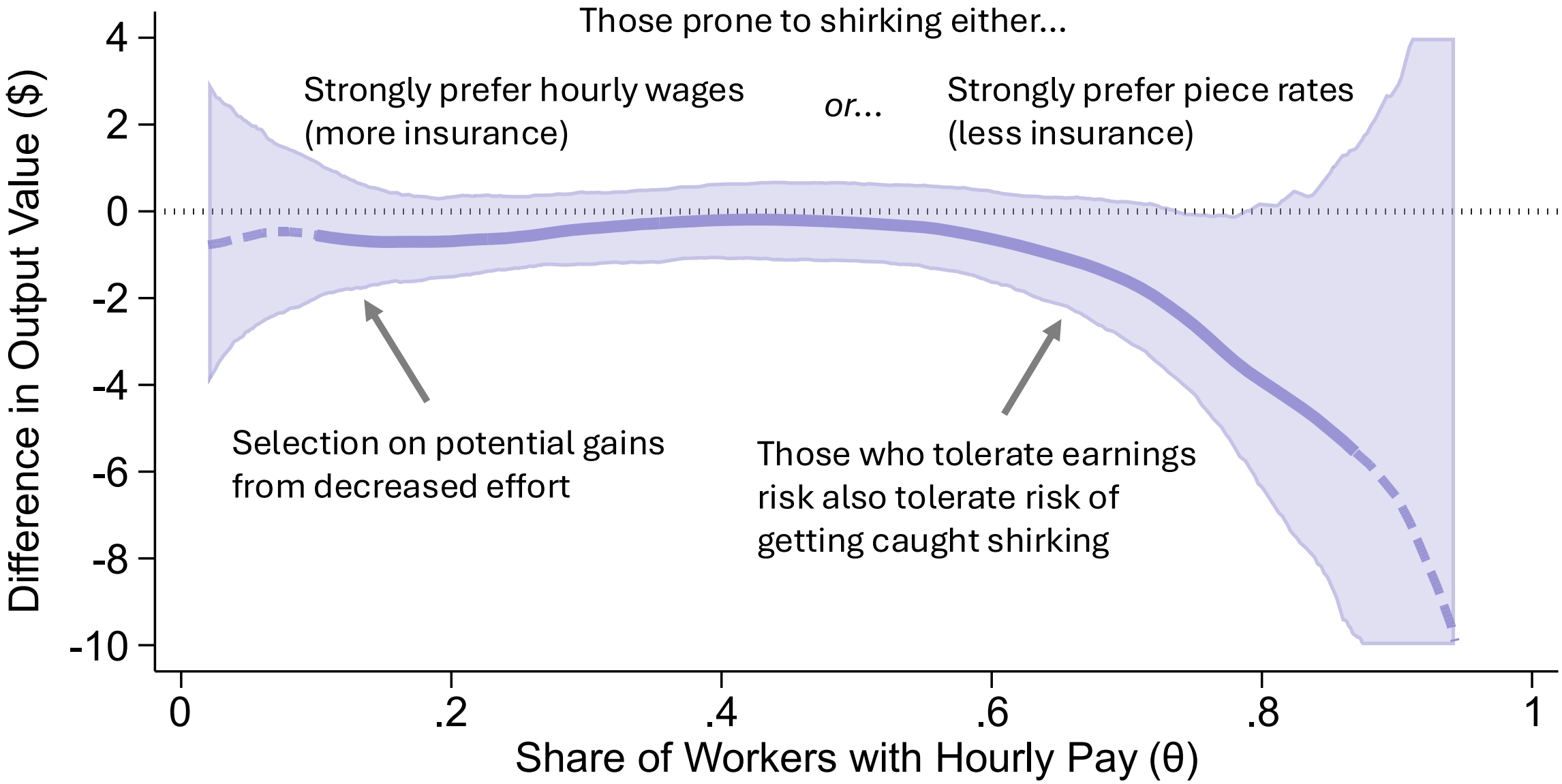


Moral Hazard: Estimates of Marginal Treatment Effects



Reference Dependence Raw Output Value

Moral Hazard: Estimates of Marginal Treatment Effects



Reference Dependence Raw Output Value

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1. Experimental Design and Results
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Policy Solutions

- Should government implement policies to encourage fixed wage contracts?
  - + Welfare benefit to marginal fixed-wage worker
  - – Fiscal cost from inducing that worker to shirk
- I calculate the **Marginal Value of Public Funds (MVPF)** for piece-rate tax

$$MVPF = \frac{Welfare\ Benefit}{Net\ Govt\ Expenditure}$$

Policy Solutions

- Should government implement policies to encourage fixed wage contracts?
  - + Welfare benefit to marginal fixed-wage worker
  - – Fiscal cost from inducing that worker to shirk

- I calculate the **Marginal Value of Public Funds (MVPF)** for piece-rate tax

$$MCPF = \frac{Welfare\ Loss}{Net\ Govt\ Revenue}$$

- MVPF of revenue-raising policies ≈ Marginal *Cost* of Public Funds (MCPF)

Socially efficient tax ⇒ ***lower*** MCPF

Policy: Piece-rate Tax

Marginal Cost of Public Funds (MCPF) for a piece-rate tax:

$$MCPF_{tax} = \frac{Welfare\ Loss}{Net\ Govt\ Revenue}$$

$Welfare\ Loss$: The aggregate amount workers would pay to avoid the tax

$Net\ Govt\ Revenue$: The aggregate revenue raised including fiscal externalities

Policy: Piece-rate Tax

Marginal Cost of Public Funds (MCPF) for a piece-rate tax:

$$MCPF_{tax} = \frac{Transfer - Insurance\ Value}{Net\ Govt\ Revenue}$$

$Welfare\ Loss$: The aggregate amount workers would pay to avoid the tax

- $Transfer$ from piece-rate employers to government
- $Insurance\ Value$ to workers who move from piece-rate to hourly pay

$Net\ Govt\ Revenue$: The aggregate revenue raised including fiscal externalities

Policy: Piece-rate Tax

Marginal Cost of Public Funds (MCPF) for a piece-rate tax:

$$MCPF_{tax} = \frac{Transfer - Insurance\ Value}{Transfer - Tax\ Loss\ from\ MH}$$

$Welfare\ Loss$: The aggregate amount workers would pay to avoid the tax

- $Transfer$ from piece-rate employers to government
- $Insurance\ Value$ to workers who move from piece-rate to hourly pay

$Net\ Govt\ Revenue$: The aggregate revenue raised including fiscal externalities

- $Transfer$ from piece-rate employers to government
- $Tax\ Loss\ from\ MH$ when hourly workers shirk and pay less income tax

Subsidy

Policy: Piece-rate Tax

Marginal Cost of Public Funds (MCPF) for a piece-rate tax:

$$MCPF_{tax} = \frac{\int_{\theta^{\rho}}^{1} \rho MV_0(\theta)d\theta - \int_{\theta^{EQ}}^{\theta^{\rho}} \left(MV_1(\theta) - \bar{w}(\theta)\right)d\theta}{\int_{\theta^{\rho}}^{1} \rho MV_0(\theta)d\theta + \int_{\theta^{EQ}}^{\theta^{\rho}} \frac{\tau}{1-\tau} MH(\theta)d\theta}$$

$Welfare\ Loss$: The aggregate amount workers would pay to avoid the tax

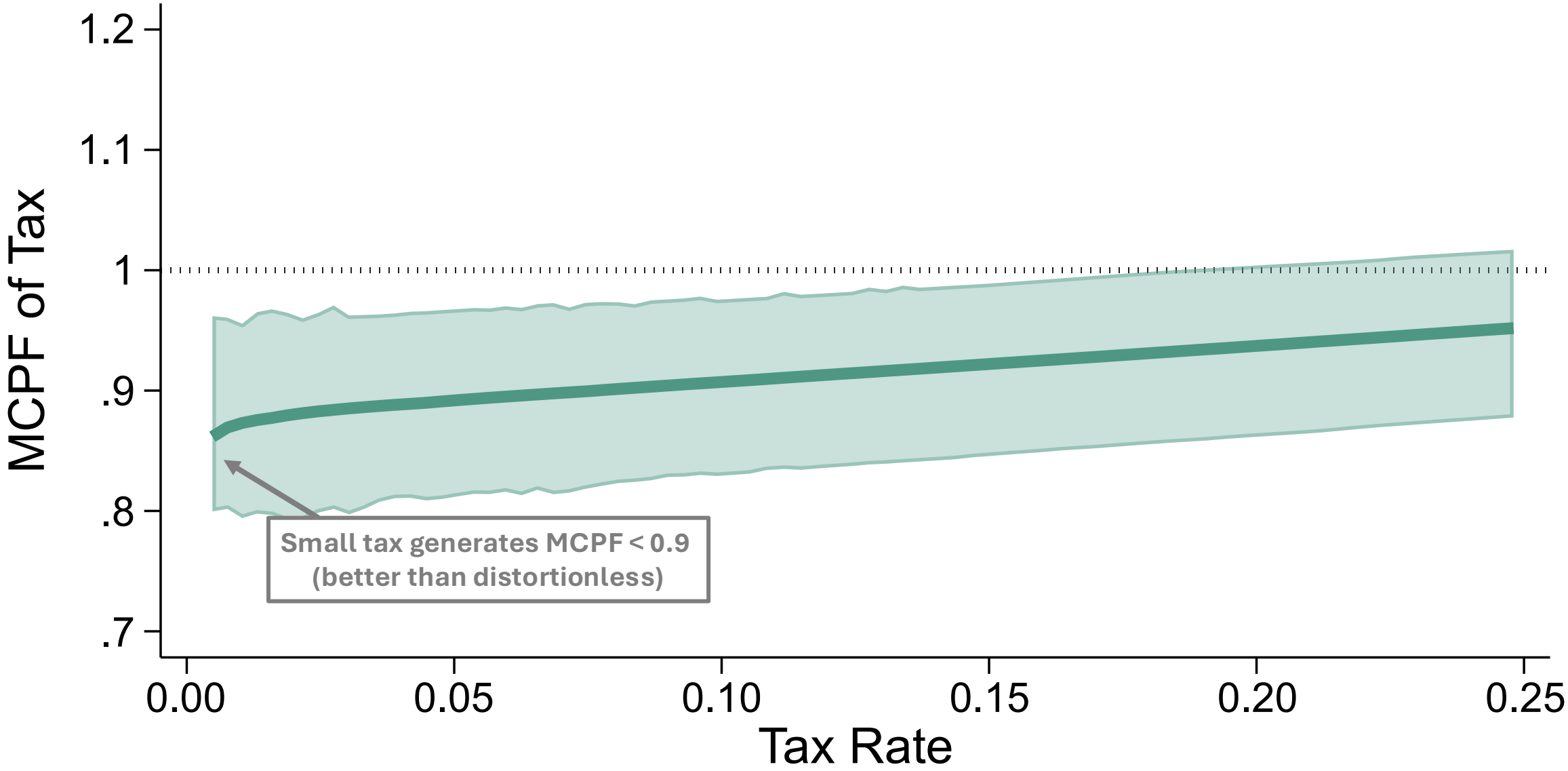
- $Transfer$ from piece-rate employers to government
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$Net\ Govt\ Revenue$: The aggregate revenue raised including fiscal externalities

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- $Tax\ Loss\ from\ MH$ when hourly workers shirk and pay less income tax

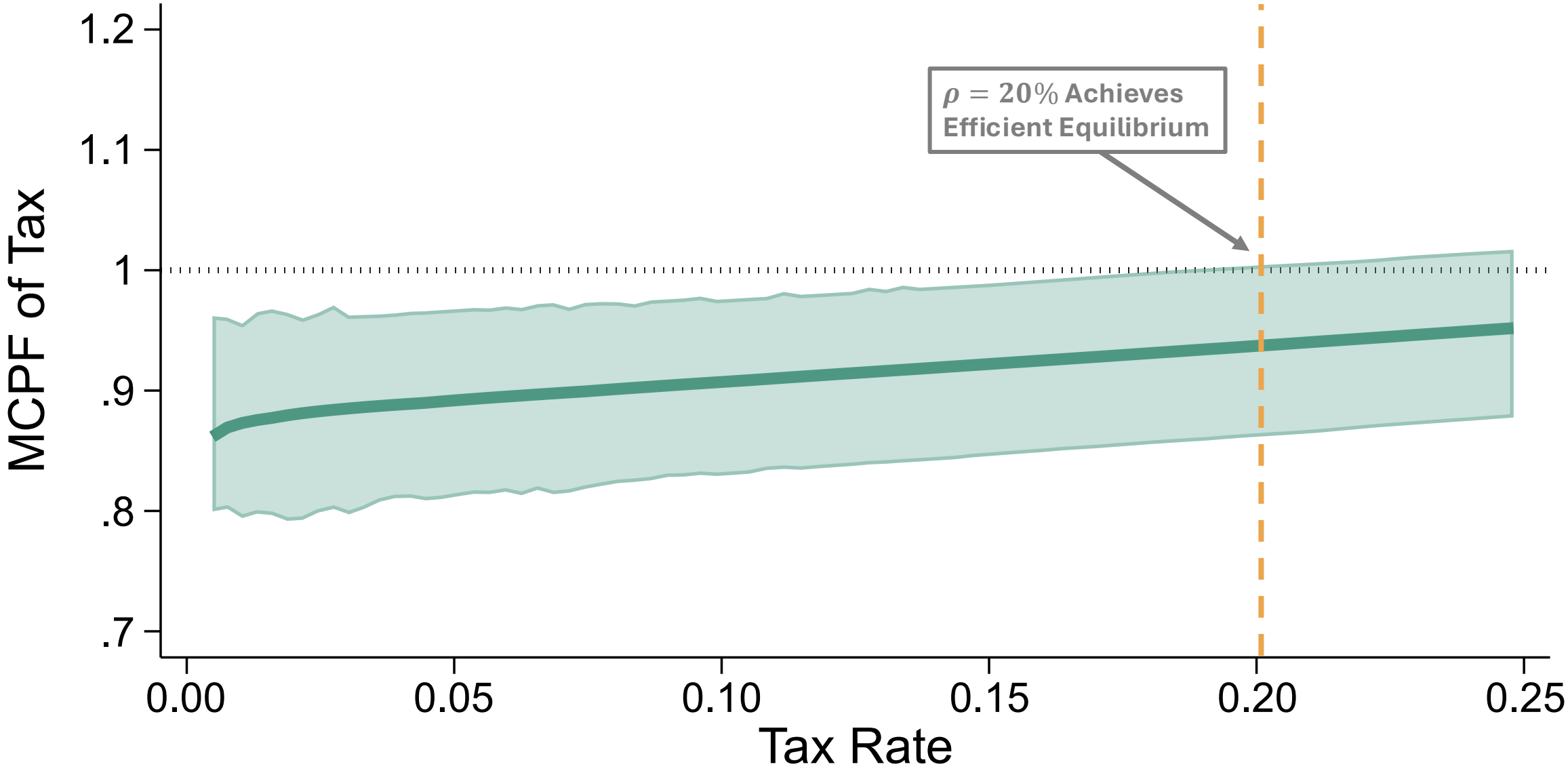
Subsidy

MCPF of Piece Rate Tax



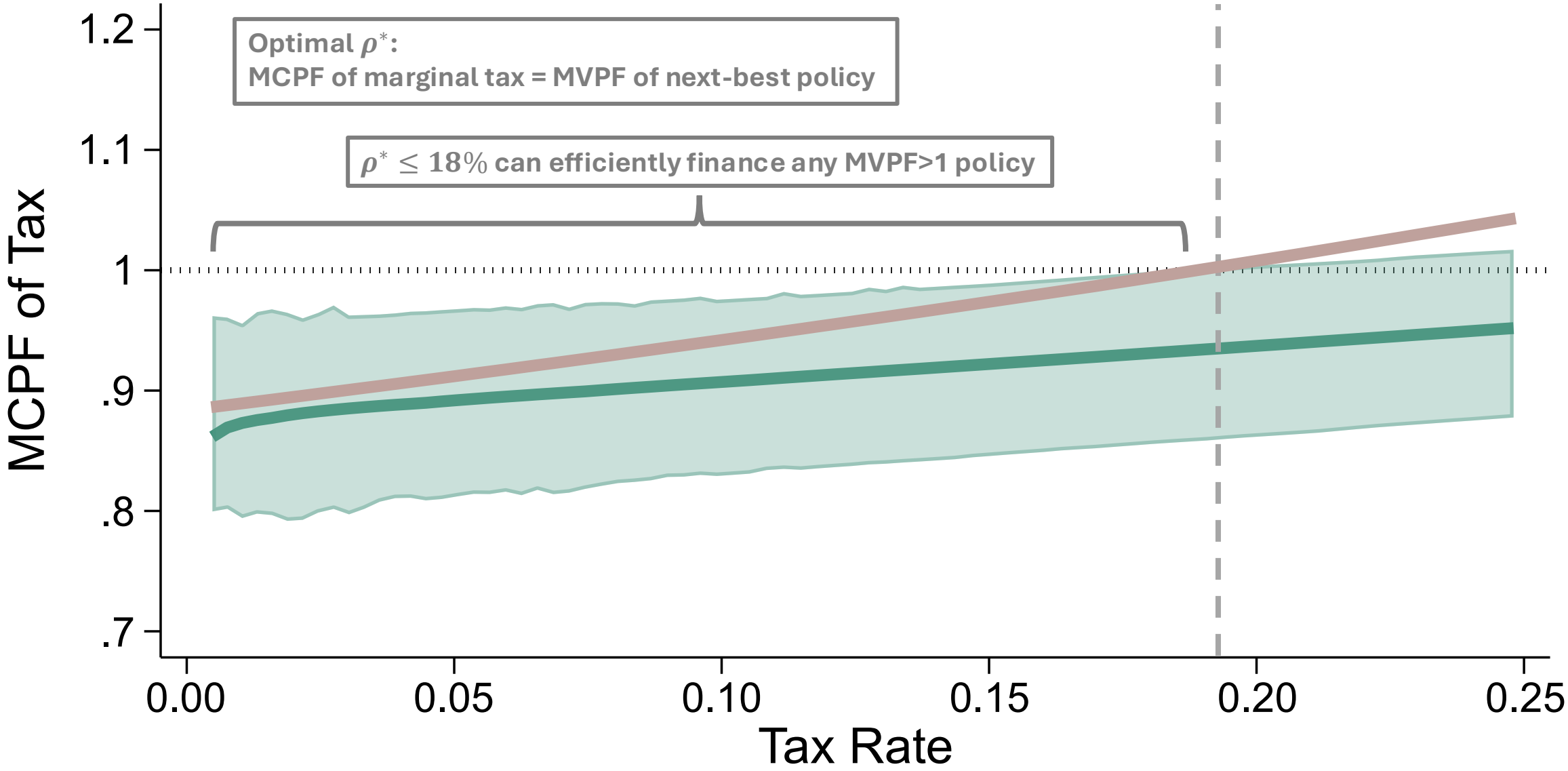
Subsidy

MCPF of Piece Rate Tax



Subsidy

MCPF of Piece Rate Tax



Subsidy

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Generalizing to Other Labor Markets

- Results are specific to online workers performing a data-entry task.
- Division of labor ⇒ applied research on worker incentives usually comes from highly specialized settings
  - Rideshare drivers (Angrist et al., 2021; Cook et al., 2021)
  - Agricultural workers (Brune et al., 2022; Bandiera et al., 2010)
  - Cashiers (Mas and Moretti 2009)
  - Windshield repair (Lazear 2000)
  - Call centers (Mas and Pallais 2017; Nagin et al. 2002)
- Even so, I try to make the experimental task as generalizable as possible
  - 66% of US workers type on the job
  - Workers don't know it's an experiment (debriefed afterwards)
- Methods can be applied elsewhere, especially short-term labor markets

Extensions: Why do these jobs have unpredictable pay?













Having full-time contractual employment does come with some certainty. You know you have an income every month…Now who knows?

– Skhumbuzo Ngubo, Uber Driver



Application to Rideshare Drivers (Angrist et al., 2021)

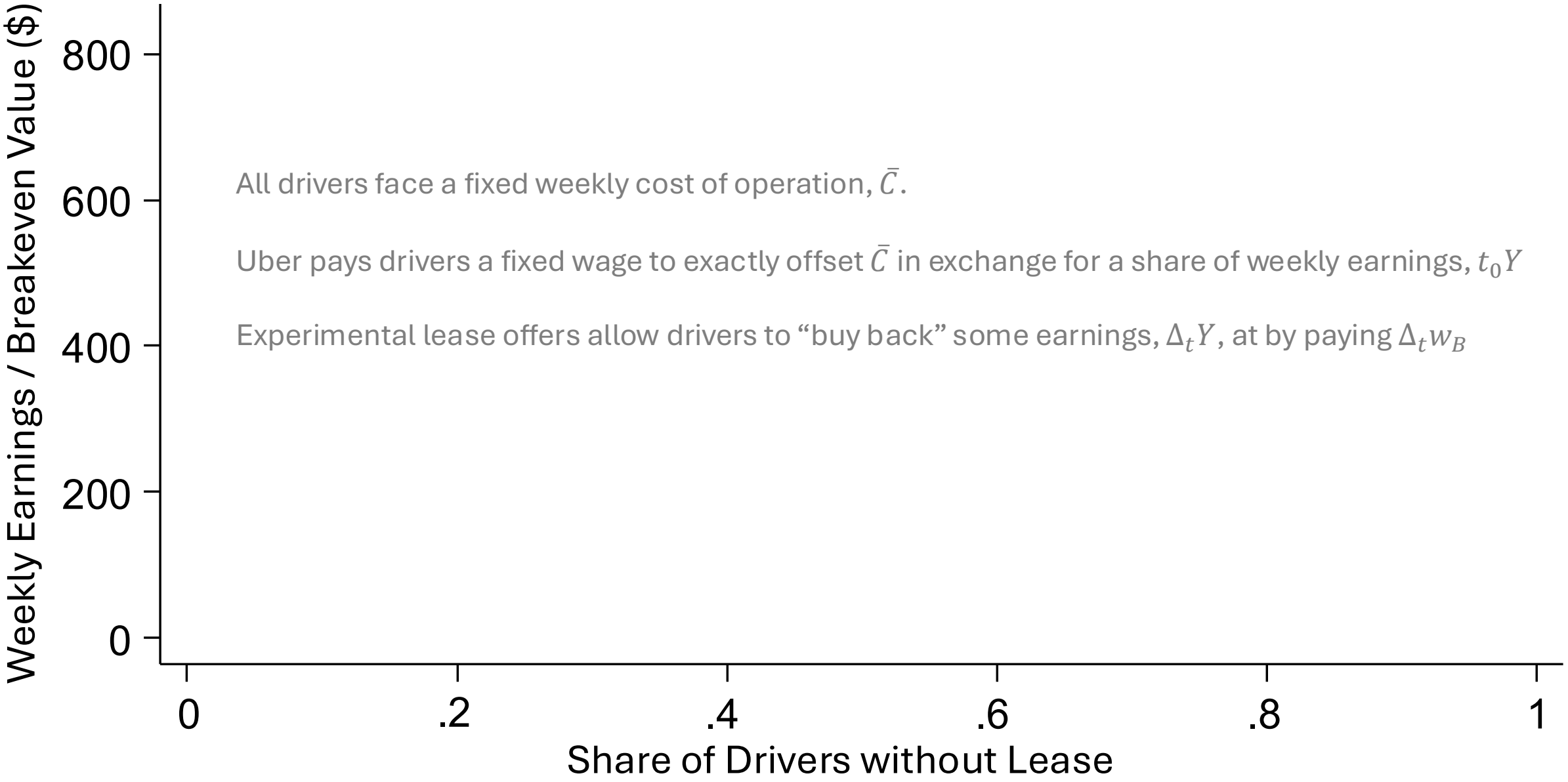
- RCT offers drivers larger commissions on ride fares in exchange for fixed "taxi-style" lease payments

  ⇒ Driver can "buy back" some labor product

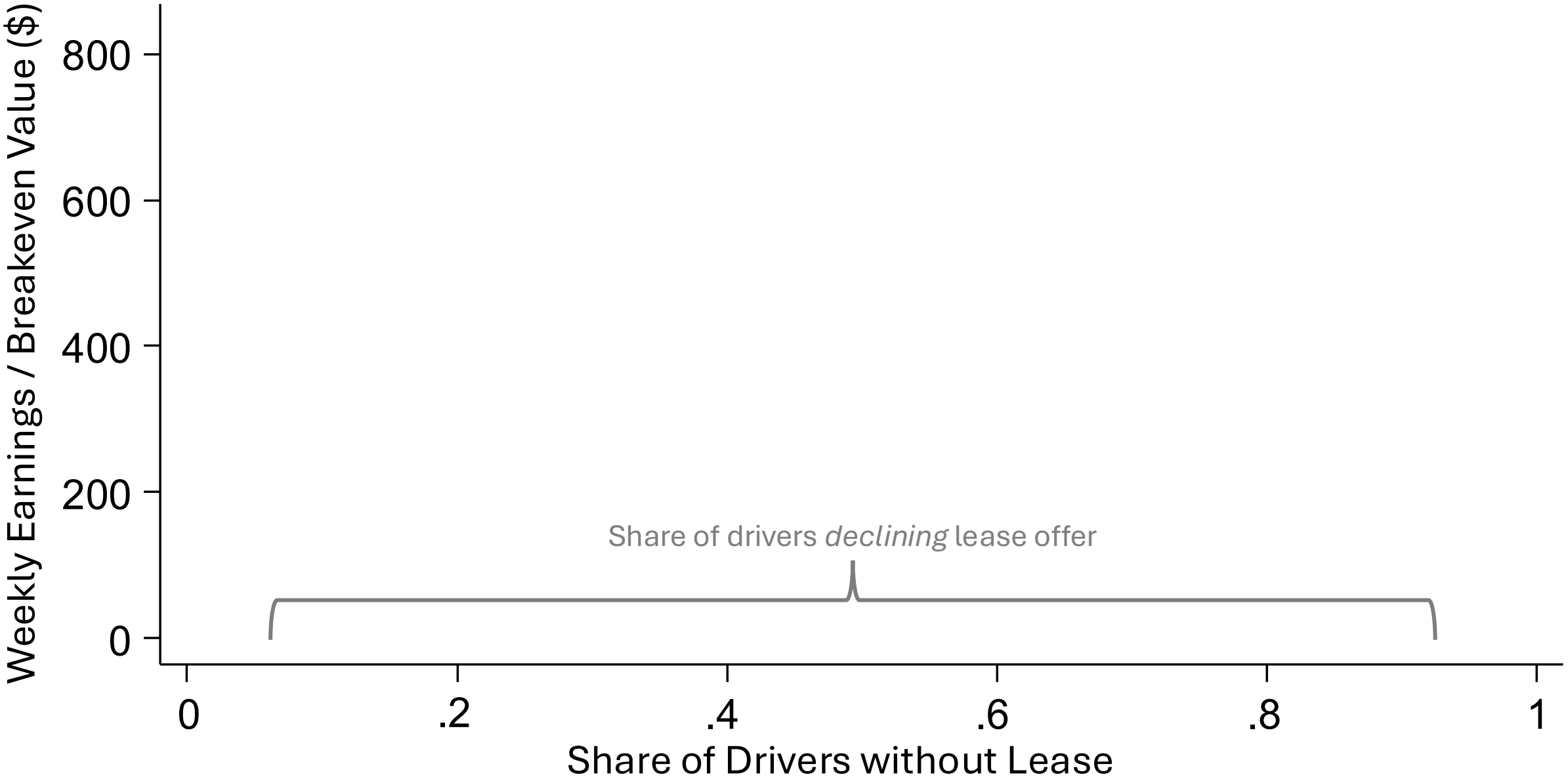
- Fixed-wage option ≈ "lease-free" driving (i.e., *decline* taxi offer)

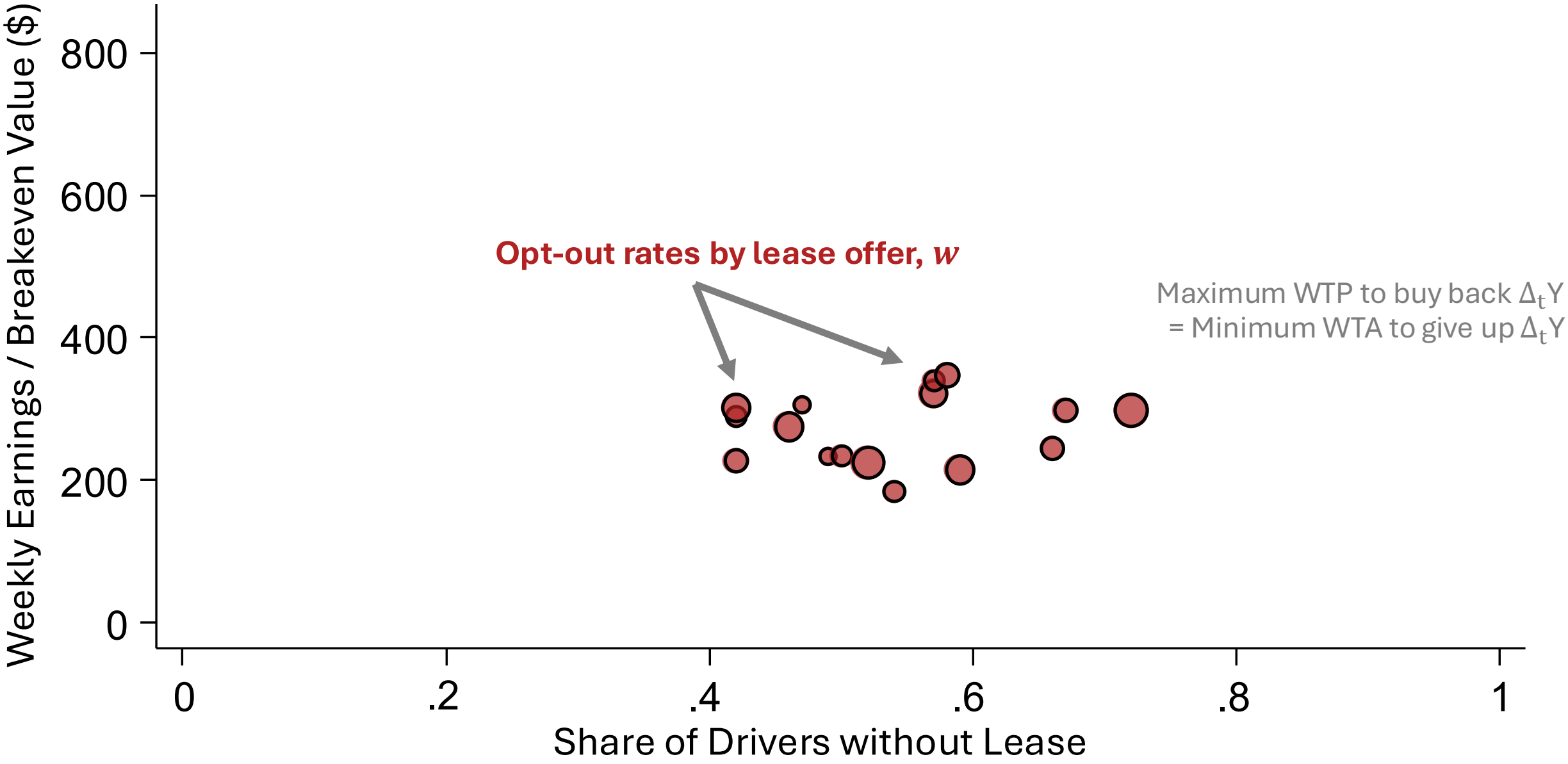


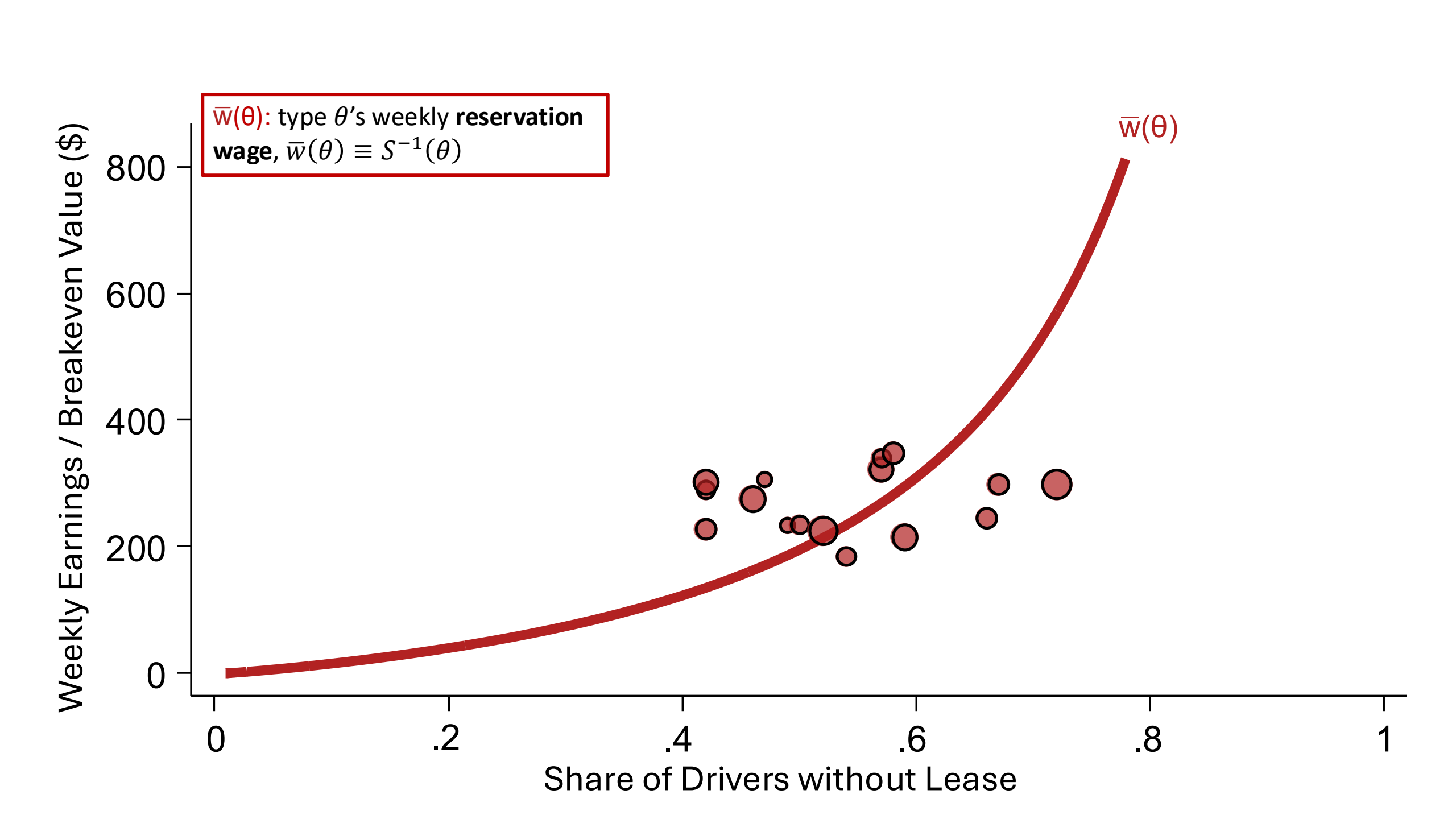
- Evidence of moral hazard and adverse selection
  - MH: lease take-up increases labor supply (ISE = 1.2)
  - AS: more productive drivers have higher WTP for leases ($\Delta_{Y_0} = \$21$)
- AS+MH ⇒ over-reliance on commissions? Welfare implications?
  - Calibrate my model using estimates from Angrist et al. (2021)

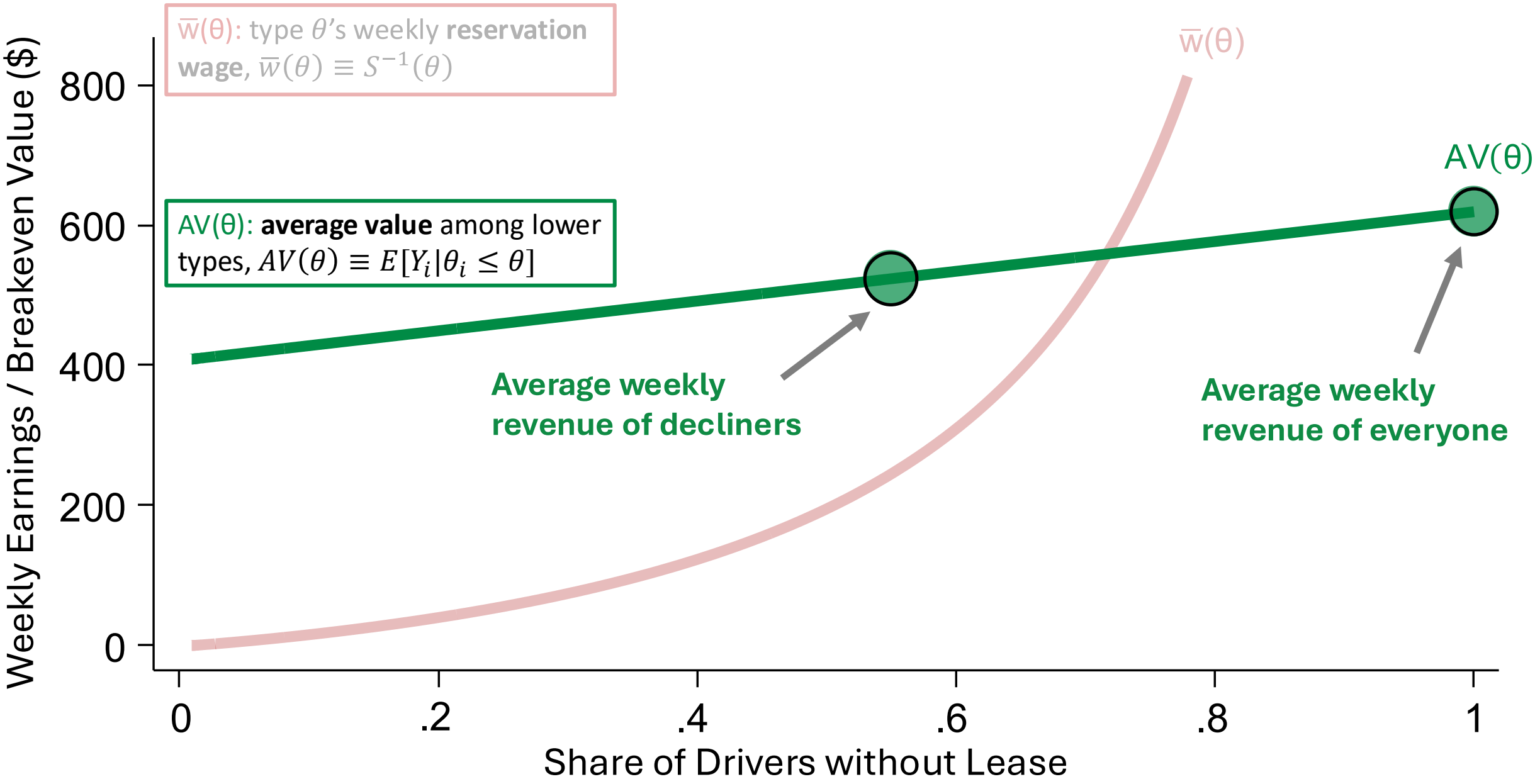

Weekly Earnings / Breakeven Value ($)
800
600
400
200
0
All drivers face a fixed weekly cost of operation, $\bar{C}$.
Uber pays drivers a fixed wage to exactly offset $\bar{C}$ in exchange for a share of weekly earnings, $t_0 Y$
Experimental lease offers allow drivers to "buy back" some earnings, $\Delta_t Y$, at by paying $\Delta_t w_B$
0
.2
.4
.6
.8
1
Share of Drivers without Lease

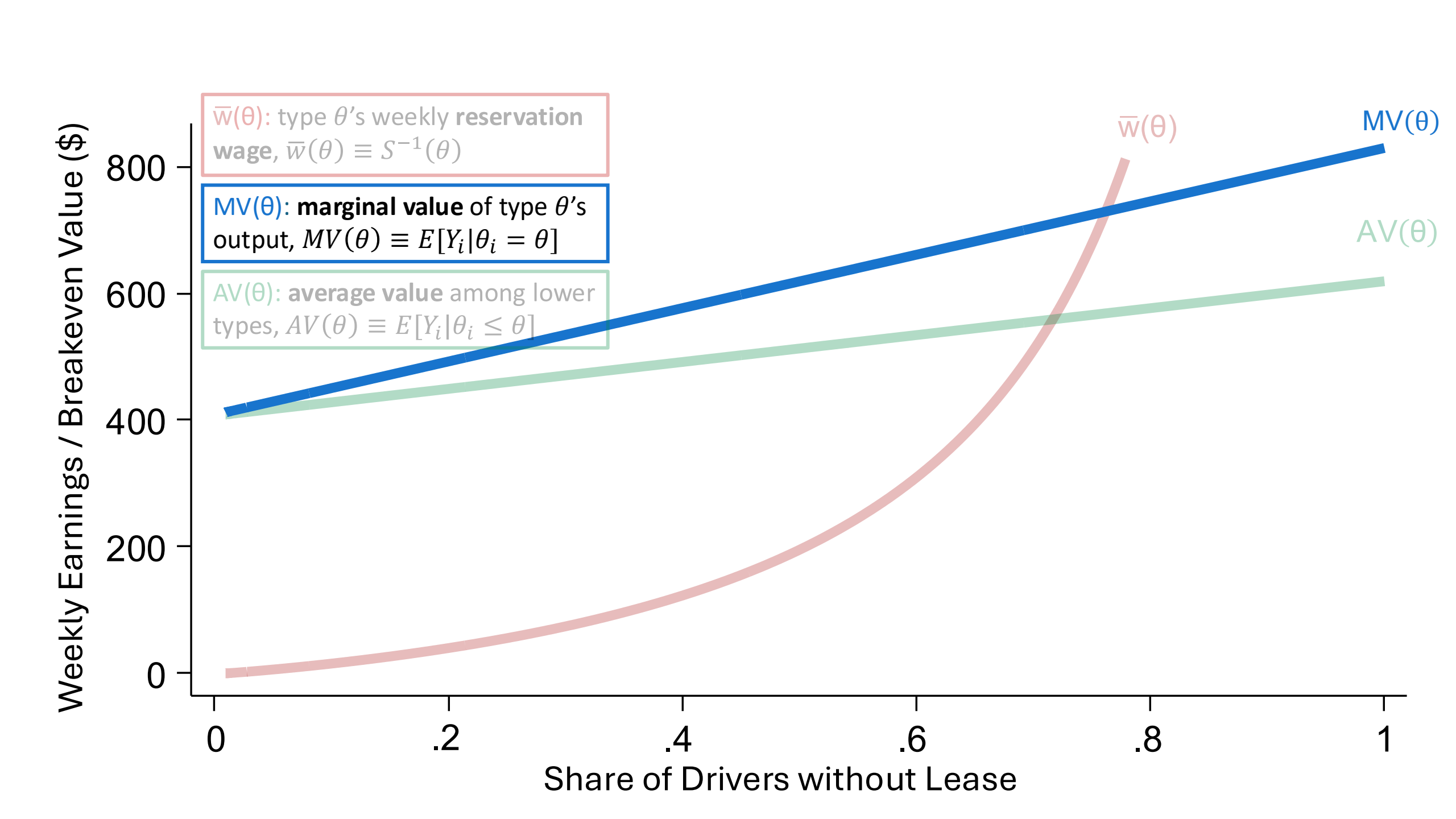

Weekly Earnings / Breakeven Value ($)
800
600
400
200
0
$w \equiv \frac{\text{Lease Payment}}{\Delta \text{ Driver's \% of Fare}}$
0
.2
.4
.6
.8
1
Share of Drivers without Lease

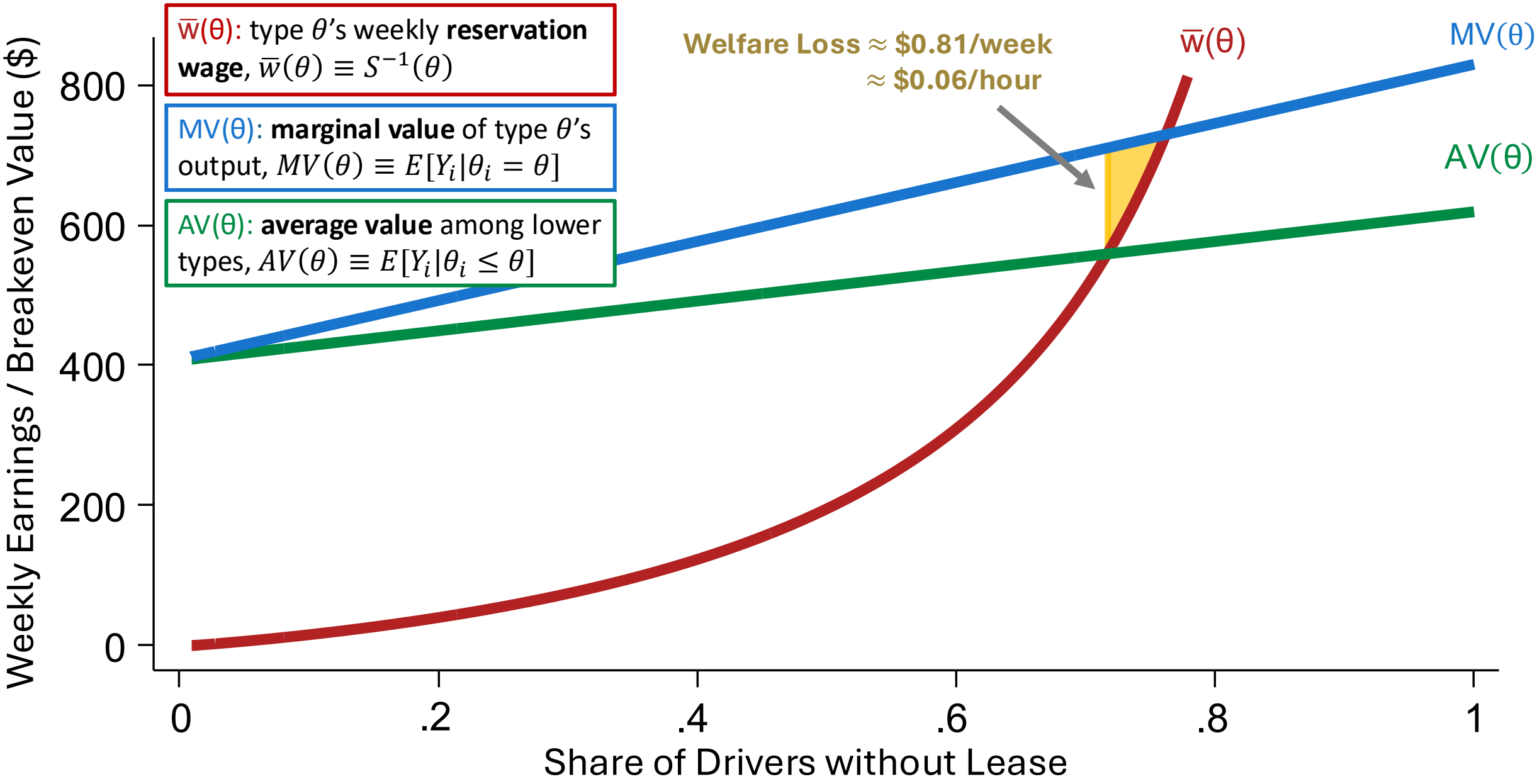

Weekly Earnings / Breakeven Value ($)
800
600
400
200
0
Share of drivers *declining* lease offer
0
.2
.4
.6
.8
1
Share of Drivers without Lease

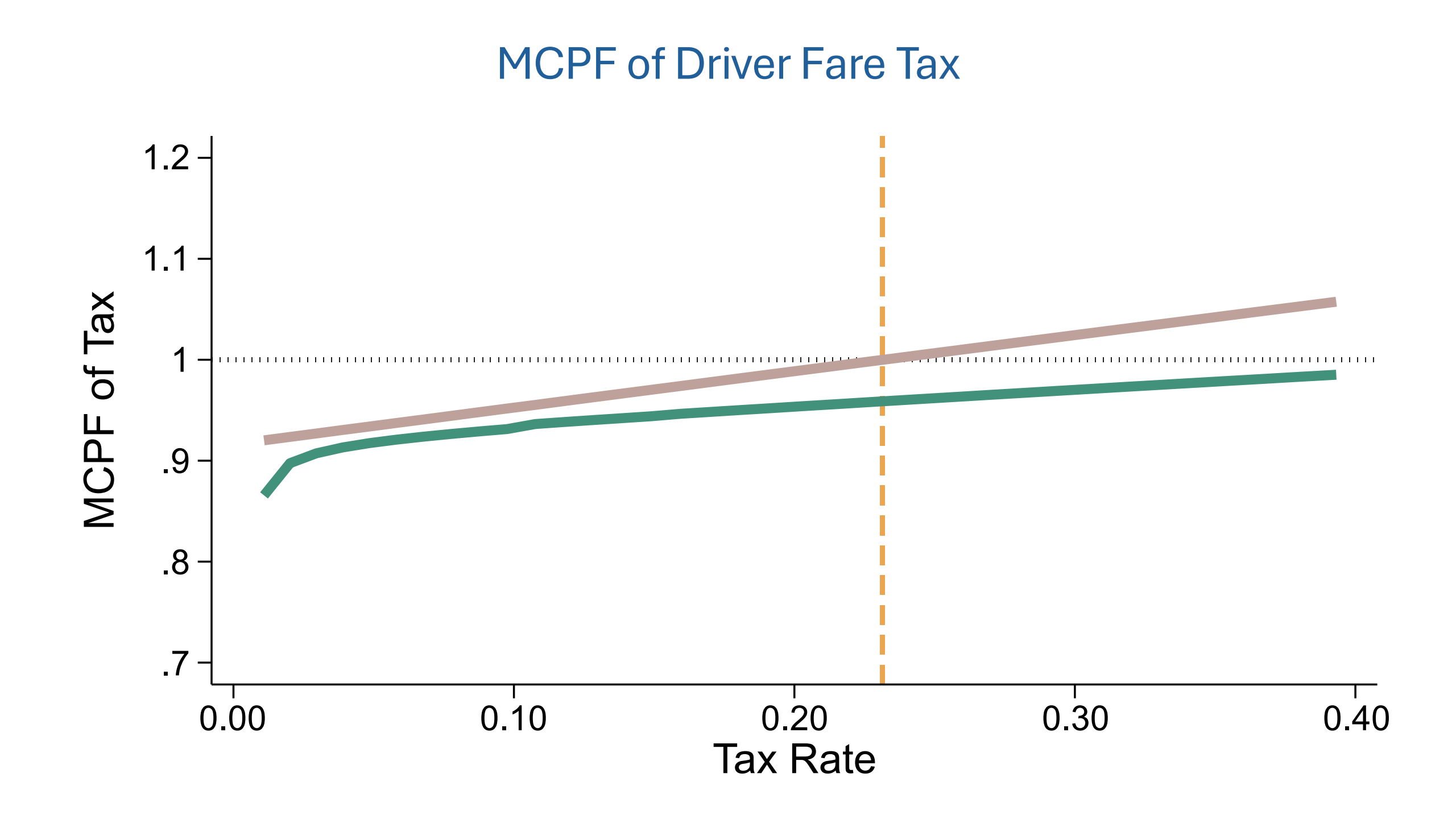

Weekly Earnings / Breakeven Value ($)
800
600
400
200
0
0
.2
.4
.6
.8
1
Share of Drivers without Lease
Opt-out rates by lease offer, w
Maximum WTP to buy back Δ_tY
= Minimum WTA to give up Δ_tY


w̄(θ): type θ's weekly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
w̄(θ)
Weekly Earnings / Breakeven Value ($)
800
600
400
200
0
0
.2
.4
.6
.8
1
Share of Drivers without Lease


w̄(θ): type θ's weekly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
AV(θ): **average value** among lower types, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$
w̄(θ)
AV(θ)
Average weekly revenue of decliners
Average weekly revenue of everyone
Weekly Earnings / Breakeven Value ($)
800
600
400
200
0
0
.2
.4
.6
.8
1
Share of Drivers without Lease


$\bar{w}(\theta)$: type θ's weekly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
MV(θ): **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$
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$\bar{w}(\theta)$
MV(θ)
AV(θ)
Weekly Earnings / Breakeven Value ($)
800
600
400
200
0
0
.2
.4
.6
.8
1
Share of Drivers without Lease


w̄(θ): type θ's weekly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$
MV(θ): **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$
AV(θ): **average value** among lower types, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$
Welfare Loss ≈ $0.81/week
≈ $0.06/hour
w̄(θ)
MV(θ)
AV(θ)
Weekly Earnings / Breakeven Value ($)
800
600
400
200
0
0
.2
.4
.6
.8
1
Share of Drivers without Lease


MCPF of Driver Fare Tax
MCPF of Tax
1.2
1.1
1
.9
.8
.7
0.00
0.10
0.20
0.30
0.40
Tax Rate

Application to Rideshare Drivers

- Findings
  - Sub-optimal equilibrium relies heavily on unpredictable fares.
  - Welfare loss of $1.32 per week driven.
  - Taxing fare-based compensation mitigates adverse selection
  - Can generate government revenue at with minimal social welfare costs.
- Limitations
  - Linear extrapolation
  - Experimental sample consists of *existing* Uber drivers.
  - Excludes drivers with highest $\overline{w}$—they're already working as taxi drivers!



[After eliminating tips, server positions] have been harder to fill. Many veteran servers weren't interested, saying they could make double elsewhere with tips…As a result, many of the people who work in the dining room started with little to no restaurant experience.

–Owner, Liholiho Yacht Club

(Early) Work in Progress: Adverse Selection and Tips



- Joint with Max Thon (UZurich)
- RCT in large restaurant chain in Germany
- Use linked survey + administrative wage data
  1. "Would you accept a **fixed payment of $$w$** in exchange for **forfeiting your tips** each week?"
  2. AV(w) = average realized tips among workers willing to accept "buyout" w.

- Research questions:
  - Can adverse selection explain restaurant server's reliance on tips?
  - What are the social returns to eliminating vs. increasing taxes on tips?

Conclusion

- Experimental and theoretical framework for asymmetric information in wage contracts
  - Shows how moral hazard and adverse selection affect market equilibrium and welfare
  - Model correspondence to MTE estimands ⇒ semi-parametric identification

- Apply framework to data-entry workers using an RCT that varies wage *offers*
  - Results imply an underprovision of fixed-wage contracts

- Policy prescription: increase taxes on piece-rate compensation (e.g., tips?)
  - $\tau \leq 19\%$ raises \$1 of government revenue at an estimated social cost of \$0.88 or less

- Extensions: use framework to estimate welfare loss in other markets
  - Experimentally vary compensation offers for other types of work (e.g., rideshare drivers)
  - Alternatively, use linked survey + administrative wage/productivity data

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Any outcome-contingent contract can be vulnerable to moral hazard and adverse selection!

Thank you!

PAPER



SLIDES



Appendix

Detecting Moral Hazard and Adverse Selection

Existing methods of testing for moral hazard and adverse selection:

1. Test for correlation between take-up and realized risk (Chiappori and Salanie 2000)
   - Cannot separate adverse selection from moral hazard
2. Compare take-up and risk across exogenous price changes in existing markets (Einav et al. 2010, Einav et al. 2013; Hackmann et al. 2012)
   - "Under-the-lamppost" problem: Cannot observe *unraveled* contracts, which are unprofitable
3. Construct *hypothetical* contract choices from surveys on subjective beliefs (Herbst and Hendren 2024; Hendren 2013, 2017)
   - Strong parametric assumptions to predict choices across "missing" contracts

Detecting Moral Hazard and Adverse Selection

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   - Strong parametric assumptions to predict choices across "missing" contracts

My strategy: if I can't observe missing contracts, offer them myself!

- Advantage #1: I can experimentally randomize contract prices (i.e., wages)
- Advantage #2: I can offer unraveled (i.e., unprofitable) contracts

Experimental Design: Single Wage Offer

Comparing piece-rate workers who faced different offer sets identifies ***selection into treatment:***

$$\frac{E[Y_i|W_i=0]-E[Y_i|H_i=0,W_i=1]}{\pi}=\frac{(\pi E[Y_{0i}|H_i=1]+(1-\pi)E[Y_{0i}|H_i=0])-E[Y_{0i}|H_i=0]}{\pi}=E[Y_{0i}|H_i=1]\ -E[Y_{0i}|H_i=0]$$

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Identifying Selection Across Multiple Wage Offers

W_i: Randomized treatment-offer group

- $W_i = L$: Offered a choice between low hourly wage and piece rate
- $W_i = H$: Offered a choice between high hourly wage and piece rate
- π^W: Take-up of wage offer $W \in \{L, H\}$

Selection on Y_0 into offer H among those who would reject offer L:

$$E\left[Y_{0i} \middle| H_i^H = 1, H_i^L = 0\right] - E\left[Y_{0i} \middle| H_i^H = 0\right]$$

$$= \frac{1-\pi^L}{\pi^H - \pi^L} E[Y_i | H_i = 0, W_i = L] - E[Y_i | H_i = 0, W_i = H]$$

Selection on Y_1 into offer L among those who would accept offer H:

$$E\left[Y_{1i} \middle| H_i^L = 1\right] - E[Y_{1i} | H_i^H = 1, H_i^L = 0]$$

$$= \frac{\pi^H}{\pi^H - \pi^L} E[Y_i | H_i = 1, W_i = L] - E[Y_i | H_i = 1, W_i = H]$$

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You will be shown a series of handwritten sentences. On each page, your task is to type the sentence into the text box below.

Here is one example of a completed sentence:



The quick brown fox jumps over the lazy dog.

Your answers should be as accurate as possible. Please be mindful of capitalization, spacing, and punctuation. When you've completed a sentence, click the "→" button to move on.

You will have 5 minutes to complete as many sentences as you can. You cannot start over, and you can only perform this task once.

→

Before you begin the task, we'd like to offer you a choice of how to earn your bonus payment. Please select your preferred bonus compensation from the options below:

Get paid a $0.03 bonus for each sentence you correctly complete.	Get paid a flat bonus of $1.00.
◉	○



Time Remaining: 05:00

For performing this task, you will receive **$1.00**, plus your chosen bonus of **$0.03 per correct sentence**.

Click "Begin Task" to begin your 5-minute typing task.

Once you begin, the timer will start. Do not press the back button or refresh your browser—doing so may invalidate your results!

Begin Task

Score: 0
Earnings: $1.00

Time Remaining: 03:34



A new leaf sprouted on the old p

→

Score: 8
Earnings: $1.24

Time is up.

Thank you for participating!

You completed 31 sentences correctly.

Please click here to receive your $1.93 payment.

Score: 31
Earnings: $1.93

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Balance Test

	(1) Any Hourly Wage Offer	(2) Experimental Wage Offer	(3) Output Value
Number of Previous Tasks/1000	0.00460	−0.0446	0.195***
	(0.00316)	(0.0718)	(0.0321)
Age	0.0000335	0.00421	−0.0599***
	(0.000520)	(0.0109)	(0.00477)
Female	−0.00469	0.156	0.567***
	(0.0118)	(0.257)	(0.113)
Minority	0.00793	0.00323	−1.040***
	(0.0117)	(0.263)	(0.114)
Employed	0.0173	−0.234	0.144
	(0.0135)	(0.287)	(0.127)
Student	0.0106	0.366	−0.538***
	(0.0162)	(0.355)	(0.156)
F-statistic	0.707	0.872	30.456
p-value	0.733	0.568	0.000
N	3030	2728	3030

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Hourly Wage Take-up

	(1) Accepted Offer	(2) Accepted Offer	(3) Accepted Offer	(4) Accepted Offer
Log Hourly Wage Offer	1.202***	1.205***	1.216***	1.248***
	(0.0555)	(0.0555)	(0.0561)	(0.0579)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
N	2728	2728	2728	2728

$$\Pr(H_i = 1) = \text{inv logit}\left(\alpha \ln W_i + \boldsymbol{\nu X_i}\right)$$

Suppose supply and average value are linear in the wage offer:

$$Y_i = \xi_0 + \alpha H_i + \beta_0(1 - H_i)\times W_i + \beta_1 H_i \times W_i + \gamma H_i \times W_i^P + \epsilon_i$$
$$H_i = \gamma + \delta w + \nu_i$$

Let $\overline{w}_i$ be worker i's hourly reservation wage ⇒ CEFs of hourly take-up and output value:

$$\Pr(\overline{w}_i \leq w) = \gamma + \delta w$$
$$\mathrm{E}[Y_i|\overline{w}_i \leq w] = \xi_0 + \alpha + \beta_1 H_i \times w$$
$$\mathrm{E}[Y_i|\overline{w}_i > w] = \xi_0 + \beta_0(1 - H_i)\times w$$

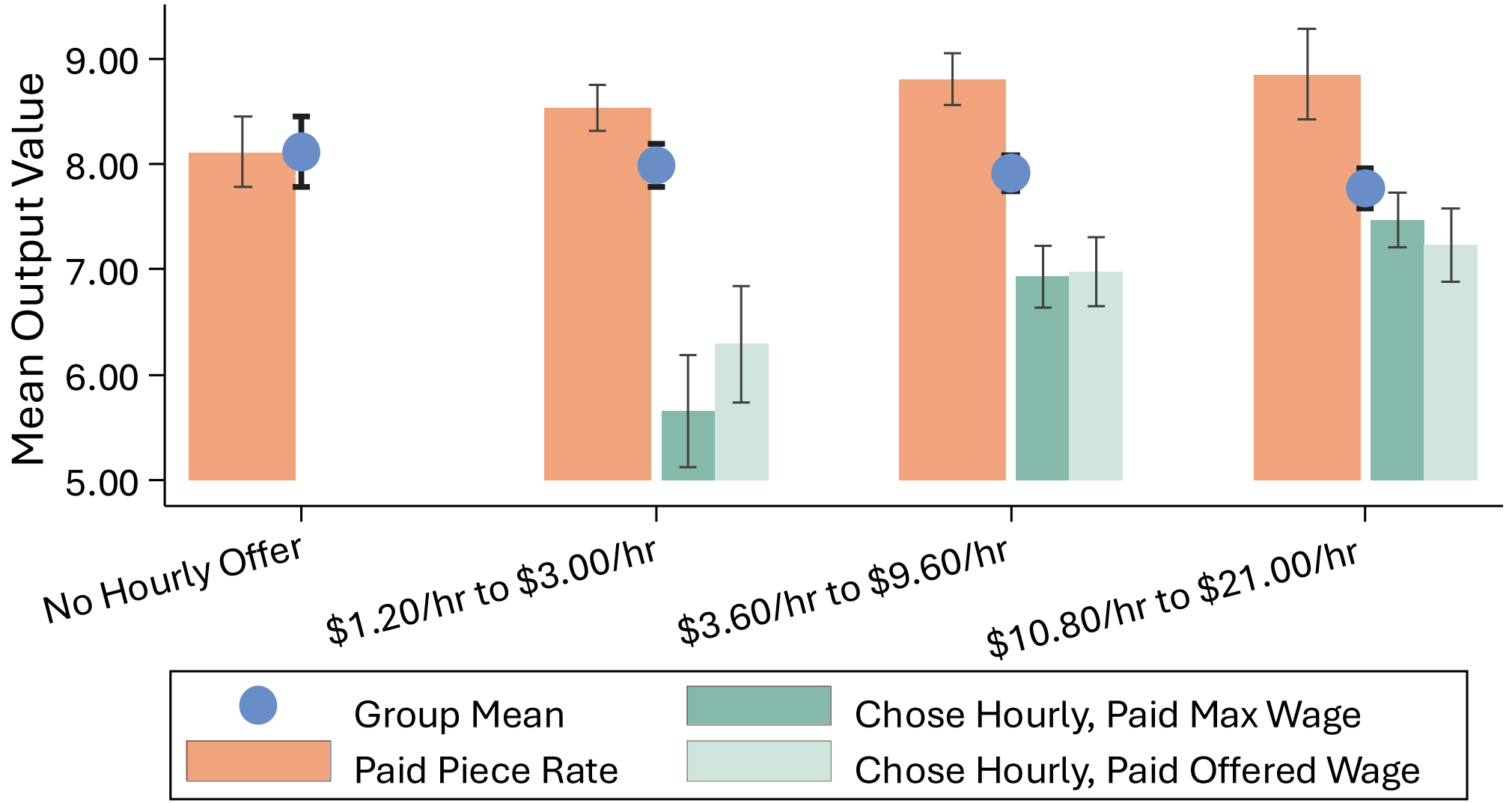
The *marginal* potential output values $E[Y_{1i}|\overline{w}_i = w]$ & $E[Y_{0i}|\overline{w}_i = w]$ are given by

$$E[Y_{1i}|\overline{w}_i = w] = \frac{\partial(\mathrm{E}[Y_i|\overline{w}_i \leq w]S(w))}{\partial S(w)} = \xi_0 + \alpha + \frac{\gamma\beta_1}{\delta} + 2\beta_1 w$$

$$E[Y_{0i}|\overline{w}_i = w] = -\frac{\partial\left(\mathrm{E}[Y_i|\overline{w}_i > w]\big(1 - S(w)\big)\right)}{\partial S(w)} = \xi_0 + \frac{(\gamma - 1)\beta_0}{\delta} + 2\beta_0 w$$

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Raw Output Value by Offer & Contract



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2SLS Estimates of Treatment Effects on Raw Output

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	-0.503^{**}	-0.498^{**}	-0.485^{**}	-0.362^{**}
	(0.206)	(0.200)	(0.199)	(0.185)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
Dep. Mean	7.91	7.91	7.91	7.91
R-squared	0.037	0.080	0.095	0.232
N	3030	3030	3030	3030

$$\tilde{Y}_i = \delta H_i + \boldsymbol{\eta} \boldsymbol{X_i} + \epsilon_i$$

$$\tilde{Y}_i \equiv Y_i - \gamma H_i \times W_i^P$$

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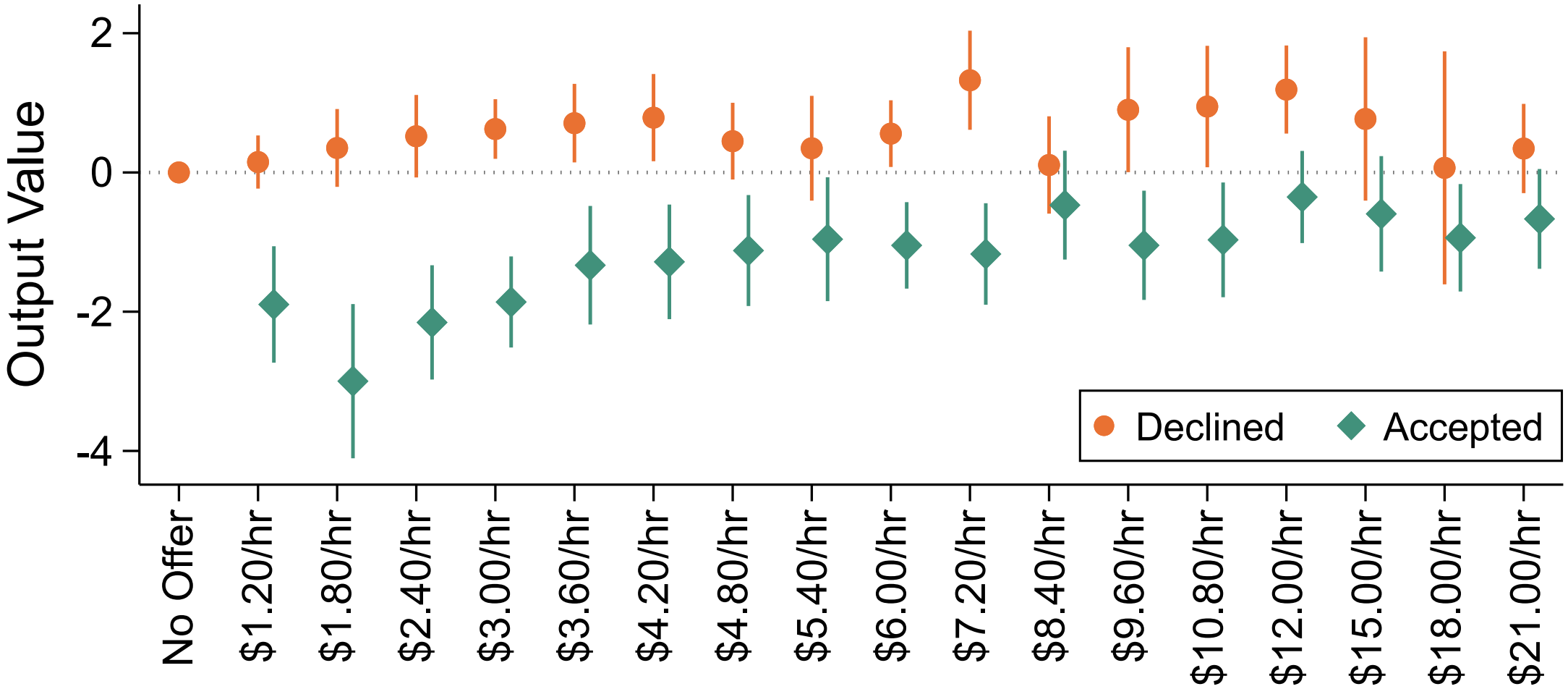
Selection on Raw Output by Log Wage Offer

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	−2.522***	−2.425***	−2.383***	−2.195***
	(0.325)	(0.318)	(0.318)	(0.297)
Declined × Log Hourly Wage Offer	0.172*	0.196**	0.213**	0.233***
	(0.0961)	(0.0932)	(0.0925)	(0.0855)
Accepted × Log Hourly Wage Offer	0.616***	0.565***	0.564***	0.497***
	(0.116)	(0.112)	(0.113)	(0.104)
Accepted × Log Effective Hourly Wage	−0.0796	−0.0583	−0.0583	−0.0153
	(0.121)	(0.117)	(0.118)	(0.110)
Control Group (No Hourly Offer)	−0.330	−0.296	−0.242	−0.228
	(0.226)	(0.219)	(0.219)	(0.201)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
Dep. Mean	7.91	7.91	7.91	7.91
R-squared	0.081	0.122	0.138	0.272
N	3030	3030	3030	3030

$$Y_i = \alpha H_i + \beta_0(1 - H_i) \times W_i + \beta_1 H_i \times W_i + \gamma H_i \times W_i^P + \boldsymbol{\xi} \boldsymbol{X_i} + \epsilon_i$$

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Selection on Raw Output by Log Wage Offer



$$Y_i = \sum_{w \in \mathbb{W}} [(\beta_0^w (1 - H_i) + \beta_1^w H_i) \times \mathbf{1}\{W_i = w\}] + \gamma H_i \times W_i^P + \boldsymbol{\xi} \boldsymbol{X_i} + \epsilon_i$$

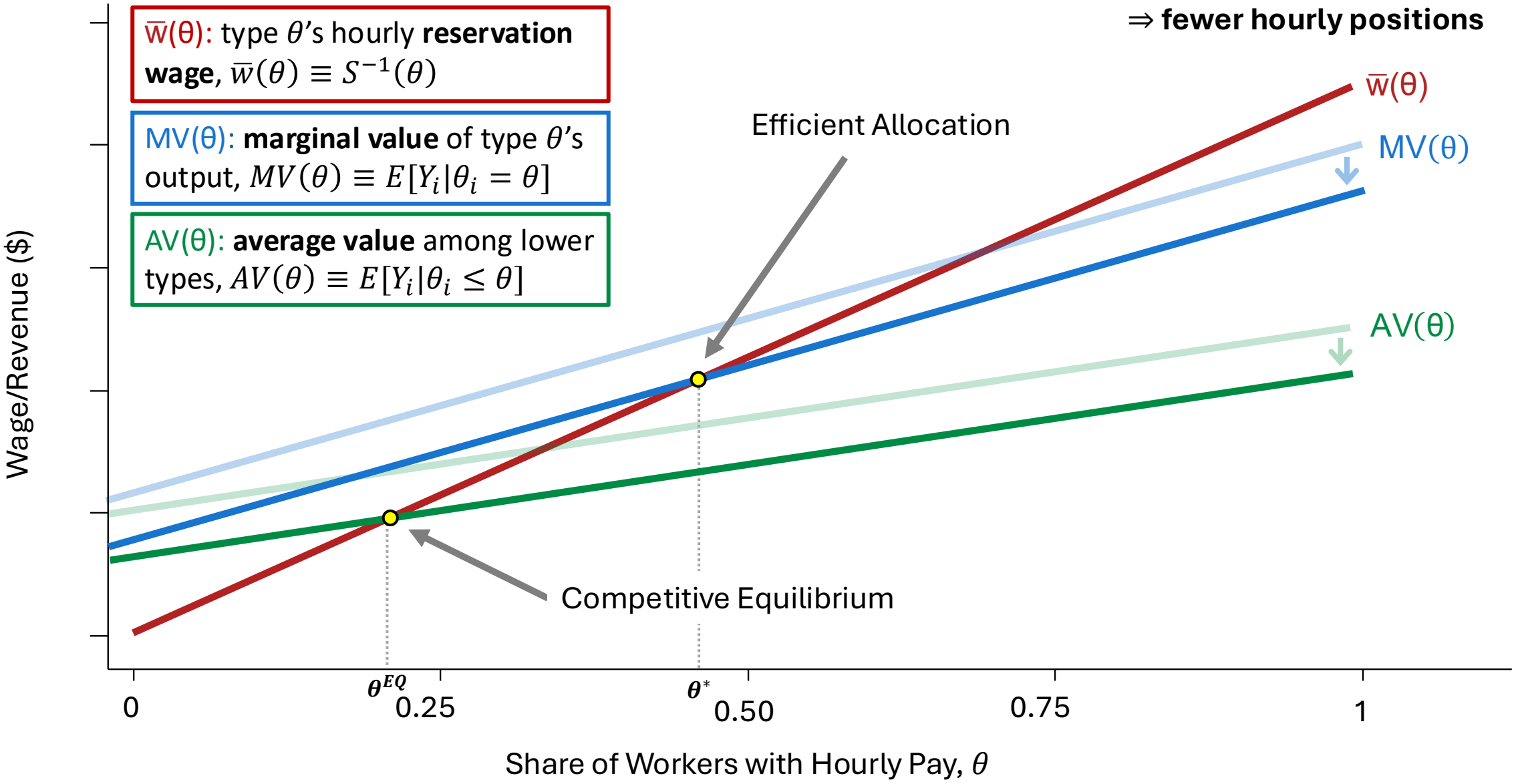
Back

Input monitoring costs reduce firm's value of fixed-wage hire
⇒ **fewer hourly positions**

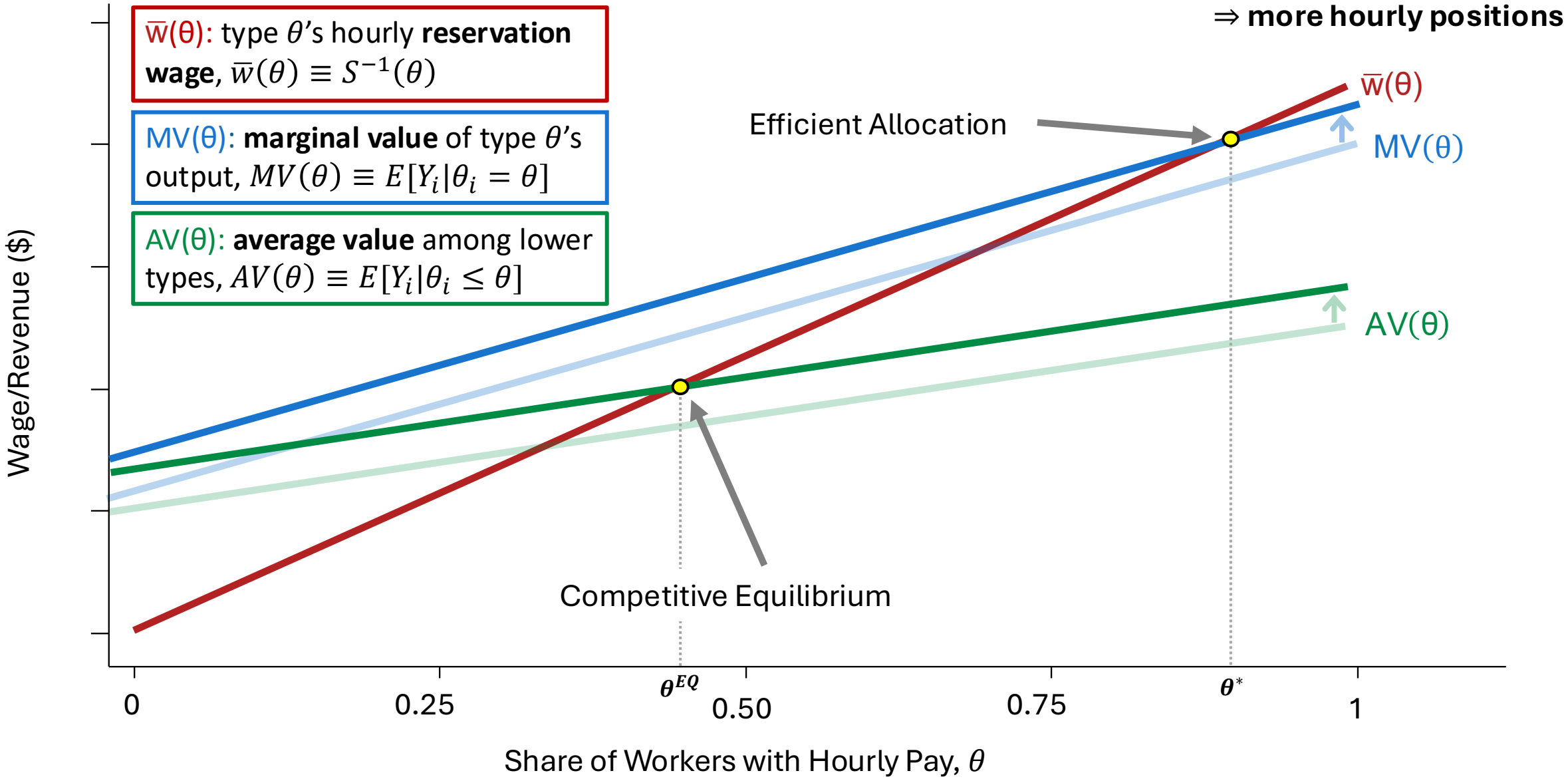
$\bar{w}(\theta)$: type θ's hourly **reservation wage**, $\bar{w}(\theta) \equiv S^{-1}(\theta)$

MV(θ): **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$

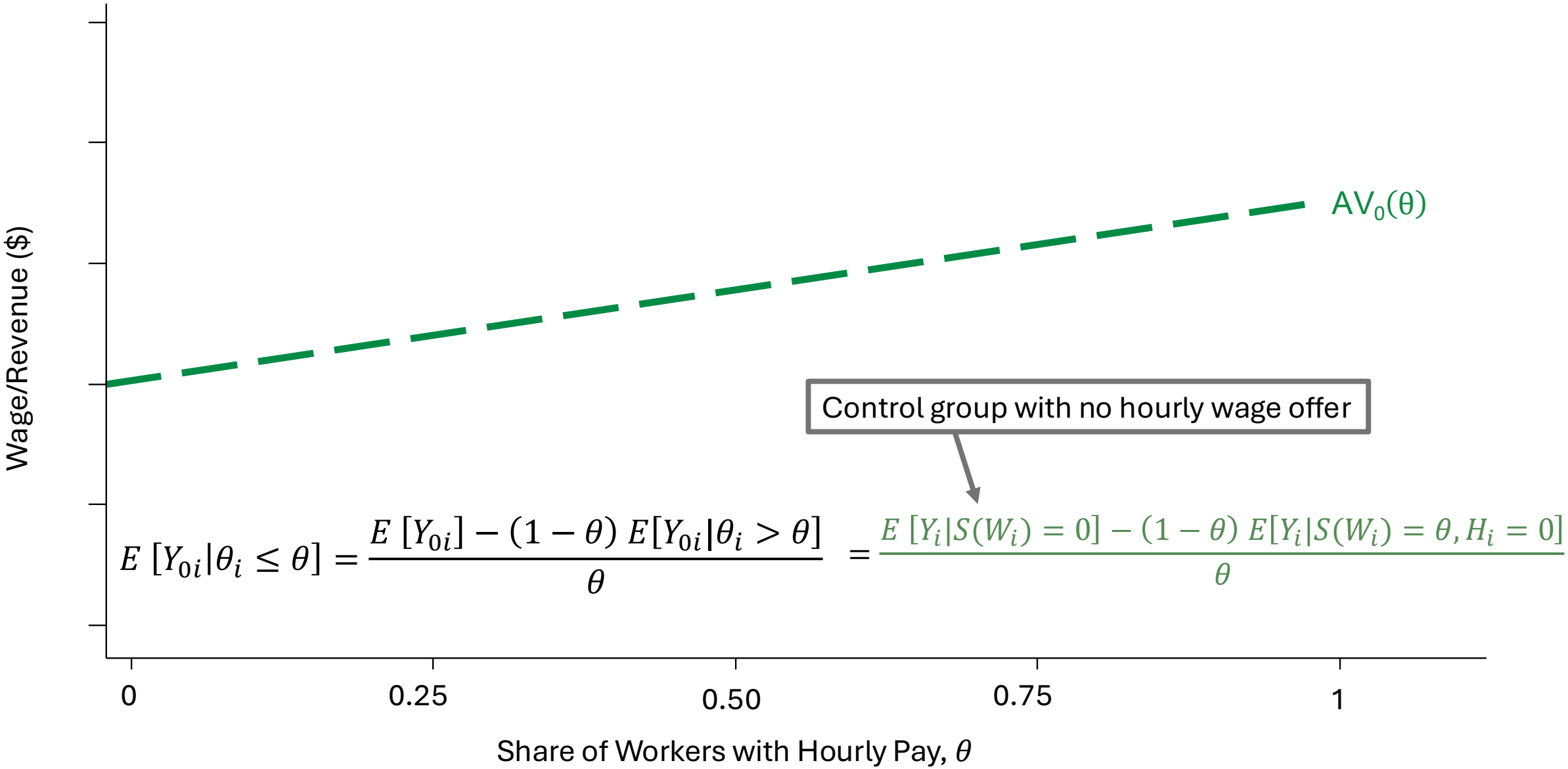
AV(θ): **average value** among lower types, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$



Output monitoring costs increase the firm's value of fixed-wage hire
⇒ **more hourly positions**



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Wage/Revenue ($)
$AV_0(\theta)$
Control group with no hourly wage offer
$E\ [Y_{0i}|\theta_i \leq \theta] = \frac{E\ [Y_{0i}] - (1-\theta)\ E[Y_{0i}|\theta_i > \theta]}{\theta} = \frac{E\ [Y_i|S(W_i) = 0] - (1-\theta)\ E[Y_i|S(W_i) = \theta, H_i = 0]}{\theta}$
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ

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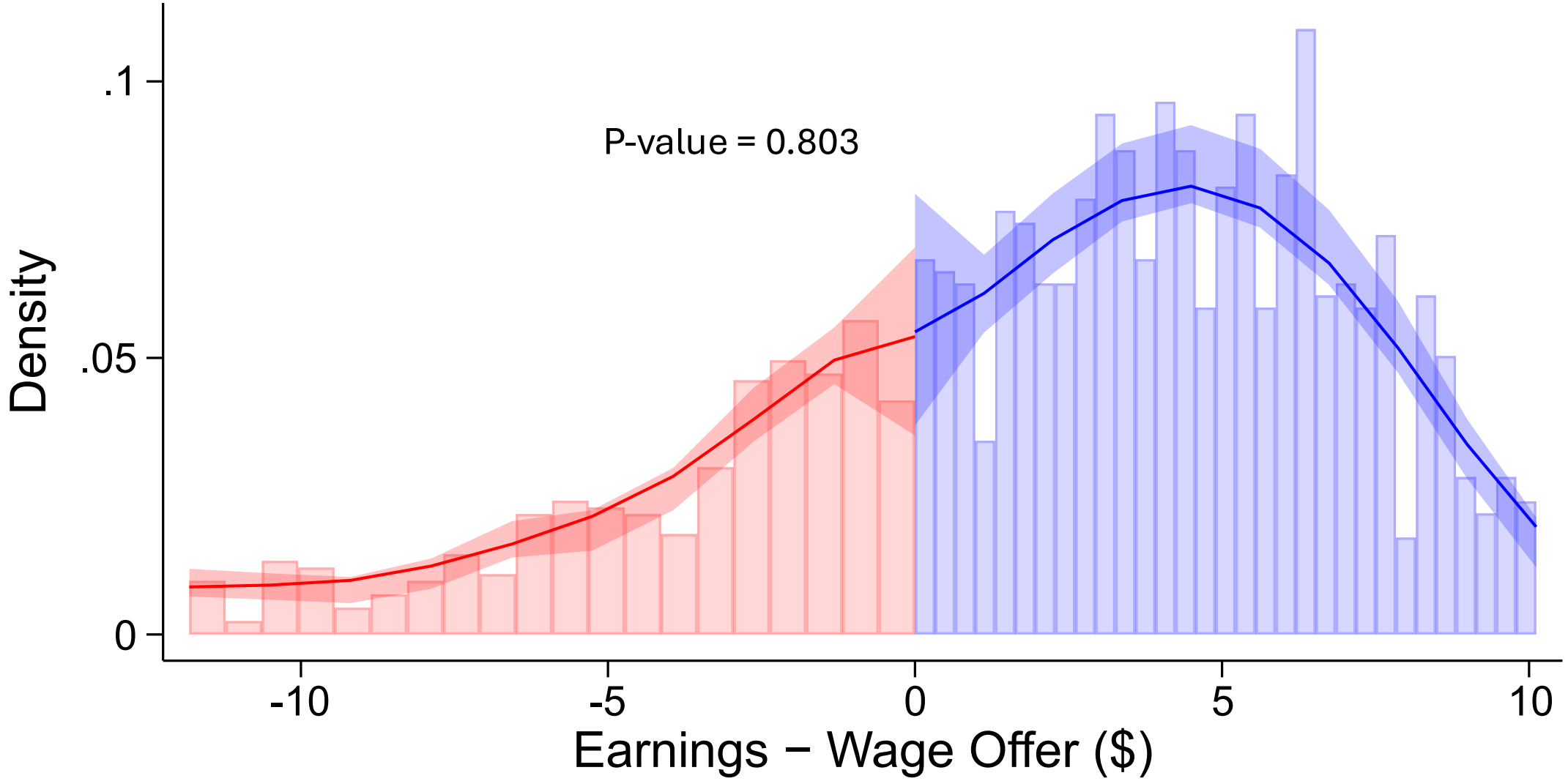
Estimation Details

Estimation Follows Carneiro et al. (2011):

1. Estimate $S(w) \equiv \Pr(H_i = 1 | W_i = w)$ using logit regression
2. Condition on X using double-residual regression (Robinson 1988)
   - X: Controls for number of previous tasks, task start time, and employment status
   - Assumes additive separability of X: $E\left[Y_{Ji} \middle| X_i = x, \theta_i = \theta\right] = \xi_J \tilde{X}_i + MV_J(\theta)$
3. Separately estimate $E[Y_i | S(W_i) = \theta, H_i = 1]$ and $E[Y_i | S(W_i) = \theta, H_i = 0]$
   - Local quadratic regressions of $\tilde{Y}$ on $S(W_i)$ for hourly and piece-rate workers (bw=0.2)
4. Differentiate with respect to θ

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No Evidence of Earnings Bunching at Wage Offer



Estimated density using optimal bandwidths of 3.93 (left) and 3.38 (right). Methods from Cattaneo et al. (2020).

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Policy: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{subsidy} = \frac{Welfare\ Benefit}{Net\ Govt\ Expenditure}$$

$Welfare\ Benefit$: The aggregate amount workers are willing to pay for subsidy

$Net\ Govt\ Expenditure$: The aggregate amount spent including fiscal externalities

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Policy: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{subsidy} = \frac{Transfer + Insurance\ Value}{Net\ Govt\ Expenditure}$$

$Welfare\ Benefit$: The aggregate amount workers are willing to pay for subsidy

- $Transfer$ from government to hourly workers
- $Insurance\ Value$ to workers who move from piece-rate to hourly pay

$Net\ Govt\ Expenditure$: The aggregate amount spent including fiscal externalities

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Policy: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{subsidy} = \frac{Transfer + Insurance\ Value}{Transfer + Tax\ Loss\ from\ MH}$$

$Welfare\ Benefit$: The aggregate amount workers are willing to pay for subsidy

- $Transfer$ from government to hourly workers
- $Insurance\ Value$ to workers who move from piece-rate to hourly pay

$Net\ Govt\ Expenditure$: The aggregate amount spent including fiscal externalities

- $Transfer$ from government to hourly workers
- $Tax\ Loss\ from\ MH$ when hourly workers shirk and pay less income tax

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Policy: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{subsidy} = \frac{\delta\theta^{\delta} + \int_{\theta^{EQ}}^{\theta^{\delta}} \left(MV_1(\theta) - \bar{w}(\theta)\right) d\theta}{\delta\theta^{\delta} - \int_{\theta^{EQ}}^{\theta^{\delta}} \frac{\tau}{1-\tau} MH(\theta) d\theta}$$

$Welfare\ Benefit$: The aggregate amount workers are willing to pay for subsidy

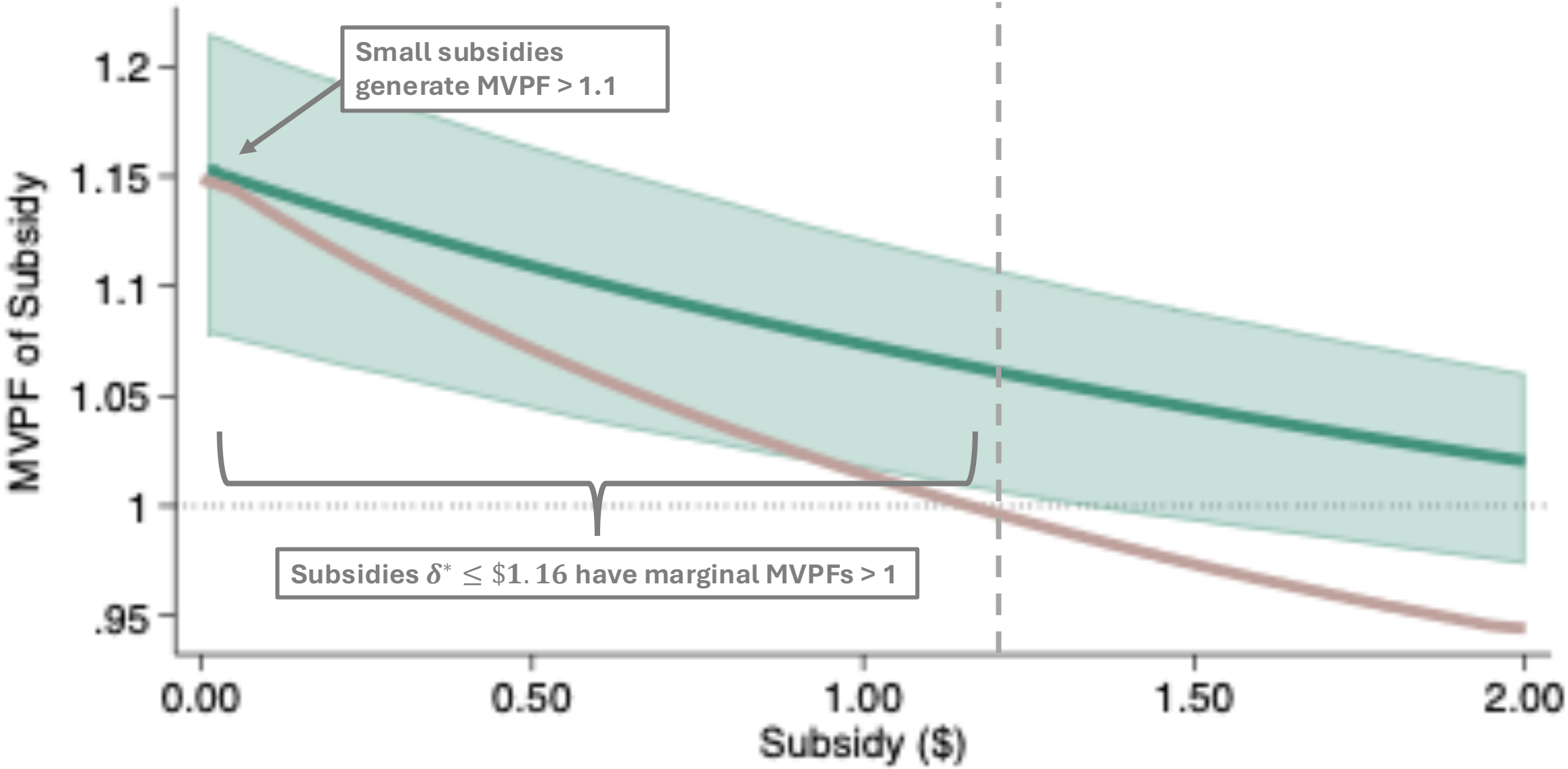
- $Transfer$ from government to hourly workers
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$Net\ Govt\ Expenditure$: The aggregate amount spent including fiscal externalities

- $Transfer$ from government to hourly workers
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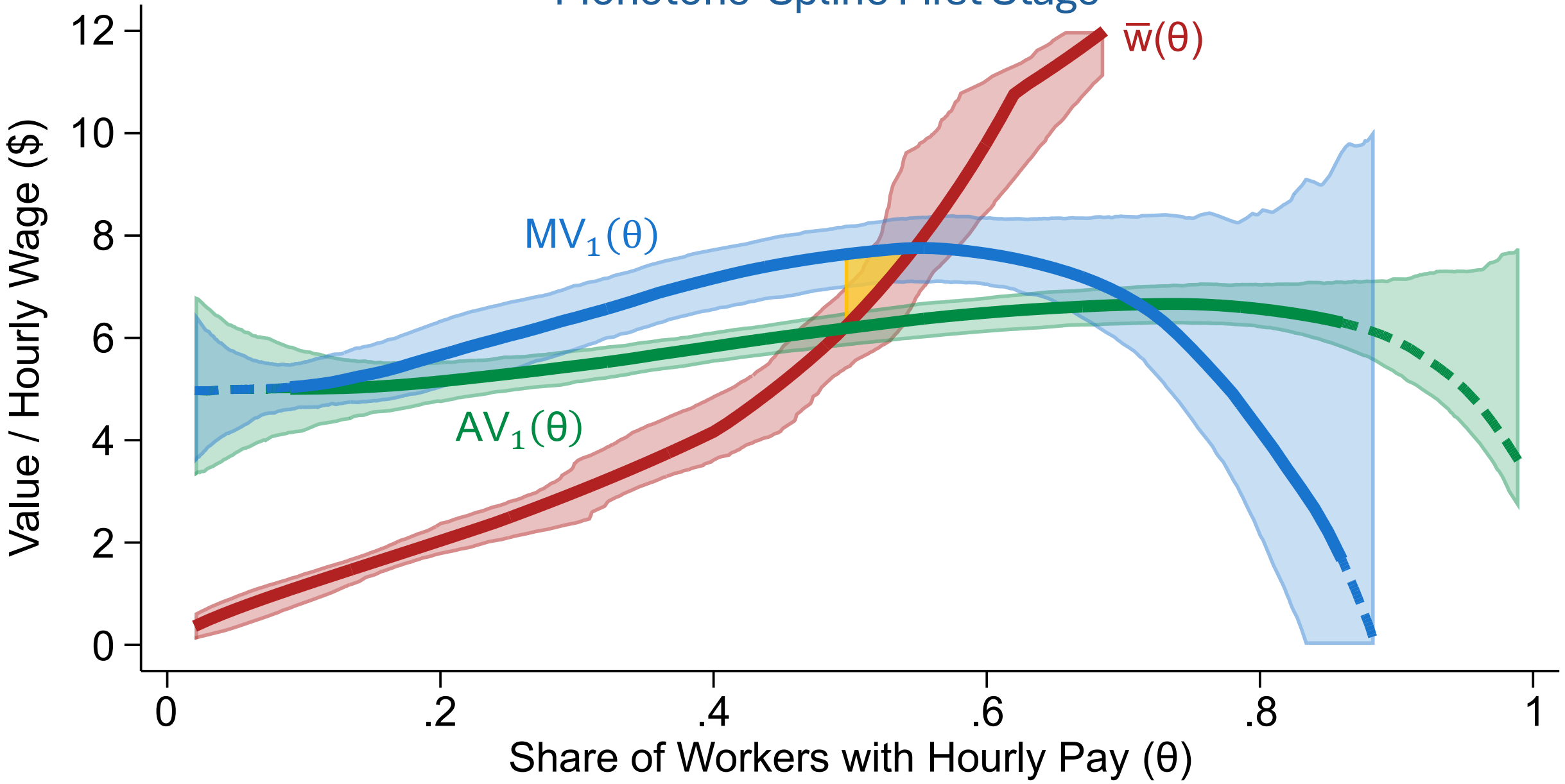
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MVPF of Hourly Wage Subsidy



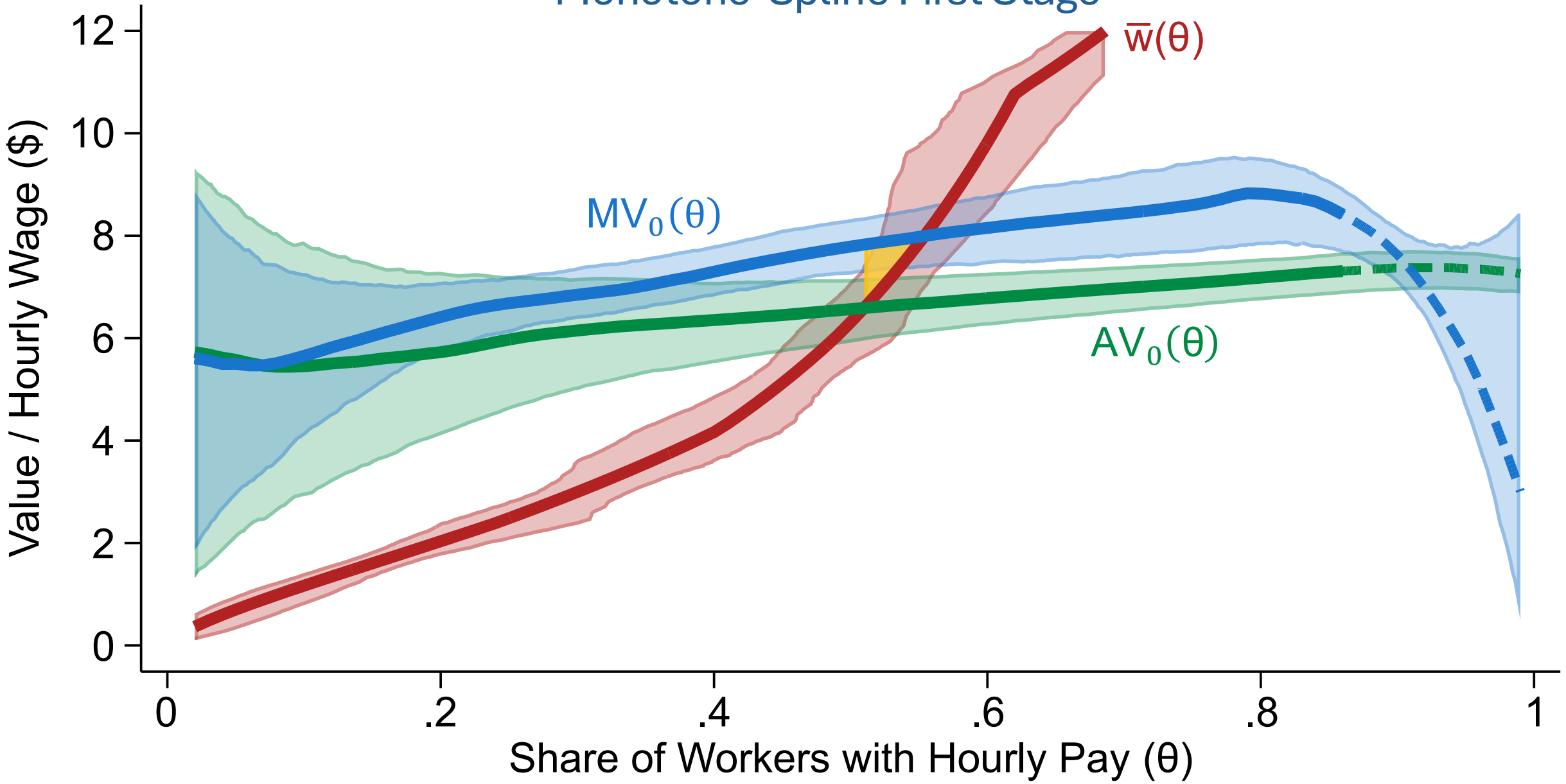
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Equilibrium with Adverse Selection and Moral Hazard: Monotone-Spline First Stage



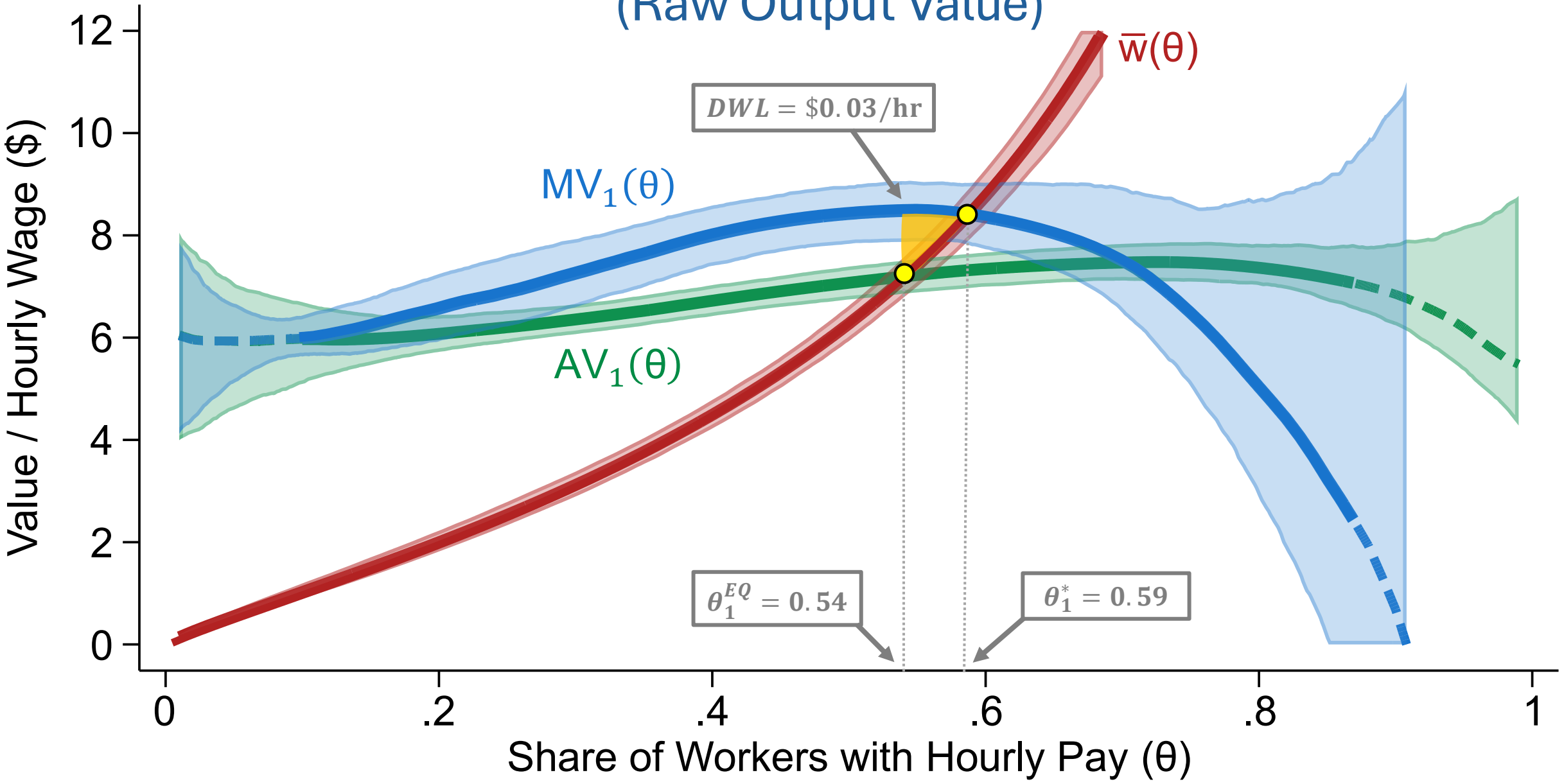
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Equilibrium with Adverse Selection and No Moral Hazard: Monotone-Spline First Stage



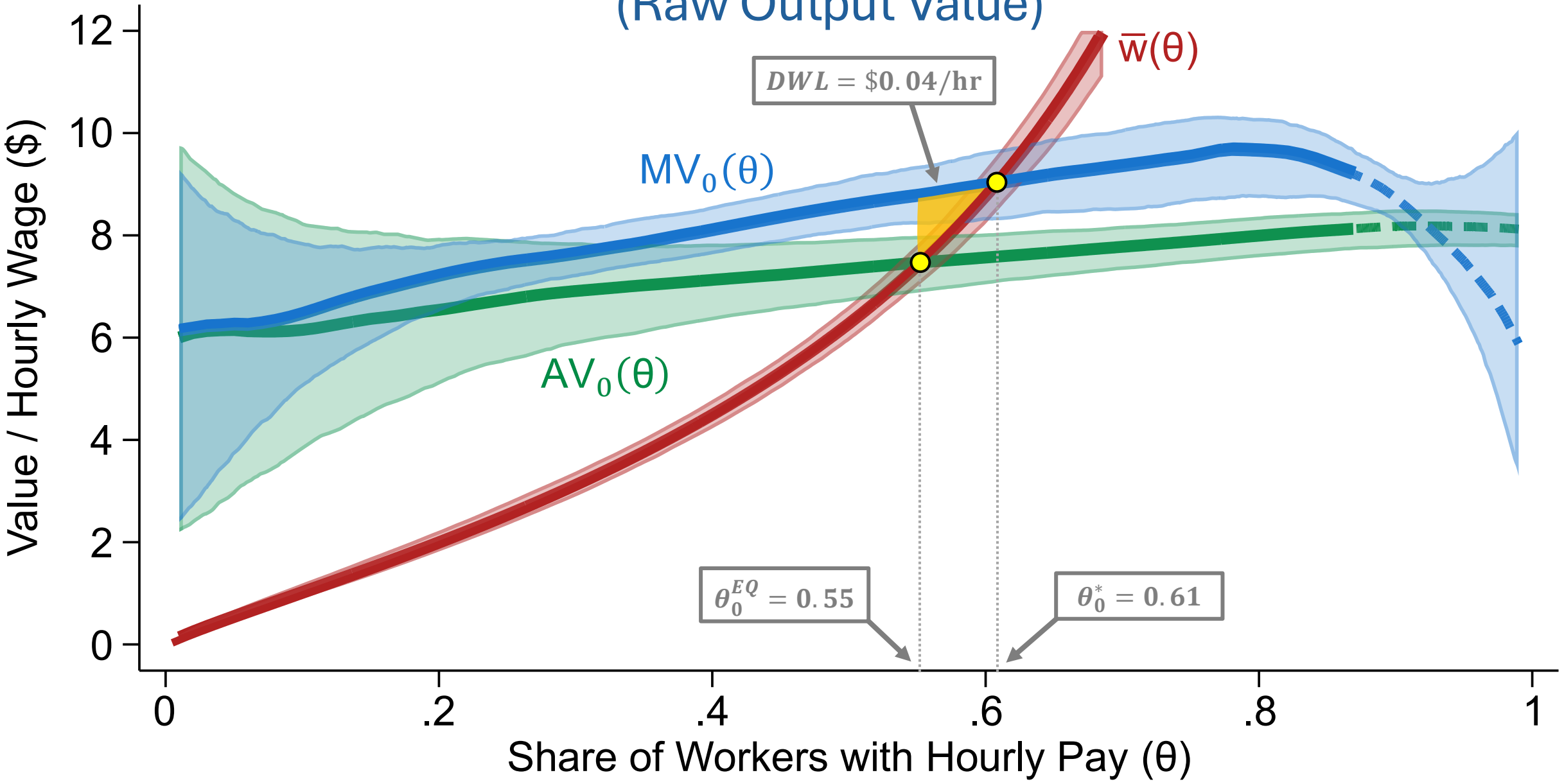
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Equilibrium with Adverse Selection and Moral Hazard (Raw Output Value)



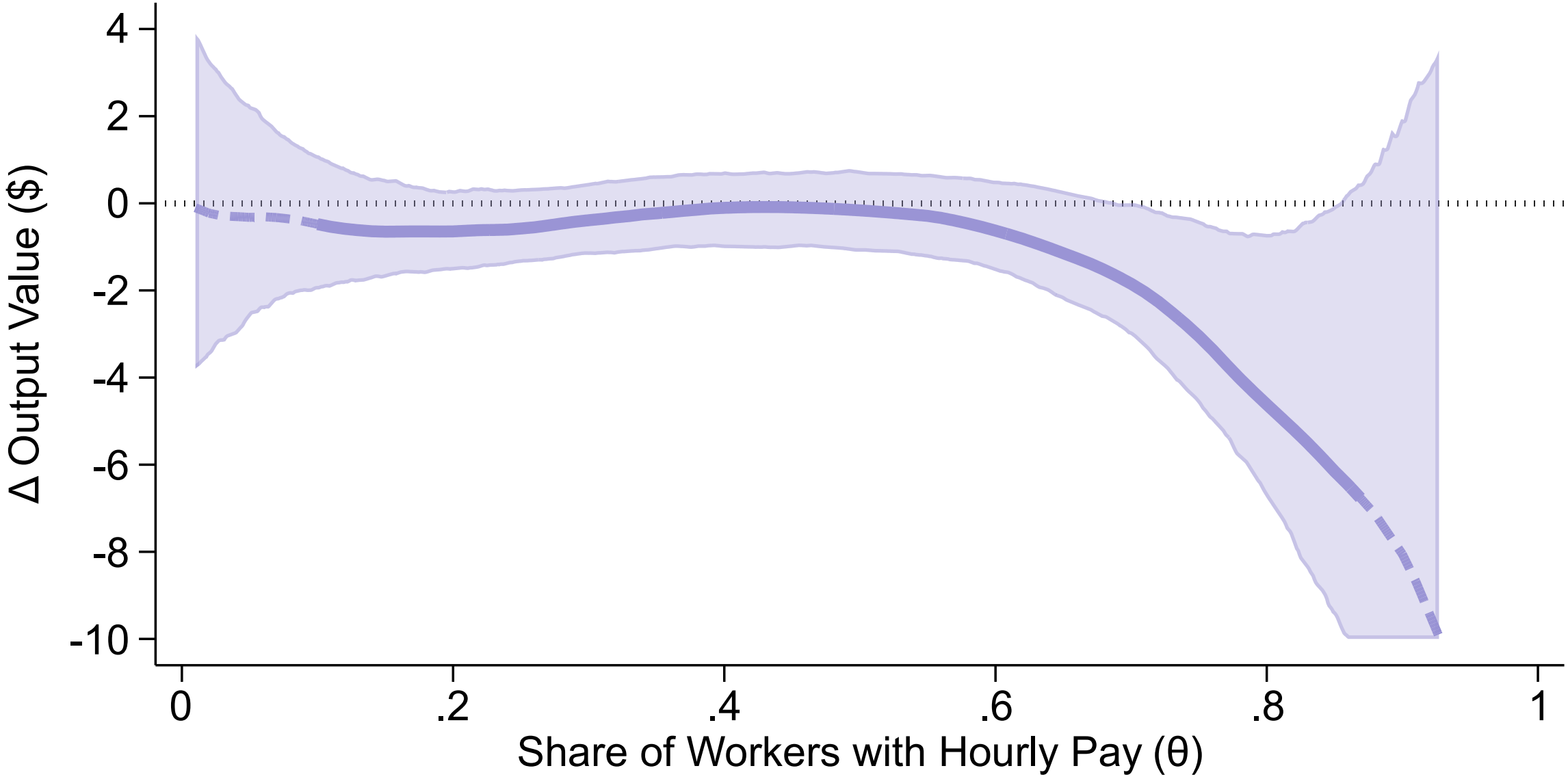
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Equilibrium with Adverse Selection and No Moral Hazard (Raw Output Value)



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Moral Hazard: Estimates of Marginal Treatment Effects (Raw Output Value)



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